



The TikZpingus package

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<https://github.com/EagleoutIce/tikzpingus>

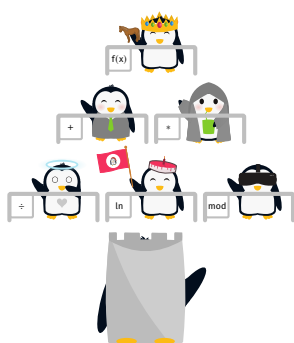
Version v1.4 – 2026/02/27

Motivation

For my slides at university, I started to use the famous $\text{\LaTeX}$ -package *tikzducks* a few years ago. Yet, it seemed somewhat of a necessity to extend the range of available “cute” animals in $\text{\LaTeX}$. Therefore, I started writing this package: *tikzpingus*.⁽¹⁾

Please note: While *tikzpingus* is certainly inspired by *tikzducks*, it does offer a different set of features (e.g., multiple wing positions, ...).

I would be happy for any feedback or issues on the *tikzpingus*-GitHub.



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⁽¹⁾ Why “pingu” and not “pengu”? Well, this is the third try on achieving cute penguins without using any templates or vector formats as a basis. As a German, the short form “pingu” was merely a typo that originated from the German word “pinguin” for “penguin”. It somewhat stuck...

All body type Options

See [subsection 2.7](#) for more info.



none



normal



chubby



legacy



tilt-right



tilt-left

Use `body type=<option>` to set the *body type* of a penguin.

All eyes Options

See [subsection 2.6](#) for more info. You can also set eyes individually with `/pingu/left eye` and `/pingu/right eye`.



none



normal



vertical



shiny



wink



shock



devil



sad



angry



hearts



critical

Use `eyes=<option>` to set the *eyes* of a penguin.

All wings Options

See [subsection 2.5](#) for more info. You can also set wings individually with `/pingu/left wing` and `/pingu/right wing`.



none



normal



wave



raise



grab



shock

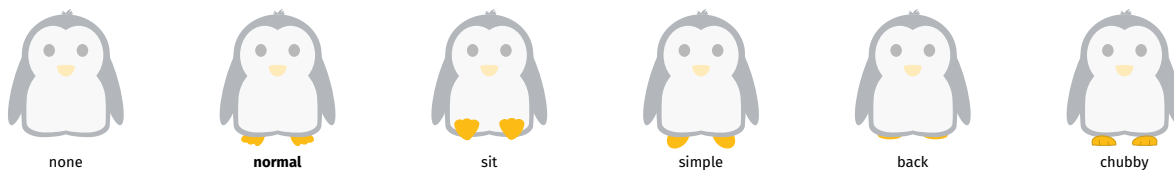


hug

Use `wings=<option>` to set the *wings* of a penguin.

All feet Options

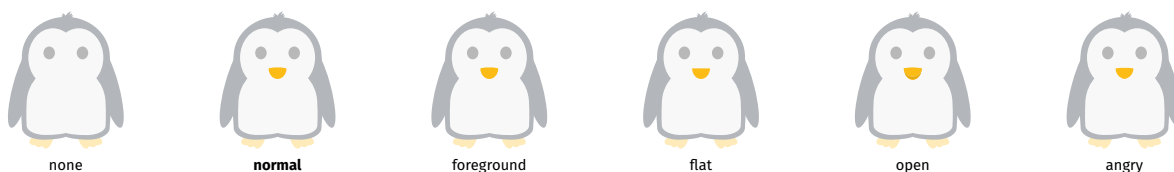
See [subsection 2.7](#) for more info. You can also set feet individually with `/pingu/left foot` and `/pingu/right foot`.



Use `feet=<option>` to set the *feet* of a penguin.

All bill Options

See [subsection 2.7](#) for more info.



Use `bill=<option>` to set the *bill* of a penguin.

All hairstyle Options

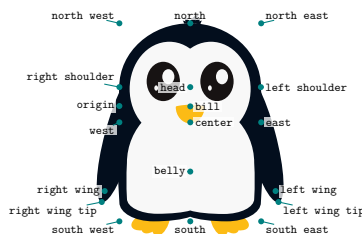
See [subsection 2.7](#) for more info.



Use `hairstyle=<option>` to set the *hairstyle* of a penguin.

All anchor Options

See [subsection 2.2](#) for more info. *head top* and *feet* are aliases of *north* and *south*; any other point works as well.



Use `anchor=<option>` to move that point of a penguin onto the origin.

Gadget Overview

These automatically link to the respective documentation sections.



second shirt



cape



horse left/right



devil fork left/right



laptop left/right



plank left/right



sword left/right



magnifier left/right



microphone left/right



drum stick left/right



hammer left/right



cross left/right



staff left/right



present left/right



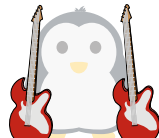
german flag left/right



pride flag left/right



flag left/right



guitar left/right



broom left/right



light-staff left/right



lightsaber left/right



sign post left/right



cane left/right



flower left/right



fish left/right



lollipop left/right



banner



mask



shoes



deer hat



basket left/right



snowball left/right



santa beard



headphone



horse behind



bee



jack o lantern



cake-hat



crown





sneaker



blush



spear left/right



cloak



shirt



scarf

1 Introduction

1.1 Dependencies

As this package is constantly work in progress, the concrete dependencies may change any time. At the moment, it loads [TikZ](#), which loads a lot of other packages (e.g. [xcolor](#)), and [etoolbox](#). Furthermore, the following TikZ-Libraries are in use:⁽²⁾ *intersections*, *shadings*, *patterns.meta*, *decorations.pathmorphing*, and *shapes.symbols*.

Please note: This document is typeset with [xlistings](#) and [code-link](#), which link every key and command in the listings to its explanation. They are only used for this document (see the git submodule `doc/xlistings`) and are *not* required by *tikzpingus*.

1.2 Copyright

Copyright © *Florian Sihler*. Permission is granted to copy, distribute and/or modify this software under the terms of LPPL Version 1.3c (please note that this changed on 2026-02-27 from GPL v3, versions **after** v1.3 are licensed with LPPL 1.3c). The shown example penguins are purely fictional characters, any resemblance to real penguins or real persons is purely coincidental and no copyright infringement is intended.

2 Usage

If you just want a penguin, import the package and start with the following:

One small penguin

```
\begin{tikzpicture}
  \pingu
\end{tikzpicture}
```



There are *a lot* of configuration-options which can be passed as an optional argument via the known `<key>=<value>`-style. See [section](#) at the start for a complete gadget overview.

Happy penguin with cup!

```
\begin{tikzpicture}
  \pingu[left wing wave, right wing grab,
    eyes shiny, cup]
\end{tikzpicture}
```



Please note, that “left” and “right” have been chosen from the penguin-perspective.

`\pingu` [penguin keys]

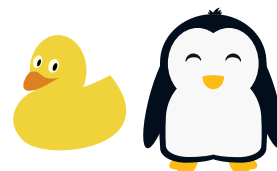
The command to draw the penguin! If you want to use `\pengu` instead, you can use `\let\pengu\pingu` in your preamble to create an alias.

⁽²⁾ A lot of the libraries loaded are important only for specific extras. I plan on cleaning them up.

Besides the keys defined by this package, you can use the keys of TikZ and pgf as well (the duck was generated by the lovely [tikzducks](#) package):

The Reunion

```
\begin{tikzpicture}
  \duck
  \pingu[xshift=2.7cm, yshift=14mm,
    eyes wink]
\end{tikzpicture}
```

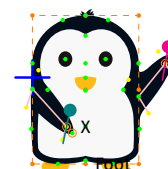


2.1 Using the Coordinates

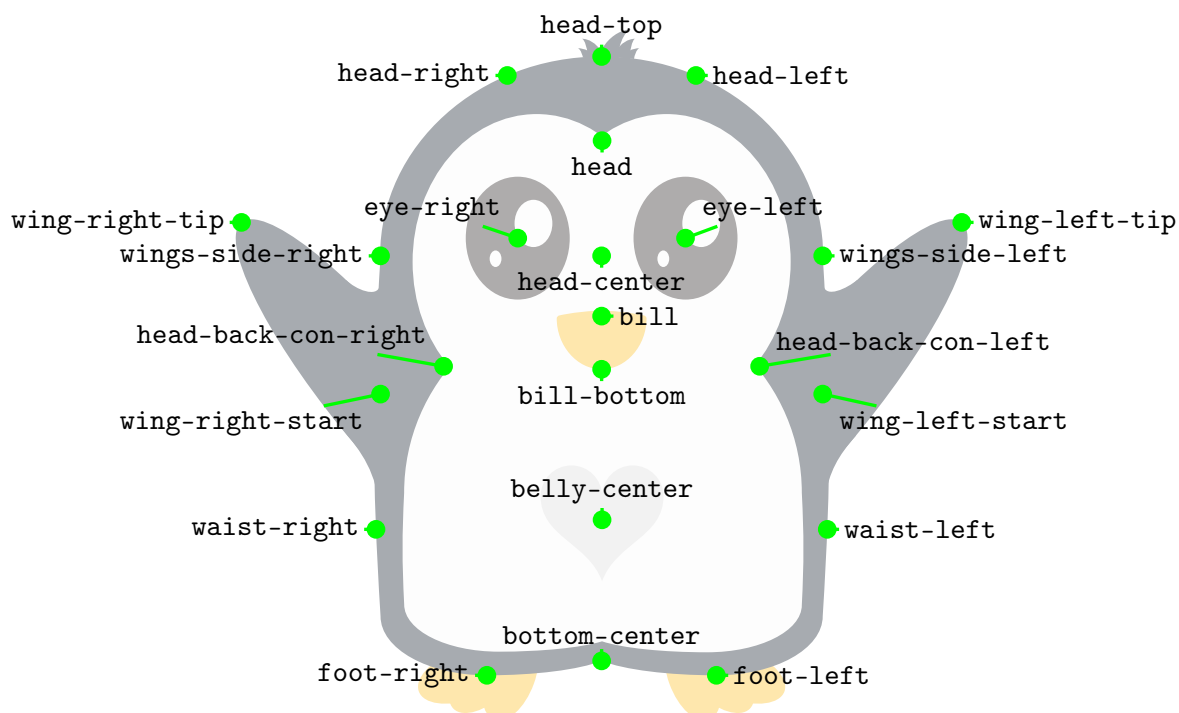
While there are a lot of gadgets available already, every penguin is accompanied by several adaptive coordinates to place custom items, texts, ... (the anchors are described in [subsection 2.2](#) and their code in [subsection C.11](#)). They can be visualized by the `/pingu/meta-dots` option. Furthermore, some extras create further coordinates themselves! All coordinates are available with `<pingu-name>-<coordinate>`. While the default name of a penguin is "pingu", it can be changed with the name option:

Lotta dots

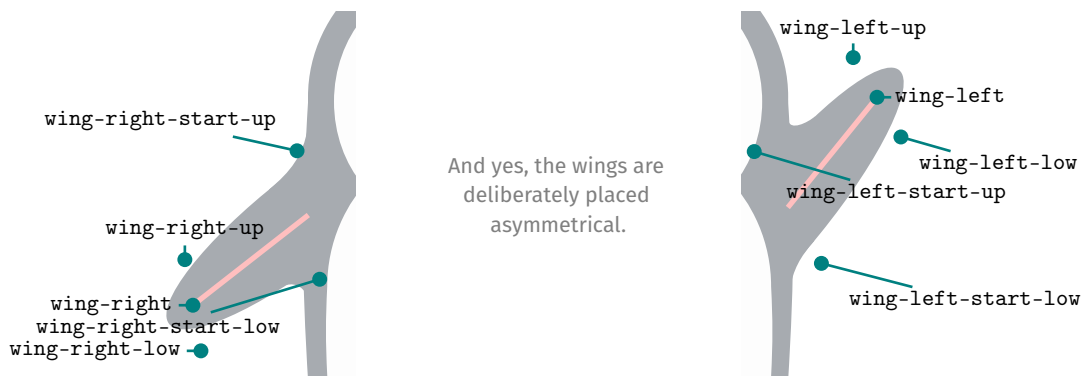
```
\begin{tikzpicture}
  \pingu[meta dots, left wing wave,
    right wing grab, name=paula]
  \node at (paula-belly-center) {X};
  \node at (paula-foot-left) {Foot};
\end{tikzpicture}
```



Lets look at those coordinates in more detail (all labels are to be prefixed by `</pingu/name>-`):



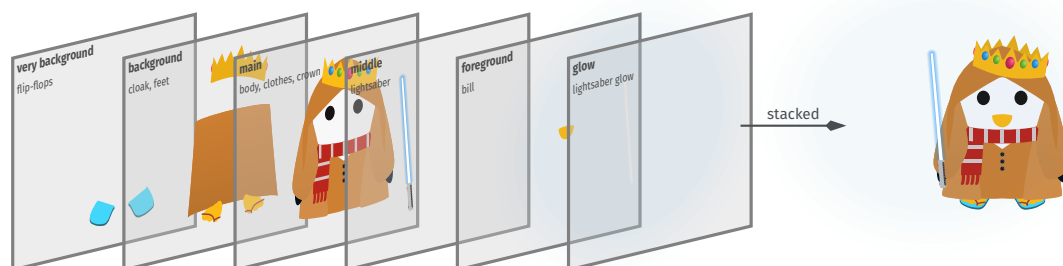
The Wings This view excluded a lot of special data collected on the wings! While there is more information stored for each wing, the following five coordinates are the most important to place items into penguins hand:



The Body Similarly to the wing position, different body types can change the coordinates (left the `/pingu/body_type chubby` and right the `/pingu/body_type legacy`):



The Layers The parts of a penguin which are behind or in front of the rest are drawn in five layers, which belong to the package and are the same for every penguin. The figure shows them for one penguin; each plane contains only what is drawn in this layer (the penguin on the right is the sum of all planes). When a penguin is finished, its five layers are put into one box of a sixth layer, the stack layer, one above the other, and are empty again. So a picture needs six layers, no matter how many penguins it contains, and a penguin which is drawn later is above all parts of an earlier one. With `/pingu/:back`, the order of the five layers is reversed. Only the glow of a lightsaber is not stacked: it has a seventh layer, which is shared by all penguins of the picture and lies above all of them.



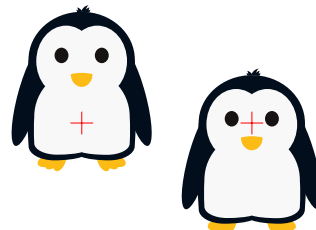
The penguin of the figure: `\pingu[flip-flops,cloak,scarf,shirt,crown,bill above,lightsaber right]`

2.2 Anchors and Placement

The origin of a penguin sits at its right shoulder. `/pingu/anchor` moves the penguin so that another of its points lands on the origin instead:

Two anchors

```
\begin{tikzpicture}
  \pingu[anchor=belly]
  \pingu[anchor=head,xshift=3cm]
  \foreach \x in {0,3cm}
    \draw[red] ([xshift=\x]-2mm,0) -- +(4mm,0)
               ([xshift=\x]0,-2mm) -- +(0,4mm);
\end{tikzpicture}
```



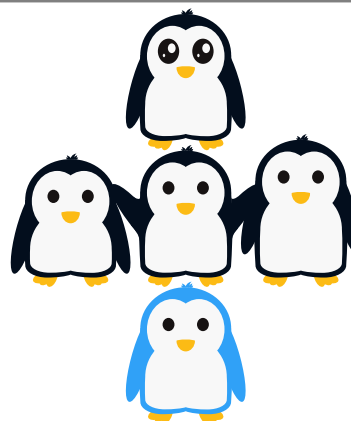
Besides the names listed at `/pingu/anchor` any point works, e.g. `anchor={(1cm,-5mm)}`.

Anchors refer to the *silhouette of the body*; gadgets, wings, and feet may stick out of it, so the anchors do not change with the extras a penguin wears. Its nine outer anchors are coordinates as well (prefixed by `</pingu/name>-`): center, north, south, east, west, north east, north west, south east, and south west. The anchor in use is available as `</pingu/name>-anchor` and, together with the silhouette, shown by `/pingu/meta-dots`.

Those outer anchors are what `/pingu/right of`, `/pingu/left of`, `/pingu/above of`, and `/pingu/below of` use, each taking the name of the other penguin and, after a colon, the distance to keep:

A row of penguins

```
\begin{tikzpicture}
  \pingu[name=a,wings wave]
  \pingu[name=b,right of=a:5mm,large size]
  \pingu[name=c,left of=a:5mm,small size]
  \pingu[name=d,above of=a:2mm,eyes shiny]
  \pingu[name=e,below of=a:2mm,body=pingu@blue]
\end{tikzpicture}
```



`/pingu/right of` and `/pingu/left of` keep both penguins on the same ground line, `/pingu/above of` and `/pingu/below of` center them horizontally. An `/pingu/anchor` passed *after* one of them keeps the target point but changes which point of the new penguin is moved onto it.

2.3 Colors

A lot of options allow for a color to be passed. In general, you can provide any color that TikZ is happy with! Yet, there are some predefined pingu-colors shipped with this package:

● **pingu@main**
r: 3, g: 14, b: 29

● **pingu@black**
r: 23, g: 19, b: 19

● **pingu@silver**
r: 192, g: 192, b: 192

● **pingu@bronze**
r: 205, g: 127, b: 50

○ **pingu@white**
r: 248, g: 248, b: 248

● **pingu@yellow**
r: 252, g: 187, b: 21

● **pingu@lightblue**
r: 174, g: 229, b: 243

● **pingu@blue**
r: 48, g: 161, b: 247

● **pingu@green**
r: 129, g: 204, b: 41

● **pingu@red**
r: 217, g: 40, b: 28

● **pingu@purple**
r: 244, g: 48, b: 93

Furthermore, there is the special color “!hide” (`\@pingu@none`) which is available for most⁽³⁾ extras and wing-items. This color prohibits the compartments from being drawn. To be more precise, the package defines the macro `\@pingu@none`, which is matched against the selected color.

As an example, lets take a look at the `/pingu/cup`-extra, which provides an additional key `/pingu/cup straw` to color the straw:

Cup without a straw

```
\begin{tikzpicture}
  \pingu[wings grab, cup=pingu@purple,
        cup straw=pingu@blue]
  \pingu[wings grab, cup, xshift=2.7cm,
        cup straw=!hide]
\end{tikzpicture}
```



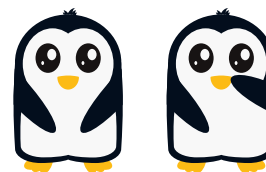
As you can see, using `!hide`, the straw will not be drawn.

2.4 Setting the defaults

You do not have to re-state every key. With `\pingudefaults` and `\pingudefaultsappend` (similar, but extends the current options) you can set default-options for all penguins to come:

Change the mainstream

```
\begin{tikzpicture}
  \pingudefaults{wings grab, eyes shiny}
  \pingu
  \pingu[left wing shock, xshift=2.7cm]
\end{tikzpicture}
```



`\pingudefaults{penguin keys}`

Sets the given *penguin keys* as defaults for the local group. Overwrites any previous defaults (see `\pingudefaultsappend`).

`\pingudefaultsappend{penguin keys}`

Appends the given *penguin keys* as defaults for the local group. To overwrite these keys, see `\pingudefaults`.

2.5 Changing the wings

As already demonstrated, it is possible to change the wing positions! All selected wing-items will adapt to the wing-position (although not all wing-items will make sense with every wing-position). Currently, there are the following wing-positions: “none”, “normal”, “wave”, “raise”, “grab”, “shock”, and “hug”. “none” is a special wing-position: it omits the drawing of wings (teaser: every selection has a none-option, which prohibits the part from being drawn)!

For each valid wing-position you can use `wings <position>` to change both wings or `left wing <position>` and `right wing <position>` to change only one wing respectively. The default wing-position is “normal”. If you supply multiple options for a wing, only the last one survives.⁽⁴⁾ This is shown in Box “Wing-Showcase”.

⁽³⁾ Why just “most”? Well, this package is work in progress and I have added the option late, so I may have forgotten to patch some keys.

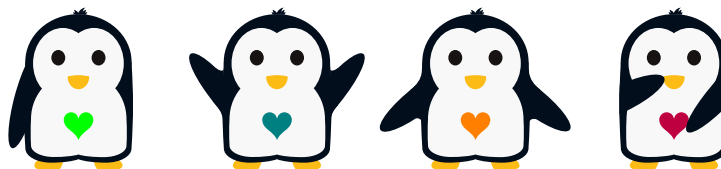
⁽⁴⁾ For the sake of completeness: `wings <position>`, `left wing <position>`, and `right wing <position>` are just alternatives I prefer: `wings=<position>`, `left wing=<position>` and `right wing=<position>`.

Wing-Showcase

```

\begin{tikzpicture}
  \pingu[left wing none, heart=green]
  \pingu[wings wave, heart=teal, xshift=3.5cm]
  \pingu[wings hug, heart=orange, xshift=7cm]
  \pingu[left wing grab, right wing shock, heart=purple, xshift=10.5cm]
\end{tikzpicture}

```



2.6 Changing the eyes

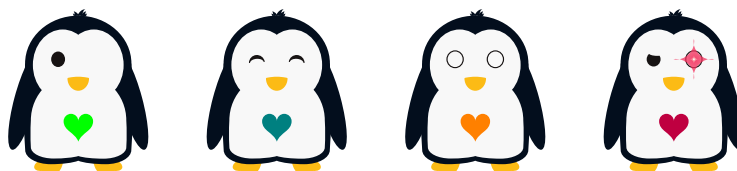
Just like the wings, there are a couple of different eye-styles to choose from: "none", "normal", "vertical", "shiny", "wink", "shock", "devil", "sad", "angry", "hearts", and "critical". Similar to the wings, there is a "none" and a "normal"-option (which is the default). Furthermore, the convenient selectors `eyes <style>`, `left eye <style>`, and `right eye <style>` exist as well. All of this is showcased in Box "Eye-Showcase".

Eye-Showcase

```

\begin{tikzpicture}
  \pingu[left eye none, heart=green]
  \pingu[eyes wink, heart=teal, xshift=3.5cm]
  \pingu[eyes shock, heart=orange, xshift=7cm]
  \pingu[left eye devil, right eye angry, heart=purple, xshift=10.5cm]
\end{tikzpicture}

```



2.7 Changing other components

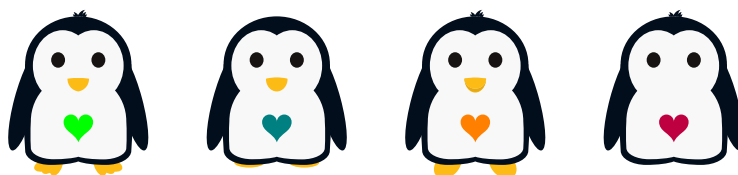
Just like for the wings and the eyes, you can change the following body parts:

- The *body type* itself
Select from: "none", "normal", "chubby", "legacy", "tilt-right", and "tilt-left".
- The *feet* (again with separate left and right)
Select from: "none", "normal", "sit", "simple", "back", and "chubby".
- The *bill* (does not have left and right, as there is just one)
Select from: "none", "normal", "foreground", "flat", "open", and "angry".
- The *hairstyle* (does not have left and right)
Select from: "none" and "normal".

For each selection, “none” will prohibit the drawing, and “normal” is the default chosen. See Box “Bodyparts-Showcase” for an example.

Bodyparts-Showcase

```
\begin{tikzpicture}
  \pingu[bill angry, heart=green]
  \pingu[feet back, hairstyle none, heart=teal, xshift=3.5cm]
  \pingu[bill open, feet simple, heart=orange, xshift=7cm]
  \pingu[feet none, bill none, heart=purple, xshift=10.5cm]
\end{tikzpicture}
```



2.8 Predefined Styles

While the penguin options offer the modification of basically every drawing routine (through other styles like @block), it is tedious to change them every time. So I have started to create some predefined styles, that do change some of the penguins appearance (and are completely new, so beware of bugs). The following list presents an incomplete overview:

- `:line`, draw everything with a line.



- `:hide`, do not draw main pingu.

- `:fill`, fill main penguin.



- `:mix`, mix the main penguin with a given color.



- `:ghost parts`, draw components with transparency.



- `:toast`, one slice of cuteness.



- `:ghost`, draw all layers with transparency.



- `:mix-draw`, mix the main penguin completely with a given color.



- `:devil`, set main “devil”-components.



- `:mix-all`, mix everything with a given color.



- `:back`, flip the penguin (swaps left & right).



Currently, only some of the styles do affect other items. As an example, consider `:line`, that changes the draw-style of wing-items and extras:

Line Penguin

```
\begin{tikzpicture}
  \pingu[:line, princess crown, silver medal]
\end{tikzpicture}
```



2.9 Randomness

Each selection (like the wings or the eyes) can receive a special command `!random`. If given, the penguin will receive a randomly picked component. Please note, that *none* (the component removing it) will never be picked. The first line in the example in Box “Random Penguin” sets the seed.

Random Penguin

```
\pgfmathsetseed{\number\pdfandomseed}
\begin{tikzpicture}
  \pingu[wings=!random,eyes=!random,
    left foot=!random,
    bill=!random,
    hairstyle=!random]
\end{tikzpicture}
```



In a more general fashion, there is a `/pingu/random from` key for completely random penguins.

```
/pingu/random from = <list>
```

You can pass any list of penguin keys and exactly one of them will be selected. You can nest `/pingu/random from`-calls. Please note, that the items are not separated by comma but in braces. The first line in the example sets the seed:

```
\pgfmathsetseed{\number\pdfandomseed}
\begin{tikzpicture}
  \pingu[random from={{eye patch left}{eye patch right}{halo
    ,halo raise=4mm}},random from={{right eye color=
    pingu@blue}{random from={{bow tie}{gold medal}}}},
    random from={{eyes=!random}{wings=!random}},body type=
    legacy]
\end{tikzpicture}
```



2.10 Extras

An extra is considered everything, that is attached to the main penguin and not to the wings (as those items may be placed separately for both wings). Most extras are activated with the format `<extra>=<color>` (the `<color>` option is not mandatory) and try to adapt with other extras that have been placed (yet you can place multiple hats if you really like to). A lot of the extras do offer more keys to customize their appearance. They are explained in the full reference (Appendix A).

Consider the somewhat overkill-example of “Lord-Gadget, the penguin”.

Lord-Gadget, the penguin

```
\begin{tikzpicture}
\pingu[crown 2d=pingu@bronze,
      medal=pingu@purple, tie,
      eye patch left=teal,
      eye patch right=orange,
      right wing wave, sunglasses,
      glow thick=yellow]
\end{tikzpicture}
```



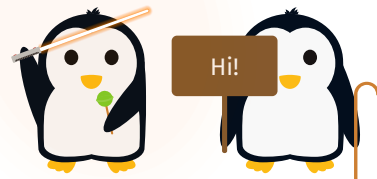
2.11 Wing-Items

Wing items are basically just like extras, but they can be selected separately for the left and right wing. Furthermore, they adapt their *default* appearance to the active wing positions (subsection 2.5). Currently there are the following wing items: *cane*, *staff*, *flower*, *present*, *snowball*, *basket*, *devilfork*, *flag*, *lollipop*, *fish*, *broom*, *hammer*, *drumstick*, *sword*, *microphone*, *plank*, *magnifier*, *horse*, *handcast*, *lightsaber*, *lightstaff*, *signpost*, *vrcontroller*, *laptop*, and *guitar*. They are selected using `<wing item> <left/right>`.

Additionally, they can be customized by `/pingu/left item angle` and `/pingu/right item angle`, as well as `/pingu/left item flip` and `/pingu/right item flip`. Lets consider an example...

Penguin with full wings!

```
\begin{tikzpicture}[scale=.75]
\pingu[lightsaber right=orange,
      lollipop left,
      right item angle=70,
      right wing raise, left wing grab]
\pingu[cane left, right item flip,
      sign post right={Hi!}, xshift=35mm]
\end{tikzpicture}
```



2.12 Clothing

Clothing is the newest extension to the collection, at and the moment there is not one “real” clothing, that really adapts to the penguins-position. I am working on the *cloak*-Clothing at the moment:

Pengu-Clothes

```
\begin{tikzpicture}[scale=.75]
\pingu[cloak]
\end{tikzpicture}
```



3 A Collection of Examples

To showcase the capabilities of this package, I have created a couple of example-penguins.

The Waddlers!

```
\begin{tikzpicture}
```

```
\pingu[wings hug, eyes wink, flip-flops, conical hat=
pingu@bronze!40!black, body type=legacy, tie=pingu@purple,
drum stick left, drum stick right,
drum stick right handle shift=-0.5mm, right wing wave,
drums, drums center text={The\\fontseries{sb}\selectfont
Waddlers!}]
\pingu[xshift=3.5cm, right wing grab, left wing wave,
eyes wink, guitar right, right item angle=20,
left wing grab, body type=legacy, bow tie=pingu@yellow!60!
orange, headphones=pingu@blue, headphones inner=
pingu@blue!5!pingu@white, sneaker, blush]
\pingu[xshift=1.75cm, yshift=-3cm, right wing grab,
eyes shiny, body type=legacy, small, shoes=pingu@purple,
heart, wool hat=pingu@purple!80!black, bill=open,
microphone right]
```

```
\end{tikzpicture}
```



Lord Double-Dagger

```
\begin{tikzpicture}
```

```
\pingu[body type=legacy, wings=wave, sword left, sword right,
sword left length=5mm, sword left width=.25mm,
left wing item angle=-40, sword left handle length=2.65mm
, right eye wink, eye patch left, bill open, cape=
pingu@bronze!70!pingu@blue!50!black, deer hat=
pingu@bronze!70!pingu@blue!50!black, belt=pingu@black,
belt knot color=pingu@bronze!70!pingu@blue!50!black]
```

```
\end{tikzpicture}
```



A Winter Walk

```
\begin{tikzpicture}[scale=.85]
```

```
\pingu[xshift=-2.3cm, yshift=.1cm, body type=legacy, scarf=
pingu@blue, scarf stripes=pingu@white, scarf end=right,
wool hat=pingu@blue!70!pingu@black, snowball right,
right wing wave, left wing hug, eyes shiny, blush]
\pingu[body type=legacy, body=pingu@black!88!blue, deer hat,
scarf=orange, scarf end=left, cup=pingu@red, cup steam,
cup straw=!hide, wings grab, eyes shiny]
\pingu[xshift=2.3cm, yshift=-.7cm, small, body=pingu@black!62
!gray, santa hat, santa beard, wings wave, eyes wink,
bill open]
```

```
\end{tikzpicture}
```



A Full Reference

Please note, that all preview-penguins have been reduced in scale to 50 % to save space and make the documentation more concise.

Aliases may set custom defaults. Those defaults are not listed as they may change.

A.1 Penguin Keys

`/pingu/name = <text>`

(pingu)

Sets the name of the penguin. This name is used for all the automatically generated coordinates (see [subsection 2.1](#)).

`/pingu/scale = <floating point>`

(active scale)

Changes the scale for the penguin. This is not supported by all items by default (as some scales have to be re-calculated according to their rotation). Yet, it should work with most.

Furthermore, this value can be used to make the penguin independent of the outer scaling.

`/pingu/meta-dots = <true/false>`

(true)

Can be used to enable and disable the meta dots ([subsection 2.1](#)). Passed true by default.

also written as `meta dots`

A.1.1 Anchors and Placement

`/pingu/anchor = <anchor or point>`

(origin)

Move the penguin so that the given anchor ends up at the origin (see [subsection 2.2](#)). Valid names are *origin* (the right shoulder, the default), *center*, *north*, *south*, *east*, *west*, *north east*, *north west*, *south east*, *south west*, *head*, *head top*, *bill*, *belly*, *feet*, *left shoulder*, and *right shoulder*. Anything else has to be a point ($\langle x \rangle$, $\langle y \rangle$) in the coordinate system of the penguin (the parentheses are optional).

```
\begin{tikzpicture}
  \pingu[anchor=belly]
  \draw[red] (-2.5mm,0) -- (2.5mm,0) (0,-2.5mm) -- (0,2.5mm);
\end{tikzpicture}
```



`/pingu/right of = <name:distance>`

Place the penguin right of the penguin with the given `/pingu/name`, on the same ground line. Without a distance (`right of=a`) both penguins touch.

```
\begin{tikzpicture}
  \pingu[name=a]
  \pingu[right of=a:5mm,body=pingu@blue]
\end{tikzpicture}
```



`/pingu/left of = <name:distance>`

Like `/pingu/right of`, but to the left.

```
\begin{tikzpicture}
  \pingu[name=a]
  \pingu[left of=a:5mm,body=pingu@blue]
\end{tikzpicture}
```



`/pingu/above of = <name:distance>`

Place the penguin above the penguin with the given `/pingu/name`, centered on it.

```
\begin{tikzpicture}
  \pingu[name=a]
  \pingu[above of=a:5mm,body=pingu@blue]
\end{tikzpicture}
```



`/pingu/below of = <name:distance>`

Like `/pingu/above of`, but below.

```
\begin{tikzpicture}
  \pingu[name=a]
  \pingu[below of=a:5mm,body=pingu@blue]
\end{tikzpicture}
```



A.1.2 The Feet

`/pingu/left foot = <foot-selector>`

(normal)

Change the style of the left foot. All valid values are listed in subsection 2.7.

```
\begin{tikzpicture}
  \pingu[left foot=simple]
\end{tikzpicture}
```



`/pingu/left foot color = <color>`

(pingu@yellow)

```
\begin{tikzpicture}
  \pingu[left foot color=green]
\end{tikzpicture}
```



shortcuts for `/pingu/left foot` (the `color` is passed to `/pingu/left foot color`): `left foot none`, `left foot normal`, `left foot sit`, `left foot simple`, `left foot back`, `left foot chubby`

`/pingu/right foot = <foot-selector>`

(normal)

Change the style of the right foot. All valid values are listed in subsection 2.7.

```
\begin{tikzpicture}
  \pingu[right foot=simple]
\end{tikzpicture}
```



`/pingu/right foot color = <color>`

(pingu@yellow)

```
\begin{tikzpicture}
  \pingu[right foot color=green]
\end{tikzpicture}
```



shortcuts for `/pingu/right foot` (the *color* is passed to `/pingu/right foot color`): *right foot none*, *right foot normal*, *right foot sit*, *right foot simple*, *right foot back*, *right foot chubby*

`/pingu/feet = <foot-selector>`

Change the style of both feet by calling `/pingu/left foot` and `/pingu/right foot` with the same value.

```
\begin{tikzpicture}
  \pingu[feet=simple]
\end{tikzpicture}
```



`/pingu/feet color = <color>`

Sets the color of both feet (using `/pingu/left foot color` and `/pingu/right foot color`).

```
\begin{tikzpicture}
  \pingu[feet color=green]
\end{tikzpicture}
```



shortcuts for `/pingu/feet` (the *color* is passed to `/pingu/feet color`): *feet none*, *feet normal*, *feet sit*, *feet simple*, *feet back*, *feet chubby*

`/pingu/feet front = <true/false>`

(false)

If set to true, the feet will be drawn in front of the body:

```
\begin{tikzpicture}
  \pingu[feet front]
\end{tikzpicture}
```



A.1.3 The Body

/pingu/body *main* = <color>

(pingu@main)

Set the main color of the penguin. This will affect /pingu/*hair* as well, as this chooses its default value from the main color.

```
\begin{tikzpicture}
  \pingu[body main=green]
\end{tikzpicture}
```



/pingu/body *head* = <color>

(pingu@main)

Set the color of the penguin head.

```
\begin{tikzpicture}
  \pingu[body head=green]
\end{tikzpicture}
```



/pingu/body = <color>

Sets the color of the main penguin and the head, by calling /pingu/*body main* and /pingu/*body head* with the same value.

```
\begin{tikzpicture}
  \pingu[body=green]
\end{tikzpicture}
```



/pingu/body *front* = <color>

(pingu@white)

Sets the frontal color of the penguin.

```
\begin{tikzpicture}
  \pingu[body front=green]
\end{tikzpicture}
```



/pingu/body *type* = <body type>

(normal)

Change the active body type. All valid values are listed in [subsection 2.7](#):

```
\begin{tikzpicture}
  \pingu[body type=legacy]
\end{tikzpicture}
```



A.1.4 The Size

`/pingu/height = <length>`

(36.27708pt)

Change the height of the penguin manually. You probably should not use this key directly and refer to `/pingu/small size`, `/pingu/normal size`, and `/pingu/large size`:

```
\begin{tikzpicture}
  \pingu[height=17mm]
\end{tikzpicture}
```



`/pingu/small size`

Will use `/pingu/height` to create a small pingu:

```
\begin{tikzpicture}
  \pingu[small size]
\end{tikzpicture}
```



also written as `small`, `small height`

`/pingu/normal size`

Will use `/pingu/height` to create a normal pingu:

```
\begin{tikzpicture}
  \pingu[normal size]
\end{tikzpicture}
```



also written as `normal`, `normal height`

`/pingu/large size`

Will use `/pingu/height` to create a large pingu:

```
\begin{tikzpicture}
  \pingu[large size]
\end{tikzpicture}
```



also written as `large`, `large height`

A.1.5 The Eyes

`/pingu/left eye = <eye-selector>`

(normal)

Change the style of the left eye. All valid values are listed in [subsection 2.6](#).

```
\begin{tikzpicture}
  \pingu[left eye=wink]
\end{tikzpicture}
```



`/pingu/left eye color = <color>`

(pingu@black)

```
\begin{tikzpicture}
  \pingu[left eye color=green]
\end{tikzpicture}
```



`/pingu/left eye second color = <color>`

(pingu@white)

Change the secondary color of the left eye. It will be used in some styles selected by `/pingu/left eye` (e.g. *shiny*):

```
\begin{tikzpicture}
  \pingu[left eye=shiny,
    left eye second color=green]
\end{tikzpicture}
```



shortcuts for `/pingu/left eye` (the *color* is passed to `/pingu/left eye color`): *left eye none*, *left eye normal*, *left eye vertical*, *left eye shiny*, *left eye wink*, *left eye shock*, *left eye devil*, *left eye sad*, *left eye angry*, *left eye hearts*, *left eye critical*

`/pingu/right eye = <eye-selector>`

(normal)

Change the style of the right eye. All valid values are listed in subsection 2.6.

```
\begin{tikzpicture}
  \pingu[right eye=wink]
\end{tikzpicture}
```



`/pingu/right eye color = <color>`

(pingu@black)

```
\begin{tikzpicture}
  \pingu[right eye color=green]
\end{tikzpicture}
```



`/pingu/right eye second color = <color>`

(pingu@white)

Change the secondary color of the right eye. It will be used in some styles selected by `/pingu/right eye` (e.g. *shiny*):

```
\begin{tikzpicture}
  \pingu[right eye=shock,
    right eye second color=green]
\end{tikzpicture}
```



shortcuts for `/pingu/right eye` (the *color* is passed to `/pingu/right eye color`): *right eye none*, *right eye normal*, *right eye vertical*, *right eye shiny*, *right eye wink*, *right eye shock*, *right eye devil*, *right eye sad*, *right eye angry*, *right eye hearts*, *right eye critical*

`/pingu/eyes = <eye-selector>`

Change the style of both eyes by calling `/pingu/left eye` and `/pingu/right eye` with the same value.

```
\begin{tikzpicture}
  \pingu[eyes=wink]
\end{tikzpicture}
```



`/pingu/eyes color = <color>`

Change the main color of both eyes by calling `/pingu/left eye color` and `/pingu/right eye color` with the same value.

```
\begin{tikzpicture}
  \pingu[eyes color=green]
\end{tikzpicture}
```



`/pingu/eyes second color = <color>`

Change the secondary color of both eyes by calling `/pingu/left eye second color` and `/pingu/right eye second color` with the same value.

```
\begin{tikzpicture}
  \pingu[left eye=shock, right eye=shiny,
    eyes second color=green]
\end{tikzpicture}
```



shortcuts for `/pingu/eyes` (the `color` is passed to `/pingu/eyes color`): `eyes none`, `eyes normal`, `eyes vertical`, `eyes shiny`, `eyes wink`, `eyes shock`, `eyes devil`, `eyes sad`, `eyes angry`, `eyes hearts`, `eyes critical`

A.1.6 The Wings

`/pingu/left wing = <wing-selector>`

(normal)

Change the style of the left wing. All valid values are listed in [subsection 2.5](#).

```
\begin{tikzpicture}
  \pingu[left wing=wave]
\end{tikzpicture}
```



`/pingu/left wing color = <color>`

(pingu@main)

```
\begin{tikzpicture}
  \pingu[left wing color=green]
\end{tikzpicture}
```



shortcuts for `/pingu/left wing` (the `color` is passed to `/pingu/left wing color`): `left wing none`, `left wing normal`, `left wing wave`, `left wing raise`, `left wing grab`, `left wing shock`, `left wing hug`

`/pingu/right wing = <wing-selector>`

(normal)

Change the style of the right wing. All valid values are listed in [subsection 2.5](#).

```
\begin{tikzpicture}
  \pingu[right wing=hug]
\end{tikzpicture}
```



`/pingu/right wing color = <color>`

(pingu@main)

```
\begin{tikzpicture}
  \pingu[right wing color=green]
\end{tikzpicture}
```



shortcuts for `/pingu/right wing` (the *color* is passed to `/pingu/right wing color`): *right wing none*, *right wing normal*, *right wing wave*, *right wing raise*, *right wing grab*, *right wing shock*, *right wing hug*

`/pingu/wings = <wing-selector>`

Change the style of both wings by calling `/pingu/left wing` and `/pingu/right wing` with the same value.

```
\begin{tikzpicture}
  \pingu[wings=grab]
\end{tikzpicture}
```



`/pingu/wings color = <color>`

Change the main color of both wings by calling `/pingu/left wing color` and `/pingu/right wing color` with the same value.

```
\begin{tikzpicture}
  \pingu[wings color=green]
\end{tikzpicture}
```



shortcuts for `/pingu/wings` (the *color* is passed to `/pingu/wings color`): *wings none*, *wings normal*, *wings wave*, *wings raise*, *wings grab*, *wings shock*, *wings hug*

A.1.7 The Hair

`/pingu/hair 1 color = <color>`

(pingu@main)

Set the color of the first hair (this may be used differently by other hairstyles):

```
\begin{tikzpicture}
  \pingu[hair 1 color=green]
\end{tikzpicture}
```



`/pingu/hair 2 color = <color>`

(pingu@main)

Set the color of the second hair (this may be used differently by other hairstyles):

```
\begin{tikzpicture}
  \pingu[hair 2 color=green]
\end{tikzpicture}
```



`/pingu/hair 3 color = <color>`

(pingu@main)

Set the color of the third hair (this may be used differently by other hairstyles):

```
\begin{tikzpicture}
  \pingu[hair 3 color=green]
\end{tikzpicture}
```



`/pingu/hair 4 color = <color>`

(pingu@main)

Set the color of the fourth hair (this may be used differently by other hairstyles):

```
\begin{tikzpicture}
  \pingu[hair 4 color=green]
\end{tikzpicture}
```



`/pingu/hair 5 color = <color>`

(pingu@main)

Set the color of the fifth hair (this may be used differently by other hairstyles):

```
\begin{tikzpicture}
  \pingu[hair 5 color=green]
\end{tikzpicture}
```



`/pingu/hairs color = <color>`

Set the color of all hairs by calling `/pingu/hair 1 color`, `/pingu/hair 2 color`, `/pingu/hair 3 color`, `/pingu/hair 4 color`, and `/pingu/hair 5 color` with the same argument:

```
\begin{tikzpicture}
  \pingu[hairs color=green]
\end{tikzpicture}
```



also written as `hairs`, `hair`

/pingu/hairstyle = <hair-selector>

(normal)

Change the hairstyle (subsection 2.7):

```
\begin{tikzpicture}
  \pingu[hairstyle=none]
\end{tikzpicture}
```



also written as `hair style`

shortcuts for /pingu/hairstyle (the *color* is passed to /pingu/hairs color): `hairstyle none`, `hairstyle normal`

A.1.8 The Bill

/pingu/bill = <bill-selector>

(normal)

Change the style of the bill (subsection 2.7):

```
\begin{tikzpicture}
  \pingu[bill=flat]
\end{tikzpicture}
```



/pingu/bill color = <color>

(pingu@yellow)

```
\begin{tikzpicture}
  \pingu[bill color=green]
\end{tikzpicture}
```



/pingu/bill above = <true/false>

(false)

Makes any bill appear on the foreground layer.

```
\begin{tikzpicture}
  \pingu[bill color=green, bill above]
\end{tikzpicture}
```



shortcuts for /pingu/bill (the *color* is passed to /pingu/bill color): `bill none`, `bill normal`, `bill foreground`, `bill flat`, `bill open`, `bill angry`

A.2 Drawing Styles

/pingu/:line

Disable glows, shades and fills and enforce a line. This line will be darker than the original fill color:

```
\begin{tikzpicture}
  \pingu[:line]
\end{tikzpicture}
```



`/pingu/:fill``(pingu@main)`

Makes the whole penguin in one solid color (basically a shortcut for setting all main penguin colors to the same):

```
\begin{tikzpicture}
  \pingu[:fill,tie=white,eyes color=white]
\end{tikzpicture}
```

`/pingu/:ghost parts = <opacity>``(.5)`

Set the opacity of each penguin component individually. At the moment, this excludes some glow calculations.

```
\begin{tikzpicture}
  \pingu[:ghost parts]
\end{tikzpicture}
```

`/pingu/:ghost = <opacity>``(.5)`

Set the opacity of the complete penguin. At the moment, this excludes some glow calculations.

```
\begin{tikzpicture}
  \pingu[:ghost]
\end{tikzpicture}
```

`/pingu/:devil = <color>``(pingu@purple)`

Enable all devil components (not the wing items) and set their main color:

```
\begin{tikzpicture}
  \pingu[:devil=green]
\end{tikzpicture}
```

`/pingu/:hide`

Do not draw the main pingu:

```
\begin{tikzpicture}
  \pingu[santa hat,:hide]
\end{tikzpicture}
```

`/pingu/:toast`

One slice of cuteness: no wings, no feet and no hair, just the body:

```
\begin{tikzpicture}
  \pingu[:toast]
\end{tikzpicture}
```



/pingu/:back

Mirror the penguin, this swaps left and right, the rotation and more. Yet, at least at the time of writing, this does not swap the drawing order in each layer, but just the layers:

```
\begin{tikzpicture}
  \pingu[:back, left wing wave,
    cane left, left item angle=70]
\end{tikzpicture}
```



/pingu/:pingu = <options>

Apply the given options to all penguin-keys (for /pingu/body, /pingu/wings color, /pingu/eyes color, /pingu/bill color, /pingu/hairs color, and /pingu/feet color) allowing to re-use the original color:

```
\begin{tikzpicture}
  \pingu[:pingu={/pingu/@block/.style={fill=green}}]
\end{tikzpicture}
```



/pingu/:mix = <color-mix>

(50!white)

Mix the main penguin with given colors by adapting the individual drawing colors individually, but leave all extras intact. This can be used (for example) to gray-out the penguin:

```
\begin{tikzpicture}
  \pingu[mask, tie=red, lollipop left, :mix=50!green]
\end{tikzpicture}
```



In contrast to `:mix-draw` this changes the active colors of the penguin.

/pingu/:mix-draw = <color-mix>

(50!white)

Mix the main penguin with given colors, but leave all extras intact. This can be used (for example) to gray-out the penguin:

```
\begin{tikzpicture}
  \pingu[mask, tie=red, :mix-draw=50!green]
\end{tikzpicture}
```



In contrast to `:mix` this changes the drawing options (using /pingu/:pingu).

/pingu/:mix-all = <color-mix>

(50!white)

Mix everything of the penguin with given colors, including all extras intact. This can be used (for example) to gray-out the penguin:

```
\begin{tikzpicture}
  \pingu[mask, tie=red, lollipop left, :mix-all=50!green]
\end{tikzpicture}
```



A.3 Extras

A.3.1 The heart

`/pingu/heart = <color>` (lightgray)

```
\begin{tikzpicture}
  \pingu[heart=green]
\end{tikzpicture}
```



A.3.2 The vampire teeth

`/pingu/vampire teeth = <color>` (pingu@yellow!10!pingu@black!70!lightgray)

```
\begin{tikzpicture}
  \pingu[vampire teeth=green]
\end{tikzpicture}
```



A.3.3 The tie

`/pingu/tie = <color>` (pingu@green)

```
\begin{tikzpicture}
  \pingu[tie]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/tie` is active.

`/pingu/tie knot = <color>` (<tie>!92!black)

```
\begin{tikzpicture}
  \pingu[tie, tie knot=orange]
\end{tikzpicture}
```



`/pingu/tie length = <length>` (.625\pingu@side@h@half)

```
\begin{tikzpicture}
  \pingu[tie, tie length=1.25cm]
\end{tikzpicture}
```



`/pingu/tie offset = <length>` (.399cm)

Change the upper vertical offset of the tie:

```
\begin{tikzpicture}
  \pingu[tie, tie offset=.75cm]
\end{tikzpicture}
```



`/pingu/tie width = <length>` (.21cm)

```
\begin{tikzpicture}
  \pingu[tie, tie width=.5cm]
\end{tikzpicture}
```



```
/pingu/tie pattern = <tex-code> ({ \pingu@block{<tie>} \pingupathxtie; \pingu@block{<tie knot>} \pingupathxtieknot; })
```

Change the tie pattern.

```
/pingu/tie dots = <color> (pingu@white)
```

Change the `/pingu/tie pattern` to dots:

```
\begin{tikzpicture}
  \pingu[tie, tie dots]
\end{tikzpicture}
```



A.3.4 The sheriff star

```
/pingu/sheriff star = <color> (pingu@yellow)
```

```
\begin{tikzpicture}
  \pingu[sheriff star=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/sheriff star` is active.

```
/pingu/sheriff star content = <length> ({\resizebox{2.25mm}!{\textsf{\bfseries Sheriff}}})
```

```
\begin{tikzpicture}
  \pingu[sheriff star, sheriff star content=X]
\end{tikzpicture}
```



```
/pingu/sheriff star lines = <color> (pingu@black!92!<sheriff star>)
```

```
\begin{tikzpicture}
  \pingu[sheriff star, sheriff star lines=green]
\end{tikzpicture}
```



A.3.5 The bowtie

```
/pingu/bow tie = <color> (pingu@blue)
```

```
\begin{tikzpicture}
  \pingu[bow tie]
\end{tikzpicture}
```



also written as `bowtie`, `bow-tie`

The following keys are only in effect if `/pingu/bow tie` is active.

```
/pingu/bow tie b = <color> (<bow tie>)
```

```
\begin{tikzpicture}
  \pingu[bow tie, bow tie b=green]
\end{tikzpicture}
```



also written as `bowtie b`, `bow-tie b`

/pingu/*bow tie knot* = *<color>*

(*<bow tie>!92!black*)

```
\begin{tikzpicture}
  \pingu[bow tie, bow tie knot=green]
\end{tikzpicture}
```



also written as *bowtie knot*, *bow-tie knot*

/pingu/*bow tie offset* = *<length>*

(.315cm)

```
\begin{tikzpicture}
  \pingu[bow tie, bow tie offset=8mm]
\end{tikzpicture}
```



also written as *bowtie offset*, *bow-tie offset*

A.3.6 The cup

/pingu/*cup* = *<color>*

(pingu@green)

```
\begin{tikzpicture}
  \pingu[cup]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*cup* is active.

/pingu/*cup straw* = *<color>*

(*<cup>*)

```
\begin{tikzpicture}
  \pingu[cup, cup straw=!hide]
\end{tikzpicture}
```



/pingu/*cup steam* = *<color>*

(\@pingu@none)

```
\begin{tikzpicture}
  \pingu[cup, cup steam=green]
\end{tikzpicture}
```



/pingu/*cup steam shift* = *<length>*

(0mm)

```
\begin{tikzpicture}
  \pingu[cup, cup steam, cup steam shift=2mm]
\end{tikzpicture}
```



/pingu/*cup steam opacity* = *<factor>*

(.55)

```
\begin{tikzpicture}
  \pingu[cup, cup steam=gray, cup steam opacity=1]
\end{tikzpicture}
```



A.3.7 The medal

`/pingu/medal = <color>`

(pingu@yellow)

```
\begin{tikzpicture}
  \pingu[medal]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/medal` is active.

`/pingu/medal band = <color>`

(pingu@red)

```
\begin{tikzpicture}
  \pingu[medal, medal band=green]
\end{tikzpicture}
```



`/pingu/medal shade = <color>`

(<medal>!65!pingu@white)

Change the color of the outer medal ring:

```
\begin{tikzpicture}
  \pingu[medal, medal shade=green]
\end{tikzpicture}
```



`/pingu/medal shade width = <length>`

(.75\p@)

Change the width of the outer medal ring:

```
\begin{tikzpicture}
  \pingu[medal, medal shade=green,
    medal shade width=2mm]
\end{tikzpicture}
```



`/pingu/medal text = <text>`

Set the text displayed in the medal. The style can be changed by updating the substyle `medal text style`.

```
\begin{tikzpicture}
  \pingu[medal, medal text=XY,
    medal text style/.style={black}]
\end{tikzpicture}
```



`/pingu/gold medal = <text>`

(1)

Basically the same as the normal medal. This will activate `/pingu/medal`:

```
\begin{tikzpicture}
  \pingu[gold medal]
\end{tikzpicture}
```



`/pingu/silver medal = <text>`

(2)

Basically the same as the normal medal, but with a silver color. This will activate `/pingu/medal:`

```
\begin{tikzpicture}
  \pingu[silver medal]
\end{tikzpicture}
```



`/pingu/bronze medal = <text>`

(3)

Basically the same as the normal medal, but with a bronze color. This will activate `/pingu/medal:`

```
\begin{tikzpicture}
  \pingu[bronze medal]
\end{tikzpicture}
```



A.3.8 The rectangular medal

`/pingu/rectmedal = <color>`

(pingu@yellow!70!pingu@bronze)

```
\begin{tikzpicture}
  \pingu[rectmedal]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/rectmedal` is active.

`/pingu/rectmedal band = <color>`

(pingu@blue!90!blue!60!pingu@black)

```
\begin{tikzpicture}
  \pingu[rectmedal, rectmedal band=green]
\end{tikzpicture}
```



`/pingu/rectmedal shade = <color>`

(<rectmedal>!65!pingu@white)

Change the color of the outer rectmedal ring:

```
\begin{tikzpicture}
  \pingu[rectmedal, rectmedal shade=green]
\end{tikzpicture}
```



`/pingu/rectmedal width = <length>`

(7.5mm)

Change the width of rectmedal:

```
\begin{tikzpicture}
  \pingu[rectmedal, rectmedal width=2mm]
\end{tikzpicture}
```



```
/pingu/rectmedal height = <length>
```

(6.5mm)

Change the height of rectmedal:

```
\begin{tikzpicture}
  \pingu[rectmedal, rectmedal height=2mm]
\end{tikzpicture}
```



```
/pingu/rectmedal shade width = <length>
```

(.75\p@)

Change the width of the outer rectmedal ring:

```
\begin{tikzpicture}
  \pingu[rectmedal, rectmedal shade=green,
    rectmedal shade width=2mm]
\end{tikzpicture}
```



```
/pingu/rectmedal text = <text>
```

Set the text displayed in the rectmedal. The style can be changed by updating the substyle `rectmedal text` style.

```
\begin{tikzpicture}
  \pingu[rectmedal, rectmedal text=XY,
    rectmedal text style/.style={black}]
\end{tikzpicture}
```



```
/pingu/gold rectmedal = <text>
```

(1)

Basically the same as the normal rectmedal. This will activate `/pingu/rectmedal:`

```
\begin{tikzpicture}
  \pingu[gold rectmedal]
\end{tikzpicture}
```



```
/pingu/silver rectmedal = <text>
```

(2)

Basically the same as the normal rectmedal, but with a silver color. This will activate `/pingu/rectmedal:`

```
\begin{tikzpicture}
  \pingu[silver rectmedal]
\end{tikzpicture}
```



```
/pingu/bronze rectmedal = <text>
```

(3)

Basically the same as the normal rectmedal, but with a bronze color. This will activate `/pingu/rectmedal:`

```
\begin{tikzpicture}
  \pingu[bronze rectmedal]
\end{tikzpicture}
```



A.3.9 The eye patches

```
/pingu/eye patch left = <color>
```

```
(<body main>)
```

```
\begin{tikzpicture}
  \pingu[eye patch left]
\end{tikzpicture}
```



also written as `eyepatch left`, `eye-patch left`

```
/pingu/eye patch right = <color>
```

```
(<body main>)
```

```
\begin{tikzpicture}
  \pingu[eye patch right]
\end{tikzpicture}
```



also written as `eyepatch right`, `eye-patch right`

A.3.10 The monocle

```
/pingu/monocle left = <color>
```

```
(pingu@black)
```

```
\begin{tikzpicture}
  \pingu[monocle left]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/monocle left` is active.

```
/pingu/monocle left glass = <color>
```

```
(pingu@blue)
```

Set the color of the glass of the left monocle. The opacity of this color is set by `/pingu/monocle left opacity`.

```
\begin{tikzpicture}
  \pingu[monocle left,
    monocle left glass=green]
\end{tikzpicture}
```



also written as `monocle left fill`

```
/pingu/monocle left opacity = <factor>
```

```
(.155)
```

Set the opacity of the glass color of the left monocle (set by `/pingu/monocle left glass`):

```
\begin{tikzpicture}
  \pingu[monocle left,
    monocle left opacity=1]
\end{tikzpicture}
```



also written as `monocle left fill opacity`

```
/pingu/monocle left string = <color>
```

```
(<monocle left>)
```

Set the color of the string of the left monocle:

```
\begin{tikzpicture}
  \pingu[monocle left,
    monocle left string=green]
\end{tikzpicture}
```



```
/pingu/monocle left string length = <length>
```

```
(5.55mm)
```

```
\begin{tikzpicture}
  \pingu[monocle left, monocle left string length=1cm]
\end{tikzpicture}
```



```
/pingu/monocle left blob = <color>
```

```
(<monocle left>)
```

Set the color of the blob at the end of the string of the left monocle:

```
\begin{tikzpicture}
  \pingu[monocle left,
    monocle left blob=green]
\end{tikzpicture}
```



```
/pingu/monocle right = <color>
```

```
(pingu@black)
```

```
\begin{tikzpicture}
  \pingu[monocle right]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/monocle right` is active.

```
/pingu/monocle right glass = <color>
```

```
(pingu@blue)
```

Set the color of the glass of the right monocle. The opacity of this color is set by `/pingu/monocle right opacity`.

```
\begin{tikzpicture}
  \pingu[monocle right,
    monocle right glass=green]
\end{tikzpicture}
```



also written as `monocle right fill`

```
/pingu/monocle right opacity = <factor>
```

```
(.155)
```

Set the opacity of the glass color of the right monocle (set by `/pingu/monocle right glass`):

```
\begin{tikzpicture}
  \pingu[monocle right,
    monocle right opacity=1]
\end{tikzpicture}
```



also written as `monocle right fill opacity`

`/pingu/monocle right string = <color>`

(`<monocle right>`)

```
\begin{tikzpicture}
  \pingu[monocle right, monocle right string=green]
\end{tikzpicture}
```



`/pingu/monocle right string length = <length>`

(5.55mm)

Set the length of the right monocle string:

```
\begin{tikzpicture}
  \pingu[monocle right,
    monocle right string length=1cm]
\end{tikzpicture}
```



`/pingu/monocle right blob = <color>`

(`<monocle right>`)

Set the color of the blob at the end of the string of the right monocle:

```
\begin{tikzpicture}
  \pingu[monocle right,
    monocle right blob=green]
\end{tikzpicture}
```



A.3.11 The pants

`/pingu/pants = <color>`

(`pingu@red`)

Sets the color of the pants:

```
\begin{tikzpicture}
  \pingu[pants=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/pants` is active.

`/pingu/pants bands = <true/false>`

(`false`)

Switch the bands of the pants on and of:

```
\begin{tikzpicture}
  \pingu[pants, pants bands]
\end{tikzpicture}
```



`/pingu/pants button left = <color>`

(`pingu@black`)

Set the color of the left pant button:

```
\begin{tikzpicture}
  \pingu[pants, pants button left=green]
\end{tikzpicture}
```



```
/pingu/pants button right = <color>
```

```
(pingu@black)
```

Set the color of the right pant button:

```
\begin{tikzpicture}
  \pingu[pants, pants button right=green]
\end{tikzpicture}
```



```
/pingu/pants buttons = <color>
```

```
(pingu@black)
```

Sets `/pingu/pants button left` and `/pingu/pants button right` with the same color.

```
\begin{tikzpicture}
  \pingu[pants, pants buttons=green]
\end{tikzpicture}
```



```
/pingu/pants button left shade = <color>
```

```
(<pants button left>!70!white)
```

Set the color of the left pant button shade:

```
\begin{tikzpicture}
  \pingu[pants,
    pants button left shade=green]
\end{tikzpicture}
```



```
/pingu/pants button right shade = <color>
```

```
(<pants button right>!70!white)
```

Set the color of the right pant button shade:

```
\begin{tikzpicture}
  \pingu[pants,
    pants button right shade=green]
\end{tikzpicture}
```



```
/pingu/pants buttons shade = <color>
```

```
(pingu@black!70!white)
```

Sets `/pingu/pants button left shade` and `/pingu/pants button right shade` with the same color.

```
\begin{tikzpicture}
  \pingu[pants, pants buttons shade=green]
\end{tikzpicture}
```



```
/pingu/pants no buttons
```

Remove the buttons from the pants:

```
\begin{tikzpicture}
  \pingu[pants, pants no buttons]
\end{tikzpicture}
```



```
/pingu/pants extra height = <length>
```

(1.5mm)

Raise the pants:

```
\begin{tikzpicture}
  \pingu[pants, pants extra height=6mm]
\end{tikzpicture}
```



also written as `pants without buttons`

A.3.12 The glow

```
/pingu/glow = <color>
```

(pingu@white)

Active a glow around the penguin:

```
\begin{tikzpicture}
  \pingu[glow=green]
\end{tikzpicture}
```



```
/pingu/glow thick = <color>
```

Will pass on the color to `/pingu/glow` and use a `/pingu/glow width function` with a thicker line width:

```
\begin{tikzpicture}
  \pingu[glow thick=green]
\end{tikzpicture}
```



```
/pingu/glow solid = <color>
```

Will pass on the color to `/pingu/glow` and use a `/pingu/glow width function` combined with `/pingu/glow function` to create a solid glow:

```
\begin{tikzpicture}
  \pingu[glow solid=green, wings wave]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/glow` is active.

```
/pingu/glow steps = <list>
```

(1,1.1,1.2,1.3,1.4,1.5)

Comma separated list of discrete intervals for the glow calculation:

```
\begin{tikzpicture}
  \pingu[glow=green, glow steps={.3,.5,1}]
\end{tikzpicture}
```



```
/pingu/glow function = <function>
```

```
(.1/\i)
```

Function using the token `i` to refer to the current `/pingu/glow steps`. Its evaluation will be used to determine the opacity of the current step:

```
\begin{tikzpicture}
  \pingu[glow=green,
        glow function={.5/\i}]
\end{tikzpicture}
```



```
/pingu/glow width function = <function>
```

```
(2.85mm-1.65*\i mm)
```

Function using the token `i` to refer to the current `/pingu/glow steps`. Its evaluation will be used to determine the width of the current step:

```
\begin{tikzpicture}
  \pingu[glow=green,
        glow width function={5mm-\i mm}]
\end{tikzpicture}
```



A.3.13 The eye frame

```
/pingu/eye frame = <color>
```

```
(pingu@black)
```

This is more of a test extra that adds a frame around both eyes:

```
\begin{tikzpicture}
  \pingu[eye frame=green]
\end{tikzpicture}
```



also written as `eyeframe`, `eye-frame`

A.3.14 The glasses

```
/pingu/glasses = <color>
```

```
(pingu@black)
```

Display glasses for the penguin:

```
\begin{tikzpicture}
  \pingu[glasses=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/glasses` is active.

```
/pingu/glasses left fill = <color>
```

```
(!hide)
```

Sets the fill color of the left glass. The opacity is determined by `/pingu/glasses left opacity`.

```
\begin{tikzpicture}
  \pingu[glasses,
        glasses left fill=green]
\end{tikzpicture}
```



```
/pingu/glasses right fill = <color>
```

```
(!hide)
```

Sets the fill color of the right glass. The opacity is determined by `/pingu/glasses right opacity`.

```
\begin{tikzpicture}
  \pingu[glasses,
    glasses right fill=green]
\end{tikzpicture}
```



```
/pingu/glasses fill = <color>
```

```
(\@pingu@none)
```

Change the color of both glasses by calling `/pingu/glasses left fill` and `/pingu/glasses right fill` with the same value.

```
\begin{tikzpicture}
  \pingu[glasses, glasses fill=green]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[glasses,
    glasses left fill=green,
    glasses left opacity=1]
\end{tikzpicture}
```



```
/pingu/glasses right opacity = <factor>
```

```
(.825)
```

Sets the fill opacity of the right glass:

```
\begin{tikzpicture}
  \pingu[glasses,
    glasses right fill=green,
    glasses right opacity=1]
\end{tikzpicture}
```



```
/pingu/glasses opacity = <factor>
```

```
(.825)
```

Change the opacity of both glasses by calling `/pingu/glasses left opacity` and `/pingu/glasses right opacity` with the same value.

```
\begin{tikzpicture}
  \pingu[glasses,
    glasses fill=teal,
    glasses opacity=1]
\end{tikzpicture}
```



```
/pingu/glasses line width = <length>
```

```
(1.125\p@)
```

```
\begin{tikzpicture}
  \pingu[glasses, glasses line width=1mm]
\end{tikzpicture}
```



```
/pingu/glasses bend left = <angle>
```

(32.5)

```
\begin{tikzpicture}
  \pingu[glasses, glasses bend left=45]
\end{tikzpicture}
```



```
/pingu/glasses bend right = <angle>
```

(32.5)

```
\begin{tikzpicture}
  \pingu[glasses, glasses bend right=45]
\end{tikzpicture}
```



```
/pingu/glasses bend = <angle>
```

(32.5)

Sets `/pingu/glasses bend left` and `/pingu/glasses bend right` with the same value.

```
\begin{tikzpicture}
  \pingu[glasses, glasses bend=45]
\end{tikzpicture}
```



```
/pingu/glasses upper bend = <angle>
```

(8)

Sets the bending of the upper part of the glasses.

```
\begin{tikzpicture}
  \pingu[glasses, glasses upper bend=30]
\end{tikzpicture}
```



```
/pingu/sun glasses = <color>
```

(pingu@black)

Configure the `/pingu/glasses` to display sunglasses. The color is passed on to `/pingu/glasses fill`

```
\begin{tikzpicture}
  \pingu[sun glasses=orange]
\end{tikzpicture}
```



also written as `sunglasses`

A.3.15 The rounded glasses

```
/pingu/glasses round = <color>
```

(pingu@black)

Behaves equivalent to `/pingu/glasses` but produces a round counterpart:

```
\begin{tikzpicture}
  \pingu[glasses round=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/glasses round` is active.

```
/pingu/glasses round left fill = <color>
```

```
(!hide)
```

Sets the fill color of the left glass. The opacity is determined by `/pingu/glasses round left opacity`.

```
\begin{tikzpicture}
  \pingu[glasses round,
    glasses round left fill=green]
\end{tikzpicture}
```



```
/pingu/glasses round right fill = <color>
```

```
(!hide)
```

Sets the fill color of the right glass. The opacity is determined by `/pingu/glasses round right opacity`.

```
\begin{tikzpicture}
  \pingu[glasses round,
    glasses round right fill=green]
\end{tikzpicture}
```



```
/pingu/glasses round fill = <color>
```

```
(\@pingu@none)
```

Change the color of both glasses by calling `/pingu/glasses round left fill` and `/pingu/glasses round right fill` with the same value.

```
\begin{tikzpicture}
  \pingu[glasses round, glasses round fill=green]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[glasses round,
    glasses round left fill=green,
    glasses round left opacity=1]
\end{tikzpicture}
```



```
/pingu/glasses round right opacity = <factor>
```

```
(.825)
```

Sets the fill opacity of the right glass:

```
\begin{tikzpicture}
  \pingu[glasses round,
    glasses round right fill=green,
    glasses round right opacity=1]
\end{tikzpicture}
```



```
/pingu/glasses round opacity = <factor>
```

(.825)

Change the opacity of both glasses round by calling `/pingu/glasses round left opacity` and `/pingu/glasses round right opacity` with the same value.

```
\begin{tikzpicture}
  \pingu[glasses round,
    glasses round fill=teal,
    glasses round opacity=1]
\end{tikzpicture}
```



```
/pingu/glasses round line width = <length>
```

(.75\p@)

```
\begin{tikzpicture}
  \pingu[glasses round, glasses round line width=1mm]
\end{tikzpicture}
```



```
/pingu/sun glasses round = <color>
```

(pingu@black)

Configure the `/pingu/glasses round` to display sunglasses round. The color is passed on to `/pingu/glasses round fill`

```
\begin{tikzpicture}
  \pingu[sun glasses round=orange]
\end{tikzpicture}
```



also written as `sunglasses round`

A.3.16 The devil horns

```
/pingu/devil horns = <color>
```

(pingu@purple)

```
\begin{tikzpicture}
  \pingu[devil horns=green]
\end{tikzpicture}
```



also written as `devilhorns`, `devil-horns`

A.3.17 The devil wings

```
/pingu/devil wings = <color>
```

(pingu@black)

```
\begin{tikzpicture}
  \pingu[devil wings=green]
\end{tikzpicture}
```



also written as `devilwings`, `devil-wings`

The following keys are only in effect if `/pingu/devil wings` is active.

/pingu/*devil wings* *b* = *<color>*

(*<devil wings>*)

```
\begin{tikzpicture}
  \pingu[devil wings, devil wings b=green]
\end{tikzpicture}
```



also written as *devilwings b*, *devil-wings b*

A.3.18 The belt

/pingu/*belt* = *<color>*

(pingu@bronze!80!black)

```
\begin{tikzpicture}
  \pingu[belt=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*belt* is active.

/pingu/*belt knot color* = *<color>*

(pingu@silver)

```
\begin{tikzpicture}
  \pingu[belt, belt knot color=green]
\end{tikzpicture}
```



A.3.19 The bunny ears

/pingu/*bunny ears* = *<color>*

(brown!95!pingu@main)

```
\begin{tikzpicture}
  \pingu[bunny ears=green]
\end{tikzpicture}
```



also written as *bunnyears*, *bunny-ears*

The following keys are only in effect if /pingu/*bunny ears* is active.

/pingu/*bunny ears band* = *<color>*

(*<bunny ears>*)

```
\begin{tikzpicture}
  \pingu[bunny ears, bunny ears band=green]
\end{tikzpicture}
```



also written as *bunnyears band*, *bunny-ears band*

/pingu/*bunny ears b* = *<color>*

(*<bunny ears>*!90!yellow!90!white)

```
\begin{tikzpicture}
  \pingu[bunny ears, bunny ears b=green]
\end{tikzpicture}
```



also written as *bunnyears b*, *bunny-ears b*

A.3.20 The head band

/pingu/head band = <color>

(pingu@red)

```
\begin{tikzpicture}
  \pingu[head band=green]
\end{tikzpicture}
```



also written as `headband`, `head-band`

The following keys are only in effect if /pingu/head band is active.

/pingu/head band bend = <angle>

(9)

```
\begin{tikzpicture}
  \pingu[head band, head band bend=25]
\end{tikzpicture}
```



also written as `headband bend`, `head-band bend`

/pingu/head band angle = <angle>

(17)

```
\begin{tikzpicture}
  \pingu[head band, head band angle=25]
\end{tikzpicture}
```



also written as `headband angle`, `head-band angle`

/pingu/head band upper angle = <angle>

(50)

```
\begin{tikzpicture}
  \pingu[head band, head band upper angle=25]
\end{tikzpicture}
```



also written as `headband upper angle`, `head-band upper angle`

/pingu/head band knot = <true/false>

(false)

```
\begin{tikzpicture}
  \pingu[head band, head band knot]
\end{tikzpicture}
```



also written as `headband knot`, `head-band knot`

/pingu/head band knot color = <color>

(<head band>!78!black)

If /pingu/head band knot is enabled, this setting changes the color of the knot:

```
\begin{tikzpicture}
  \pingu[head band, head band knot,
    head band knot color=green]
\end{tikzpicture}
```



also written as `headband knot color`, `head-band knot color`

```
/pingu/head band knot a color = <color>
```

```
(<head band knot color>!90!black)
```

If `/pingu/head band knot` is enabled, this setting changes the color of the left headband wing (this will, by default, affect the right wing as well):

```
\begin{tikzpicture}
  \pingu[head band, head band knot,
    head band knot a color=green]
\end{tikzpicture}
```



also written as `headband knot a color`, `head-band knot a color`

```
/pingu/head band knot b color = <color>
```

```
(<head band knot a color>)
```

If `/pingu/head band knot` is enabled, this setting changes the color of the left headband wing (this will, by default, affect the right wing as well):

```
\begin{tikzpicture}
  \pingu[head band, head band knot,
    head band knot a color=blue,
    head band knot b color=green]
\end{tikzpicture}
```



also written as `headband knot b color`, `head-band knot b color`

```
/pingu/head band bands = <true/false>
```

```
(true)
```

```
\begin{tikzpicture}
  \pingu[head band, head band bands=false]
\end{tikzpicture}
```



also written as `headband bands`, `head-band bands`

```
/pingu/head band bands a color = <color>
```

```
(<head band>!78!black)
```

If `/pingu/head band bands` is enabled, this setting changes the color of the large one of the both bands:

```
\begin{tikzpicture}
  \pingu[head band, head band bands,
    head band bands a color=green]
\end{tikzpicture}
```



also written as `headband bands a color`, `head-band bands a color`

```
/pingu/head band bands b color = <color>
```

```
(<head band>)
```

If `/pingu/head band bands` is enabled, this setting changes the color of the left headband wing (this will, by default, affect the right wing as well):

```
\begin{tikzpicture}
  \pingu[head band, head band bands,
    head band bands a color=blue,
    head band bands b color=green]
\end{tikzpicture}
```



also written as `headband bands b color`, `head-band bands b color`

A.3.21 The rook

`/pingu/rook = <color>`

(pingu@silver)

```
\begin{tikzpicture}
  \pingu[rook=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/rook` is active.

`/pingu/rook back = <color>`

(<rook>!85!black)

Change the color of the rook-costume background:

```
\begin{tikzpicture}
  \pingu[rook, rook back=green]
\end{tikzpicture}
```



`/pingu/rook hatch = <true/false>`

(true)

Toggles the opening in the rook costume:

```
\begin{tikzpicture}
  \pingu[rook, rook hatch=false]
\end{tikzpicture}
```



`/pingu/rook shade = <color>`

(<rook>!92!black)

```
\begin{tikzpicture}
  \pingu[rook, rook shade=green]
\end{tikzpicture}
```



A.3.22 The halo

`/pingu/halo = <color>`

(pingu@lightblue)

```
\begin{tikzpicture}
  \pingu[halo=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/halo` is active.

`/pingu/halo raise = <length>`

(0mm)

Define the vertical raise of the halo above the penguins head:

```
\begin{tikzpicture}
  \pingu[halo, halo raise=4mm]
\end{tikzpicture}
```



```
/pingu/halo glow = <true/false>
```

```
(\ifpengu@setup@glows@ true\else false\fi)
```

Disable or enable the glow of the halo. The glow is enabled by default, use the `noglow` package option to disable it.

```
\begin{tikzpicture}
  \pingu[halo, halo glow=false]
\end{tikzpicture}
```



```
/pingu/halo above = <true/false>
```

```
(false)
```

Draws the halo above, which is useful in case of other gadgets:

```
\begin{tikzpicture}
  \pingu[halo, halo above=true]
\end{tikzpicture}
```



A.3.23 The strawhat

```
/pingu/strawhat = <color>
```

```
(brown!50!white)
```

```
\begin{tikzpicture}
  \pingu[strawhat=green]
\end{tikzpicture}
```



also written as `straw hat`

The following keys are only in effect if `/pingu/strawhat` is active.

```
/pingu/strawhat ribbon = <color>
```

```
(gray!85!black)
```

```
\begin{tikzpicture}
  \pingu[strawhat, strawhat ribbon=green]
\end{tikzpicture}
```



also written as `straw hat ribbon`

```
/pingu/strawhat position = <angle>:(<x>,<y>)<scale>
```

```
({-26.5: (-.185cm, .14cm){1.375}})
```

Currently, this is a very cumbersome command to change various strawhat parameters at the same time:

```
\begin{tikzpicture}
  \pingu[strawhat,
    strawhat position={33:(-.8cm, .14cm){1.4}}]
\end{tikzpicture}
```



also written as `straw hat position`

A.3.24 The hat

```
/pingu/hat = <color>
```

```
(pingu@black!90!black)
```

```
\begin{tikzpicture}
  \pingu[hat=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/hat` is active.

`/pingu/hat ribbon = <color>`

`(<hat>!87!white)`

```
\begin{tikzpicture}
  \pingu[hat, hat ribbon=green]
\end{tikzpicture}
```



`/pingu/hat base = <color>`

`(<hat>)`

```
\begin{tikzpicture}
  \pingu[hat, hat base=green]
\end{tikzpicture}
```



`/pingu/hat coronal = <color>`

`(<hat>!91!white)`

```
\begin{tikzpicture}
  \pingu[hat, hat coronal=green]
\end{tikzpicture}
```



`/pingu/hat position = <angle>:(<x>,<y>)<scale>`

`(-2:(.35mm),0)1)`

Currently, this is a very cumbersome command to change various hat parameters at the same time:

```
\begin{tikzpicture}
  \pingu[hat, hat position={1:(0cm,-.09cm){1.33}}]
\end{tikzpicture}
```



A.3.25 The graduate cap

`/pingu/graduate = <color>`

`(pingu@black!95!white)`

```
\begin{tikzpicture}
  \pingu[graduate=green]
\end{tikzpicture}
```



also written as `graduate cap`

The following keys are only in effect if `/pingu/graduate` is active.

`/pingu/graduate top = <color>`

`(<graduate>!95!white)`

```
\begin{tikzpicture}
  \pingu[graduate, graduate top=green]
\end{tikzpicture}
```



also written as `graduate cap top`

`/pingu/graduate tilt = <length>`

`(2.33mm)`

```
\begin{tikzpicture}
  \pingu[graduate, graduate tilt=5mm]
\end{tikzpicture}
```



also written as `graduate cap tilt`

`/pingu/graduate tassel = <color>`

`(\@pingu@none)`

```
\begin{tikzpicture}
  \pingu[graduate, graduate tassel=green]
\end{tikzpicture}
```



also written as `graduate cap tassel`

`/pingu/graduate tassel drop = <color>`

`(<graduate tassel>)`

```
\begin{tikzpicture}
  \pingu[graduate, graduate tassel, graduate tassel drop=green]
\end{tikzpicture}
```



also written as `graduate cap tassel drop`

`/pingu/graduate tassel start = <coordinate>`

`((.55cm,3.5mm-2.33mm))`

```
\begin{tikzpicture}
  \pingu[graduate, graduate tassel, graduate tassel start={(2mm,4mm)}]
\end{tikzpicture}
```



also written as `graduate cap tassel start`

`/pingu/graduate tassel width = <length>`

`(.5mm)`

```
\begin{tikzpicture}
  \pingu[graduate, graduate tassel, graduate tassel width=2mm]
\end{tikzpicture}
```



also written as `graduate cap tassel width`

`/pingu/graduate tassel length = <length>`

`(5.5mm)`

```
\begin{tikzpicture}
  \pingu[graduate, graduate tassel, graduate tassel length=12mm]
\end{tikzpicture}
```



also written as `graduate cap tassel length`

`/pingu/graduate tassel drop size = <length>`

`(.75*<graduate tassel width>)`

```
\begin{tikzpicture}
  \pingu[graduate, graduate tassel, graduate tassel drop size=4mm]
\end{tikzpicture}
```



also written as `graduate cap tassel drop size`

```
/pingu/graduate position = <angle>:(<x>,<y>)<scale> ({-27:(-.19cm,.14cm){1.375}})
```

Currently, this is a very cumbersome command to change various graduate parameters at the same time:

```
\begin{tikzpicture}
  \pingu[graduate,
    graduate position={1:(0cm,-.09cm){1.33}}]
\end{tikzpicture}
```



also written as `graduate cap position`

A.3.26 The space helmet

```
/pingu/space helmet = <color> (pingu@black!45!white)
```

```
\begin{tikzpicture}
  \pingu[space helmet=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/space helmet` is active.

```
/pingu/space helmet glass = <color> (pingu@blue!40!pingu@yellow!50!white)
```

```
\begin{tikzpicture}
  \pingu[space helmet, space helmet glass=green]
\end{tikzpicture}
```



```
/pingu/space helmet glass opacity = <factor> (.33)
```

```
\begin{tikzpicture}
  \pingu[space helmet, space helmet glass opacity=1]
\end{tikzpicture}
```



```
/pingu/space helmet position = <angle>:(<x>,<y>)<scale> (-2:(.35mm,0)1)
```

Currently, this is a very cumbersome command to change various space helmet parameters at the same time:

```
\begin{tikzpicture}
  \pingu[space helmet, space helmet position={1:(0cm,-.09cm){1.33}}]
\end{tikzpicture}
```



A.3.27 The mitra

```
/pingu/mitra = <color> (pingu@red!67!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[mitra=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/hat` is active.

/pingu/*mitra height* = *<length>*

(1.125cm)

```
\begin{tikzpicture}
  \pingu[mitra,mitra height=1.5cm]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*mitra* is active.

/pingu/*mitra background* = *<color>*

(*<mitra>!80!pingu@black*)

```
\begin{tikzpicture}
  \pingu[mitra,mitra background=green]
\end{tikzpicture}
```



/pingu/*mitra overset* = *<length>*

(1.25mm)

```
\begin{tikzpicture}
  \pingu[mitra,mitra overset=2.5mm]
\end{tikzpicture}
```



/pingu/*mitra lower angle* = *<angle>*

(125)

```
\begin{tikzpicture}
  \pingu[mitra,mitra lower angle=140]
\end{tikzpicture}
```



/pingu/*mitra upper angle* = *<angle>*

(-35)

```
\begin{tikzpicture}
  \pingu[mitra,mitra upper angle=-45]
\end{tikzpicture}
```



/pingu/*mitra lower band* = *<color>*

(pingu@yellow!75!pingu@black)

```
\begin{tikzpicture}
  \pingu[mitra,mitra lower band=green]
\end{tikzpicture}
```



/pingu/*mitra lower band height* = *<factor>*

(.2)

```
\begin{tikzpicture}
  \pingu[mitra,mitra lower band height=.5]
\end{tikzpicture}
```



/pingu/*mitra upper band* = *<color>*

(*<mitra lower band>*)

```
\begin{tikzpicture}
  \pingu[mitra,mitra upper band=green]
\end{tikzpicture}
```



```
/pingu/mitra upper band height = <factor>
```

```
(1.5*<mitra lower band height>)
```

```
\begin{tikzpicture}
  \pingu[mitra, mitra upper band height=.5]
\end{tikzpicture}
```



```
/pingu/mitra center band = <color>
```

```
(<mitra lower band>)
```

```
\begin{tikzpicture}
  \pingu[mitra, mitra center band=green]
\end{tikzpicture}
```



```
/pingu/mitra center band width = <factor>
```

```
(2*<mitra lower band height>)
```

```
\begin{tikzpicture}
  \pingu[mitra, mitra center band width=.5]
\end{tikzpicture}
```



```
/pingu/mitra position = <angle>:(<x>,<y>)<scale>
```

```
(0:(0,0)1)
```

Currently, this is a very cumbersome command to change various mitra parameters at the same time:

```
\begin{tikzpicture}
  \pingu[mitra, mitra position={1:(0cm,-.09cm){1.33}}]
\end{tikzpicture}
```



A.3.28 The witch hat

```
/pingu/witch hat = <color>
```

```
(pingu@purple!40!black)
```

```
\begin{tikzpicture}
  \pingu[witch hat=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/witch hat` is active.

```
/pingu/witch hat ribbon = <color>
```

```
(<witch hat>!87!white)
```

```
\begin{tikzpicture}
  \pingu[witch hat, witch hat ribbon=green]
\end{tikzpicture}
```



```
/pingu/witch hat base = <color>
```

```
(<witch hat>)
```

```
\begin{tikzpicture}
  \pingu[witch hat, witch hat base=green]
\end{tikzpicture}
```



/pingu/*witch hat coronal* = *<color>*

(*<witch hat>!91!white*)

```
\begin{tikzpicture}
  \pingu[witch hat, witch hat coronal=green]
\end{tikzpicture}
```



/pingu/*witch hat band* = *<color>*

(*!hide*)

```
\begin{tikzpicture}
  \pingu[witch hat, witch hat band=green]
\end{tikzpicture}
```



/pingu/*witch hat position* = *<angle>:(<x>,<y>)<scale>*

(*{-7:(.625mm,.45mm){1.15}}*)

Currently, this is a very cumbersome command to change various witch hat parameters at the same time:

```
\begin{tikzpicture}
  \pingu[witch hat, witch hat position={1:(0cm,-.09cm){1.33}}]
\end{tikzpicture}
```



A.3.29 The conical hat

/pingu/*conical hat* = *<color>*

(*pingu@bronze*)

```
\begin{tikzpicture}
  \pingu[conical hat=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*conical hat* is active.

/pingu/*conical hat rounding* = *<length>*

(*.4pt*)

```
\begin{tikzpicture}
  \pingu[conical hat, conical hat rounding=.25pt]
\end{tikzpicture}
```



/pingu/*conical hat shade* = *<length>*

(*<conical hat>!80!pingu@black*)

```
\begin{tikzpicture}
  \pingu[conical hat, conical hat shade=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*conical hat height* is active.

/pingu/*conical hat height* = *<length>*

(*8mm*)

```
\begin{tikzpicture}
  \pingu[conical hat, conical hat height=10mm]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*conical hat width* is active.

/pingu/*conical hat width* = *<length>*

(2.25cm)

```
\begin{tikzpicture}
  \pingu[conical hat, conical hat width=3cm]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*conical hat* is active.

/pingu/*conical hat position* = *<angle>:(<x>,<y>)<scale>*

(-15:(2mm,-3mm)1)

Currently, this is a very cumbersome command to change various conical hat parameters at the same time:

```
\begin{tikzpicture}
  \pingu[conical hat,
    conical hat position={1:(-.1cm,-.275cm){1.33}}]
\end{tikzpicture}
```



A.3.30 The cap

/pingu/*cap* = *<color>*

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[cap=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*cap* is active.

/pingu/*cap padding* = *<length>*

(.8mm)

```
\begin{tikzpicture}
  \pingu[cap, cap padding=4mm]
\end{tikzpicture}
```



/pingu/*cap extra height* = *<length>*

(0pt)

```
\begin{tikzpicture}
  \pingu[cap, cap extra height=2mm]
\end{tikzpicture}
```



A.3.31 The construction helmet

/pingu/*construction helmet* = *<color>*

(pingu@yellow)

```
\begin{tikzpicture}
  \pingu[construction helmet=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*construction helmet* is active.

/pingu/*construction helmet padding* = *<length>*

(.325cm)

```
\begin{tikzpicture}
  \pingu[construction helmet, construction helmet padding=4mm]
\end{tikzpicture}
```



/pingu/*construction helmet extra height* = *<length>*

(0pt)

```
\begin{tikzpicture}
  \pingu[construction helmet, construction helmet extra height=2mm]
\end{tikzpicture}
```



/pingu/*construction helmet position* = *<angle>:(<x>,<y>)<scale>*

(-.5:(.05mm,-1.25mm)1)

Currently, this is a very cumbersome command to change various construction helmet parameters at the same time:

```
\begin{tikzpicture}
  \pingu[construction helmet,
    construction helmet position={1:(-.1cm,-.275cm){1.33}}]
\end{tikzpicture}
```



A.3.32 The detective cap

/pingu/*detective cap* = *<color>*

(pingu@bronze!60!lightgray!95!green)

```
\begin{tikzpicture}
  \pingu[detective cap=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*detective cap* is active.

/pingu/*detective cap padding* = *<length>*

(.3cm)

```
\begin{tikzpicture}
  \pingu[detective cap, detective cap padding=4mm]
\end{tikzpicture}
```



/pingu/*detective cap position* = *<angle>:(<x>,<y>)<scale>*

(-.5:(0mm,-1.25mm)1)

Currently, this is a very cumbersome command to change various detective cap parameters at the same time:

```
\begin{tikzpicture}
  \pingu[detective cap,
    detective cap position={1:(-.1cm,-.275cm){1.33}}]
\end{tikzpicture}
```



A.3.33 The drums

/pingu/drums = <color>

(brown!10!pingu@white)

```
\begin{tikzpicture}
  \pingu[drums=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/drums is active.

/pingu/drums center band = <color>

(pingu@black!35!pingu@red)

```
\begin{tikzpicture}
  \pingu[drums, drums center band=green]
\end{tikzpicture}
```



/pingu/drums center side = <color>

(pingu@black!35!pingu@red!60!pingu@silver)

```
\begin{tikzpicture}
  \pingu[drums, drums center side=green]
\end{tikzpicture}
```



/pingu/drums center text = <text>

```
\begin{tikzpicture}
  \pingu[drums, drums center text=Foo!]
\end{tikzpicture}
```



/pingu/drums left base = <color>

(<drums>!45!pingu@black)

```
\begin{tikzpicture}
  \pingu[drums, drums left base=green]
\end{tikzpicture}
```



/pingu/drums left band = <color>

(<drums left base>!35!pingu@red)

```
\begin{tikzpicture}
  \pingu[drums, drums left band=green]
\end{tikzpicture}
```



/pingu/drums left side = <color>

(pingu@black!35!pingu@red!60!pingu@silver)

```
\begin{tikzpicture}
  \pingu[drums, drums left side=green]
\end{tikzpicture}
```



/pingu/drums left color = <color>

(<drums>)

```
\begin{tikzpicture}
  \pingu[drums, drums left color=green]
\end{tikzpicture}
```



/pingu/drums right base = <color>

(<drums left base>)

```
\begin{tikzpicture}
  \pingu[drums, drums right base=green]
\end{tikzpicture}
```



/pingu/drums right band = <color>

(<drums left band>)

```
\begin{tikzpicture}
  \pingu[drums, drums right band=green]
\end{tikzpicture}
```



/pingu/drums right side = <color>

(pingu@black!35!pingu@red!60!pingu@silver)

```
\begin{tikzpicture}
  \pingu[drums, drums right side=green]
\end{tikzpicture}
```



/pingu/drums right color = <color>

(<drums>)

```
\begin{tikzpicture}
  \pingu[drums, drums right color=green]
\end{tikzpicture}
```



A.3.34 The crown

/pingu/crown = <color>

(pingu@yellow)

```
\begin{tikzpicture}
  \pingu[crown=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/crown is active.

/pingu/crown 3d = <true/false>

(true)

Toggle the 3d-Design of the crown.

```
\begin{tikzpicture}
  \pingu[crown, crown 3d=false]
\end{tikzpicture}
```



/pingu/crown back = <color>

(<crown>!93!black)

Change the back color of the crown:

```
\begin{tikzpicture}
  \pingu[crown, crown back=green]
\end{tikzpicture}
```



```
/pingu/crown front bend = <angle>
```

(16)

Change the front lower bend of the crown:

```
\begin{tikzpicture}
  \pingu[crown, crown front bend=52]
\end{tikzpicture}
```



```
/pingu/crown back bend = <angle>
```

(9)

Change the back lower bend of the crown:

```
\begin{tikzpicture}
  \pingu[crown, crown back bend=46]
\end{tikzpicture}
```



```
/pingu/crown gem shade = <true/false>
```

(true)

Toggle the gem shading of the crown.

```
\begin{tikzpicture}
  \pingu[crown, crown gem shade=false]
\end{tikzpicture}
```



```
/pingu/crown gem colors = <a><b><c><d><e><f> ({pingu@purple}{pingu@blue}{pingu@green}{pingu@purple}
{pingu@green}{pingu@blue}{pingu@purple})
```

Change the color of all the seven gems of the crown:

```
\begin{tikzpicture}
  \pingu[crown, crown gem colors={green}{green}
{green}{white}{green}{green}{green}]
\end{tikzpicture}
```



```
/pingu/crown gem ring = <color>
```

(<crown>!85!white)

Change the color of the rings around the crown:

```
\begin{tikzpicture}
  \pingu[crown, crown gem ring=green]
\end{tikzpicture}
```



```
/pingu/crown position = <angle>:(<x>,<y>)<scale>
```

(-9:(1mm,0mm)1)

Currently, this is a very cumbersome command to change various crown parameters at the same time:

```
\begin{tikzpicture}
  \pingu[crown, eyes wink,
crown position={1:(-.1cm,-.275cm){1.33}}]
\end{tikzpicture}
```



/pingu/*crown 2d* = *<color>*

(pingu@yellow)

Enables the /pingu/*crown* with the given color and disables /pingu/*crown 3d*:

```
\begin{tikzpicture}
  \pingu[crown 2d=green]
\end{tikzpicture}
```



A.3.35 The princess crown

Similar to /pingu/*crown* but smaller.

/pingu/*princess crown* = *<color>*

(pingu@yellow)

Enable the smaller crown with a specific color:

```
\begin{tikzpicture}
  \pingu[princess crown=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*princess crown* is active.

/pingu/*princess crown 3d* = *<true/false>*

(true)

Toggle the 3d-Design of the smaller crown.

```
\begin{tikzpicture}
  \pingu[princess crown, princess crown 3d=false]
\end{tikzpicture}
```



/pingu/*princess crown back* = *<color>*

(*<princess crown>!93!black*)

Change the back color of the smaller crown:

```
\begin{tikzpicture}
  \pingu[princess crown, princess crown back=green]
\end{tikzpicture}
```



/pingu/*princess crown front bend* = *<angle>*

(12)

Change the front lower bend of the smaller crown:

```
\begin{tikzpicture}
  \pingu[princess crown, princess crown front bend=52]
\end{tikzpicture}
```



/pingu/*princess crown back bend* = *<angle>*

(7)

Change the back lower bend of the smaller crown:

```
\begin{tikzpicture}
  \pingu[princess crown, princess crown back bend=46]
\end{tikzpicture}
```



```
/pingu/princess crown gem shade = <true/false>
```

(true)

Toggle the gem shading of the smaller crown.

```
\begin{tikzpicture}
  \pingu[princess crown,
    princess crown gem shade=false]
\end{tikzpicture}
```



```
/pingu/princess crown bobbles = <true/false>
```

(true)

Toggle the bobbles of the smaller crown.

```
\begin{tikzpicture}
  \pingu[princess crown, princess crown bobbles=false]
\end{tikzpicture}
```



```
/pingu/princess crown gem colors = <a><b><c><d>
{pingu@purple}{pingu@blue}{pingu@green}{pingu@blue}
{pingu@purple}
```

({pingu@purple}{pingu@blue}{pingu@green}{pingu@blue}

Change the color of all the seven gems of the smaller crown:

```
\begin{tikzpicture}
  \pingu[princess crown,
    princess crown gem colors={green}{green}{white}
    {green}{green}]
\end{tikzpicture}
```



```
/pingu/princess crown gem ring = <color>
```

(<princess crown>!85!white)

Change the color of the rings around the small crown:

```
\begin{tikzpicture}
  \pingu[princess crown,
    princess crown gem ring=green]
\end{tikzpicture}
```



```
/pingu/princess crown position = <angle>:(<x>,<y>)<scale>
```

(-9:(1mm,0mm)1)

Currently, this is a very cumbersome command to change various princess crown parameters at the same time:

```
\begin{tikzpicture}
  \pingu[princess crown, eyes wink,
    princess crown position={1:(-.19cm,-.2cm){2.2}}]
\end{tikzpicture}
```



`/pingu/princess crown 2d = <color>`

`(pingu@yellow)`

Enables the `/pingu/princess crown` with the given color and disables `/pingu/princess crown 3d`:

```
\begin{tikzpicture}
  \pingu[princess crown 2d=green]
\end{tikzpicture}
```



A.3.36 The cake hat

`/pingu/cake-hat = <color>`

`(pingu@white!92!<cake-hat top>)`

Enable a cake hat with a specific color:

```
\begin{tikzpicture}
  \pingu[cake-hat=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/cake-hat` is active.

`/pingu/cake-hat top = <color>`

`(pingu@purple)`

Change the color of the cake hat top:

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat top=green]
\end{tikzpicture}
```



`/pingu/cake-hat shade = <color>`

`(gray)`

Change the color of the heavily transparent cake hat shading:

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat shade=green]
\end{tikzpicture}
```



`/pingu/cake-hat candle = <color>`

`(pingu@purple!60!pingu@black)`

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat candle=green]
\end{tikzpicture}
```



`/pingu/cake-hat candle fire = <color>`

`(pingu@red)`

Change the color of the cake hats' candle most outer fire:

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat candle fire=green]
\end{tikzpicture}
```



```
/pingu/cake-hat candle fire 2 = <color>
```

```
(<cake-hat candle fire>!50!yellow)
```

Change the color of the cake hats' candle middle fire:

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat candle fire 2=green]
\end{tikzpicture}
```



```
/pingu/cake-hat candle fire 3 = <color>
```

```
(<cake-hat candle fire 2>!50!white)
```

Change the color of the cake hats' candle inner fire:

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat candle fire 3=green]
\end{tikzpicture}
```



```
/pingu/cake-hat candle wick = <color>
```

```
(pingu@black)
```

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat candle wick=green]
\end{tikzpicture}
```



```
/pingu/cake-hat candle shade = <color>
```

```
(gray!80!<cake-hat candle back>)
```

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat candle shade=green]
\end{tikzpicture}
```



```
/pingu/cake-hat candle back = <color>
```

```
(<cake-hat candle>!85!black)
```

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat candle back=green]
\end{tikzpicture}
```



```
/pingu/cake-hat outline = <color>
```

```
(pingu@black!80!<cake-hat>)
```

Change the color of the cake hats' outline (width by /pingu/cake-hat outline width):

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat outline=green]
\end{tikzpicture}
```



```
/pingu/cake-hat outline width = <length>
```

```
(.25pt)
```

Change the width of the cake hats' outline (color by /pingu/cake-hat outline):

```
\begin{tikzpicture}
  \pingu[cake-hat, cake-hat outline width=1mm]
\end{tikzpicture}
```



```
/pingu/cake-hat position = <angle>:(<x>,<y>)<scale> (-9:(1mm,0mm)1)
```

Currently, this is a very cumbersome command to change various cake hat parameters at the same time:

```
\begin{tikzpicture}
  \pingu[cake-hat,
    cake-hat position={1:(-.085cm,-.2cm){1.275}}]
\end{tikzpicture}
```



A.3.37 The pumpkin hat

```
/pingu/pumpkin-hat = <color> (pingu@bronze!97!white)
```

Enable a pumpkin hat with a specific color:

```
\begin{tikzpicture}
  \pingu[pumpkin-hat=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/pumpkin-hat` is active.

```
/pingu/pumpkin-hat stalk = <color> (pingu@green!95!<pumpkin-hat>!45!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[pumpkin-hat, pumpkin-hat stalk=teal]
\end{tikzpicture}
```



```
/pingu/pumpkin-hat stalk top = <color> (<pumpkin-hat stalk>!95!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[pumpkin-hat, pumpkin-hat stalk top=teal]
\end{tikzpicture}
```



```
/pingu/pumpkin-hat stripe a = <color> (pingu@black)
```

Change the color of the first stripe. By default the other stripes share this ones color:

```
\begin{tikzpicture}
  \pingu[pumpkin-hat,pumpkin-hat stripe a=green]
\end{tikzpicture}
```



```
/pingu/pumpkin-hat stripe b = <color> (<pumpkin-hat stripe a>)
```

Change the color of the second stripe. By default the third stripe share this ones color:

```
\begin{tikzpicture}
  \pingu[pumpkin-hat,pumpkin-hat stripe b=green]
\end{tikzpicture}
```



```
/pingu/pumpkin-hat stripe c = <color>
```

```
(<pumpkin-hat stripe b>)
```

Change the color of the third stripe:

```
\begin{tikzpicture}
  \pingu[pumpkin-hat,pumpkin-hat stripe c=green]
\end{tikzpicture}
```



```
/pingu/pumpkin-hat outline = <color>
```

```
(pingu@black)
```

```
\begin{tikzpicture}
  \pingu[pumpkin-hat, pumpkin-hat outline=green]
\end{tikzpicture}
```



```
/pingu/pumpkin-hat outline width = <length>
```

```
(.7pt)
```

```
\begin{tikzpicture}
  \pingu[pumpkin-hat, pumpkin-hat outline width=3pt]
\end{tikzpicture}
```



```
/pingu/pumpkin-hat position = <angle>:(<x>,<y>)<scale>
```

```
(-9:(1.65mm,.25mm)1.05)
```

Currently, this is a very cumbersome command to change various pumpkin hat parameters at the same time:

```
\begin{tikzpicture}
  \pingu[pumpkin-hat,
    pumpkin-hat position={1:(-.085cm,-.15cm){1.275}}]
\end{tikzpicture}
```



A.3.38 The jack o lantern

```
/pingu/jack o lantern = <color>
```

```
(pingu@bronze!97!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[jack o lantern=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/jack o lantern` is active.

```
/pingu/jack o lantern stalk = <color>
```

```
(pingu@green!95!<jack o lantern>!45!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[jack o lantern, jack o lantern stalk=teal]
\end{tikzpicture}
```



```
/pingu/jack o lantern stalk top = <color>
```

```
(<jack o lantern stalk>!95!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[jack o lantern, jack o lantern stalk top=teal]
\end{tikzpicture}
```



```
/pingu/jack o lantern stripe a = <color>
```

```
(pingu@black)
```

Change the color of the first stripe. By default the other stripes share this ones color:

```
\begin{tikzpicture}
  \pingu[jack o lantern,jack o lantern stripe a=green]
\end{tikzpicture}
```



```
/pingu/jack o lantern stripe b = <color>
```

```
(<jack o lantern stripe a>)
```

Change the color of the second stripe. By default the third stripe share this ones color:

```
\begin{tikzpicture}
  \pingu[jack o lantern,jack o lantern stripe b=green]
\end{tikzpicture}
```



```
/pingu/jack o lantern stripe c = <color>
```

```
(<jack o lantern stripe b>)
```

Change the color of the third stripe:

```
\begin{tikzpicture}
  \pingu[jack o lantern,jack o lantern stripe c=green]
\end{tikzpicture}
```



```
/pingu/jack o lantern back stripe a = <color>
```

```
(pingu@black)
```

Change the color of the first stripe. By default the other stripes share this ones color:

```
\begin{tikzpicture}
  \pingu[jack o lantern,jack o lantern back stripe a=green]
\end{tikzpicture}
```



```
/pingu/jack o lantern back stripe b = <color>
```

```
(<jack o lantern stripe a>)
```

Change the color of the second stripe. By default the third stripe share this ones color:

```
\begin{tikzpicture}
  \pingu[jack o lantern,jack o lantern back stripe b=green]
\end{tikzpicture}
```



```
/pingu/jack o lantern back stripe c = <color>
```

```
(<jack o lantern stripe b>)
```

Change the color of the third stripe:

```
\begin{tikzpicture}
  \pingu[jack o lantern,jack o lantern back stripe c=green]
\end{tikzpicture}
```



/pingu/jack o lantern background = <color>

(<jack o lantern>!85!pingu@black)

```
\begin{tikzpicture}
  \pingu[jack o lantern, jack o lantern background=green]
\end{tikzpicture}
```



/pingu/jack o lantern outline = <color>

(pingu@black)

```
\begin{tikzpicture}
  \pingu[jack o lantern, jack o lantern outline=green]
\end{tikzpicture}
```



/pingu/jack o lantern outline width = <length>

(.7pt)

```
\begin{tikzpicture}
  \pingu[jack o lantern, jack o lantern outline width=3pt]
\end{tikzpicture}
```



/pingu/jack o lantern width = <length>

(5.75mm)

```
\begin{tikzpicture}
  \pingu[jack o lantern, jack o lantern width=8mm]
\end{tikzpicture}
```



/pingu/jack o lantern height = <length>

(5.5mm)

```
\begin{tikzpicture}
  \pingu[jack o lantern, jack o lantern height=8mm]
\end{tikzpicture}
```



/pingu/jack o lantern position = <angle>:(<x>,<y>)<scale>

({-9:(3.15mm,.95mm)}{1.3})

Currently, this is a very cumbersome command to change various pumpkin hat parameters at the same time:

```
\begin{tikzpicture}
  \pingu[jack o lantern,
    jack o lantern position={1:(-.085cm,-.15cm){1.275}}]
\end{tikzpicture}
```



/pingu/jack o lantern helmet = <color>

(pingu@bronze!97!pingu@black)

Uses /pingu/jack o lantern and some of its keys:

```
\begin{tikzpicture}
  \pingu[jack o lantern helmet=green]
\end{tikzpicture}
```



A.3.39 The bee

/pingu/bee = <color>

(pingu@yellow!60!pingu@white)

```
\begin{tikzpicture}
  \pingu[bee=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/bee is active.

/pingu/bee body = <color>

(pingu@black!82!gray)

```
\begin{tikzpicture}
  \pingu[bee, bee body=green]
\end{tikzpicture}
```



/pingu/bee wings = <color>

(pingu@blue!8!pingu@white)

```
\begin{tikzpicture}
  \pingu[bee, bee wings=green]
\end{tikzpicture}
```



/pingu/bee mouth = <color>

(<bee body>)

```
\begin{tikzpicture}
  \pingu[bee, bee mouth=green]
\end{tikzpicture}
```



/pingu/bee eyes = <color>

(<bee body>)

```
\begin{tikzpicture}
  \pingu[bee, bee eyes=green]
\end{tikzpicture}
```



/pingu/bee blush = <color>

(pingu@red!21!<bee>)

```
\begin{tikzpicture}
  \pingu[bee, bee blush=green]
\end{tikzpicture}
```



/pingu/bee position = <angle>:(<x>,<y>)<scale>

({-27.5:(8mm,15mm){.875}})

Currently, this is a very cumbersome command to change various bee parameters at the same time:

```
\begin{tikzpicture}
  \pingu[bee, bee position={1:(0cm,-.09cm){1.33}}]
\end{tikzpicture}
```



A.3.40 The horse

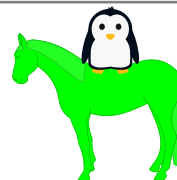
Although the horse does not have a bounding box by default (`/pingu/on horse has bounding box`), we add it so you can see everything.

`/pingu/on horse = <color>`

`(pingu@bronze!80!pingu@black)`

Place your penguin on a not-so-to-scale horse (see `/pingu/horse left` and `/pingu/horse right`):

```
\begin{tikzpicture}
  \pingu[on horse=green, on horse has bounding box]
\end{tikzpicture}
```

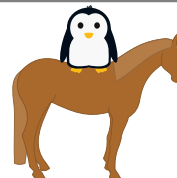


The following keys are only in effect if `/pingu/on horse` is active.

`/pingu/on horse flip = <true/false>`

`(false)`

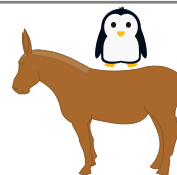
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse flip=true]
\end{tikzpicture}
```



`/pingu/on horse donkey = <true/false>`

`(false)`

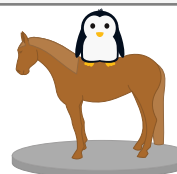
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse donkey=true]
\end{tikzpicture}
```



`/pingu/on horse has base = <true/false>`

`(false)`

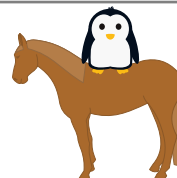
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse has base]
\end{tikzpicture}
```



`/pingu/on horse has bounding box = <true/false>`

`(false)`

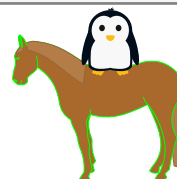
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box]
\end{tikzpicture}
```



`/pingu/on horse draw = <color>`

`(<on horse>!80!pingu@black)`

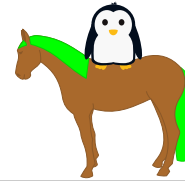
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse draw=green]
\end{tikzpicture}
```



```
/pingu/on horse mane = <color>
```

```
(<on horse>!86!pingu@white)
```

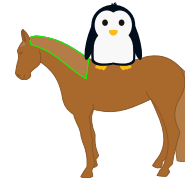
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse mane=green]
\end{tikzpicture}
```



```
/pingu/on horse mane draw = <color>
```

```
(<on horse mane>!80!pingu@black)
```

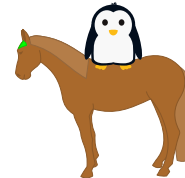
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse mane draw=
    green]
\end{tikzpicture}
```



```
/pingu/on horse thatch = <color>
```

```
(<on horse mane>)
```

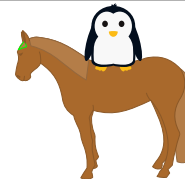
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse thatch=green
  ]
\end{tikzpicture}
```



```
/pingu/on horse thatch draw = <color>
```

```
(<on horse thatch>!80!pingu@black)
```

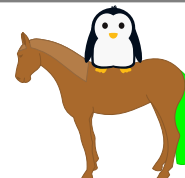
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse thatch draw=
    green]
\end{tikzpicture}
```



```
/pingu/on horse tail = <color>
```

```
(<on horse mane>)
```

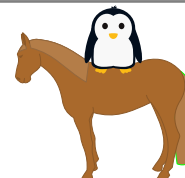
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse tail=green]
\end{tikzpicture}
```



```
/pingu/on horse tail draw = <color>
```

```
(<on horse tail>!80!pingu@black)
```

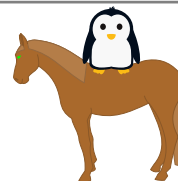
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse tail draw=
    green]
\end{tikzpicture}
```



/pingu/*on horse eyes* = *<color>*

(*<on horse draw>*)

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse eyes=green]
\end{tikzpicture}
```

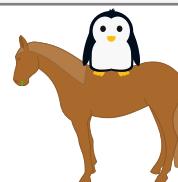


also written as *on horse eye*

/pingu/*on horse mouth* = *<color>*

(*<on horse draw>*)

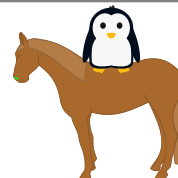
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse mouth=green]
\end{tikzpicture}
```



/pingu/*on horse nose* = *<color>*

(*<on horse draw>*)

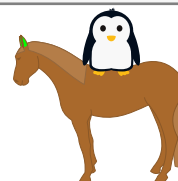
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse nose=green]
\end{tikzpicture}
```



/pingu/*on horse ears* = *<color>*

(*<on horse>*)

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse ears=green]
\end{tikzpicture}
```



/pingu/*on horse base* = *<color>*

(lightgray!90!black)

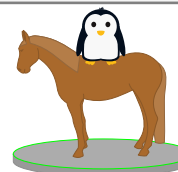
```
\begin{tikzpicture}
  \pingu[on horse, on horse has base, on horse has bounding box,
    on horse base=green]
\end{tikzpicture}
```



/pingu/*on horse base draw* = *<color>*

(*<on horse base>!91!pingu@black*)

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse has base,
    on horse base draw=green]
\end{tikzpicture}
```



```
/pingu/on horse base shade = <color>
```

```
(lightgray!80!black)
```

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse has base,
    on horse base shade=green]
\end{tikzpicture}
```



```
/pingu/on horse base shade draw = <color>
```

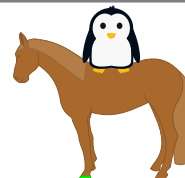
```
(<on horse base shade>!91!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[on horse, on horse has base, on horse has bounding box,
    on horse base shade draw=green]
\end{tikzpicture}
```



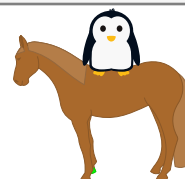
```
/pingu/on horse front left hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box,
    on horse front left hoof=green]
\end{tikzpicture}
```



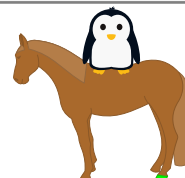
```
/pingu/on horse front right hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box,
    on horse front right hoof=green]
\end{tikzpicture}
```



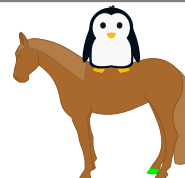
```
/pingu/on horse back left hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box,
    on horse back left hoof=green]
\end{tikzpicture}
```



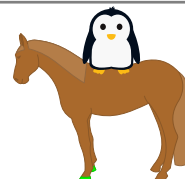
```
/pingu/on horse back right hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box,
    on horse back right hoof=green]
\end{tikzpicture}
```



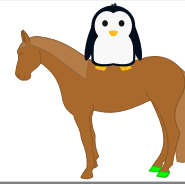
```
/pingu/on horse front hoofs = <color>
```

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse front hoofs=
    green]
\end{tikzpicture}
```



/pingu/*on horse back hoofs* = <color>

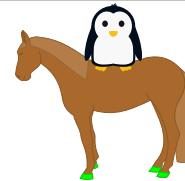
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse back hoofs=
    green]
\end{tikzpicture}
```



/pingu/*on horse hoofs* = <color>

(\@pingu@none)

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse hoofs=green]
\end{tikzpicture}
```

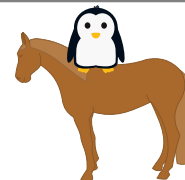


/pingu/*on horse xshift* = <length>

(Opt)

This key reacts with the /pingu/*on horse flip* option!

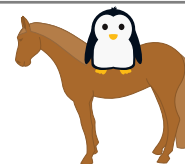
```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box,
    on horse xshift=1cm]
\end{tikzpicture}
```



/pingu/*on horse yshift* = <length>

(Opt)

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse yshift=1cm]
\end{tikzpicture}
```



/pingu/*on horse scale* = <length>

(0.75)

```
\begin{tikzpicture}
  \pingu[on horse, on horse has bounding box, on horse scale=.25]
\end{tikzpicture}
```



/pingu/*horse behind* = <color>

(pingu@bronze!80!pingu@black)

Place your penguin in front of a horse (configures /pingu/*on horse*):

```
\begin{tikzpicture}
  \pingu[horse behind=green]
\end{tikzpicture}
```



A.3.41 The VR-headset

/pingu/vr-headset = <color>	(pingu@black!92!gray)
<pre>\begin{tikzpicture} \pingu[vr-headset=green] \end{tikzpicture}</pre>	

The following keys are only in effect if /pingu/vr-headset is active.

/pingu/vr-headset band = <color>	(<vr-headset>!92!gray)
<pre>\begin{tikzpicture} \pingu[vr-headset, vr-headset band=purple] \end{tikzpicture}</pre>	

/pingu/vr-headset band top = <color>	(<vr-headset>!96!gray)
<pre>\begin{tikzpicture} \pingu[vr-headset, vr-headset band top=purple] \end{tikzpicture}</pre>	

/pingu/vr-headset hair

Change the hair to support the headset:

<pre>\begin{tikzpicture} \pingu[vr-headset, vr-headset hair] \end{tikzpicture}</pre>	
--	---

/pingu/vr-headset text = <text>	({\textit{\textsf{VR\thinspace\texttrademark}}})
<pre>\begin{tikzpicture} \pingu[vr-headset, vr-headset text={ABCD}] \end{tikzpicture}</pre>	

/pingu/vr-headset text color = <color>	(pingu@white)
<pre>\begin{tikzpicture} \pingu[vr-headset, vr-headset text color=green] \end{tikzpicture}</pre>	

A.3.42 The headphones

/pingu/headphone = <color>	(pingu@blue!80!pingu@black)
<pre>\begin{tikzpicture} \pingu[headphone=green] \end{tikzpicture}</pre>	

also written as headphones

The following keys are only in effect if /pingu/headphone is active.

/pingu/headphone left = <color>

(<headphone>!65!pingu@black)

Change the color of the left headphone (automatically sets the color of /pingu/headphone right):

```
\begin{tikzpicture}
  \pingu[headphone, headphone left=green]
\end{tikzpicture}
```



/pingu/headphone right = <color>

(<headphone left>)

```
\begin{tikzpicture}
  \pingu[headphone, headphone right=green]
\end{tikzpicture}
```



/pingu/headphone left outer = <color>

(\@pingu@none)

```
\begin{tikzpicture}
  \pingu[headphone, headphone left outer=green]
\end{tikzpicture}
```



/pingu/headphone right outer = <color>

(\@pingu@none)

```
\begin{tikzpicture}
  \pingu[headphone, headphone right outer=green]
\end{tikzpicture}
```



/pingu/headphone outer = <color>

(pingu@black)

Set /pingu/headphone left outer and /pingu/headphone right outer with the same value:

```
\begin{tikzpicture}
  \pingu[headphone, headphone outer=green]
\end{tikzpicture}
```



also written as headphones outer

/pingu/headphone left inner = <color>

(\@pingu@none)

```
\begin{tikzpicture}
  \pingu[headphone, headphone left inner=green]
\end{tikzpicture}
```



/pingu/headphone right inner = <color>

(\@pingu@none)

```
\begin{tikzpicture}
  \pingu[headphone, headphone right inner=green]
\end{tikzpicture}
```



```
/pingu/headphone inner = <color>
```

```
(pingu@black)
```

Set `/pingu/headphone left inner` and `/pingu/headphone right inner` with the same value:

```
\begin{tikzpicture}
  \pingu[headphone, headphone inner=green]
\end{tikzpicture}
```



also written as `headphones inner`

A.3.43 The santa hat

```
/pingu/santa hat = <color>
```

```
(pingu@red!87!pingu@black)
```

Merry christmas:

```
\begin{tikzpicture}
  \pingu[santa hat=pingu@red]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/santa hat` is active.

```
/pingu/santa hat second = <color>
```

```
(pingu@white!97!<santa hat>)
```

Change the wool color:

```
\begin{tikzpicture}
  \pingu[santa hat,santa hat second=green]
\end{tikzpicture}
```



```
/pingu/santa hat bobble = <color>
```

```
(<santa hat second>)
```

```
\begin{tikzpicture}
  \pingu[santa hat, santa hat bobble=green]
\end{tikzpicture}
```



A.3.44 The santa beard

```
/pingu/santa beard = <color>
```

```
(pingu@white!96!pingu@red!98!pingu@black!92!gray)
```

```
\begin{tikzpicture}
  \pingu[santa beard=brown!20!white]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/santa beard` is active.

```
/pingu/santa beard string = <color>
```

```
(<body main>!85!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[santa beard, santa beard string=green]
\end{tikzpicture}
```



A.3.45 The snowball

/pingu/snowball left = <color>	(pingu@white!90!pingu@black)
<pre>\begin{tikzpicture} \pingu[snowball left=pingu@bronze!80!pingu@black] \end{tikzpicture}</pre>	

The following keys are only in effect if /pingu/snowball left is active.

/pingu/snowball left size = <length>	(2.33mm)
<pre>\begin{tikzpicture} \pingu[snowball left, snowball left size=2mm] \end{tikzpicture}</pre>	

/pingu/snowball left xshift = <length>	(0pt)
<pre>\begin{tikzpicture} \pingu[snowball left, snowball left xshift=2mm] \end{tikzpicture}</pre>	

/pingu/snowball left yshift = <length>	(0pt)
<pre>\begin{tikzpicture} \pingu[snowball left, snowball left yshift=2mm] \end{tikzpicture}</pre>	

/pingu/snowball right = <color>	(pingu@white!90!pingu@black)
<pre>\begin{tikzpicture} \pingu[snowball right=pingu@bronze!80!pingu@black] \end{tikzpicture}</pre>	

The following keys are only in effect if /pingu/snowball right is active.

/pingu/snowball right size = <length>	(2.33mm)
<pre>\begin{tikzpicture} \pingu[snowball right, snowball right size=2mm] \end{tikzpicture}</pre>	

/pingu/snowball right xshift = <length>	(0pt)
<pre>\begin{tikzpicture} \pingu[snowball right, snowball right xshift=2mm] \end{tikzpicture}</pre>	

```
/pingu/snowball right yshift = <length>
```

(Opt)

```
\begin{tikzpicture}
  \pingu[snowball right, snowball right yshift=2mm]
\end{tikzpicture}
```



A.3.46 The basket

```
/pingu/basket left = <color>
```

(brown!95!pingu@main!90!yellow!90!white)

```
\begin{tikzpicture}
  \pingu[basket left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/basket left` is active.

```
/pingu/basket left back = <color>
```

(<basket left>!90!pingu@black)

```
\begin{tikzpicture}
  \pingu[basket left, basket left back=green]
\end{tikzpicture}
```



```
/pingu/basket left egg a = <color>
```

(pingu@basket@eggA)

```
\begin{tikzpicture}
  \pingu[basket left, basket left egg a=green]
\end{tikzpicture}
```



```
/pingu/basket left egg b = <color>
```

(pingu@basket@eggB)

```
\begin{tikzpicture}
  \pingu[basket left, basket left egg b=green]
\end{tikzpicture}
```



```
/pingu/basket left egg c = <color>
```

(pingu@basket@eggC!90!black)

```
\begin{tikzpicture}
  \pingu[basket left, basket left egg c=green]
\end{tikzpicture}
```



```
/pingu/basket left xshift = <length>
```

(Opt)

```
\begin{tikzpicture}
  \pingu[basket left, basket left xshift=2mm]
\end{tikzpicture}
```



```
/pingu/basket left yshift = <length>
```

(Opt)

```
\begin{tikzpicture}
  \pingu[basket left, basket left yshift=2mm]
\end{tikzpicture}
```



/pingu/*basket right* = *<color>*

(brown!95!pingu@main!90!yellow!90!white)

```
\begin{tikzpicture}
  \pingu[basket right=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*basket right* is active.

/pingu/*basket right back* = *<color>*

(<*basket right*>!90!pingu@black)

```
\begin{tikzpicture}
  \pingu[basket right, basket right back=green]
\end{tikzpicture}
```



/pingu/*basket right egg a* = *<color>*

(pingu@basket@eggA)

```
\begin{tikzpicture}
  \pingu[basket right, basket right egg a=green]
\end{tikzpicture}
```



/pingu/*basket right egg b* = *<color>*

(pingu@basket@eggB)

```
\begin{tikzpicture}
  \pingu[basket right, basket right egg b=green]
\end{tikzpicture}
```



/pingu/*basket right egg c* = *<color>*

(pingu@basket@eggC!90!black)

```
\begin{tikzpicture}
  \pingu[basket right, basket right egg c=green]
\end{tikzpicture}
```



/pingu/*basket right xshift* = *<length>*

(Opt)

```
\begin{tikzpicture}
  \pingu[basket right, basket right xshift=2mm]
\end{tikzpicture}
```



/pingu/*basket right yshift* = *<length>*

(Opt)

```
\begin{tikzpicture}
  \pingu[basket right, basket right yshift=2mm]
\end{tikzpicture}
```



A.3.47 The wool hat

/pingu/*wool hat* = *<color>*

(pingu@blue!57!pingu@black)

```
\begin{tikzpicture}
  \pingu[wool hat=pingu@red]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/wool hat` is active.

`/pingu/wool hat second = <color>`

`(<wool hat>!55!pingu@white)`

Change the wool color:

```
\begin{tikzpicture}
  \pingu[wool hat, wool hat second=green]
\end{tikzpicture}
```



`/pingu/wool hat bobble = <color>`

`(<wool hat second>)`

```
\begin{tikzpicture}
  \pingu[wool hat, wool hat bobble=green]
\end{tikzpicture}
```



A.3.48 The deer hat

`/pingu/deer hat = <color>`

`(pingu@bronze!57!pingu@black)`

```
\begin{tikzpicture}
  \pingu[deer hat=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/deer hat` is active.

`/pingu/deer hat b = <color>`

`(<deer hat>)`

```
\begin{tikzpicture}
  \pingu[deer hat, deer hat b=green]
\end{tikzpicture}
```



`/pingu/deer hat band = <color>`

`(<deer hat>!85!pingu@black)`

```
\begin{tikzpicture}
  \pingu[deer hat, deer hat band=green]
\end{tikzpicture}
```



A.3.49 The flip flops

`/pingu/flip-flops = <color>`

`(pingu@blue!70!pingu@white!60!cyan)`

The key is also available as `flip flops`, `flipflops`, `flip-flop`, `flip flop` and `flipflop`.

```
\begin{tikzpicture}
  \pingu[flip-flops=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/flip-flops` is active.

/pingu/*flip-flops base left* = *<color>*

(*<flip-flops>!45!pingu@black*)

```
\begin{tikzpicture}
  \pingu[flip-flops, flip-flops base left=green]
\end{tikzpicture}
```



/pingu/*flip-flops base right* = *<color>*

(*<flip-flops base left>*)

```
\begin{tikzpicture}
  \pingu[flip-flops, flip-flops base right=green]
\end{tikzpicture}
```



/pingu/*flip-flops band left* = *<color>*

(*<flip-flops base left>!35!pingu@red*)

```
\begin{tikzpicture}
  \pingu[flip-flops, flip-flops band left=green]
\end{tikzpicture}
```



/pingu/*flip-flops band right* = *<color>*

(*<flip-flops band left>*)

```
\begin{tikzpicture}
  \pingu[flip-flops, flip-flops band right=green]
\end{tikzpicture}
```



/pingu/*flip-flops color right* = *<color>*

(*<flip-flops>*)

```
\begin{tikzpicture}
  \pingu[flip-flops, flip-flops color right=green]
\end{tikzpicture}
```



A.3.50 The shoes

/pingu/*shoes* = *<color>*

(pingu@bronze!70!pingu@red!80!pingu@white!80!brown)

```
\begin{tikzpicture}
  \pingu[shoes=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*shoes* is active.

/pingu/*shoes base left* = *<color>*

(*<shoes>!45!pingu@white!90!yellow*)

```
\begin{tikzpicture}
  \pingu[shoes, shoes base left=green]
\end{tikzpicture}
```



/pingu/*shoes base right* = *<color>*

(*<shoes base left>*)

```
\begin{tikzpicture}
  \pingu[shoes, shoes base right=green]
\end{tikzpicture}
```



```
/pingu/shoes color right = <color>
```

```
(<shoes>)
```

```
\begin{tikzpicture}
  \pingu[shoes, shoes color right=green]
\end{tikzpicture}
```



A.3.51 The sneaker

```
/pingu/sneaker = <color>
```

```
(pingu@bronze!70!pingu@red!80!pingu@white!80!brown)
```

```
\begin{tikzpicture}
  \pingu[sneaker=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/sneaker` is active.

```
/pingu/sneaker base left = <color>
```

```
(<sneaker>!45!pingu@white!90!yellow)
```

```
\begin{tikzpicture}
  \pingu[sneaker, sneaker base left=green]
\end{tikzpicture}
```



```
/pingu/sneaker base right = <color>
```

```
(<sneaker base left>)
```

```
\begin{tikzpicture}
  \pingu[sneaker, sneaker base right=green]
\end{tikzpicture}
```



```
/pingu/sneaker front left = <color>
```

```
(<sneaker>!32!pingu@white!83!yellow)
```

```
\begin{tikzpicture}
  \pingu[sneaker, sneaker front left=green]
\end{tikzpicture}
```



```
/pingu/sneaker front right = <color>
```

```
(<sneaker front left>)
```

```
\begin{tikzpicture}
  \pingu[sneaker, sneaker front right=green]
\end{tikzpicture}
```



```
/pingu/sneaker side left = <color>
```

```
(<sneaker>!60!pingu@black!85!red!74!brown)
```

```
\begin{tikzpicture}
  \pingu[sneaker, sneaker side left=green]
\end{tikzpicture}
```



```
/pingu/sneaker side right = <color>
```

```
(<sneaker side left>)
```

```
\begin{tikzpicture}
  \pingu[sneaker, sneaker side right=green]
\end{tikzpicture}
```



/pingu/*sneaker band left* = *<color>*(<*sneaker*>!60!pingu@black!35!red)

```
\begin{tikzpicture}
  \pingu[sneaker, sneaker band left=green]
\end{tikzpicture}
```

/pingu/*sneaker band right* = *<color>*(<*sneaker band left*>)

```
\begin{tikzpicture}
  \pingu[sneaker, sneaker band right=green]
\end{tikzpicture}
```

/pingu/*sneaker color right* = *<color>*(<*sneaker*>)

```
\begin{tikzpicture}
  \pingu[sneaker, sneaker color right=green]
\end{tikzpicture}
```



A.3.52 The mask

/pingu/*mask* = *<color>*

(pingu@white!61!gray)

Keep the penguin safe:

```
\begin{tikzpicture}
  \pingu[mask=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*mask* is active.

/pingu/*mask band* = *<color>*(<*body main*>)

```
\begin{tikzpicture}
  \pingu[mask, mask band=green]
\end{tikzpicture}
```

/pingu/*mask line width* = *<length>*

(.565pt)

```
\begin{tikzpicture}
  \pingu[mask, mask line width=1.5pt]
\end{tikzpicture}
```

/pingu/*mask band inner* = *<color>*(<*mask band*>!60!<*mask*>)

```
\begin{tikzpicture}
  \pingu[mask, mask band inner=green]
\end{tikzpicture}
```



```
/pingu/mask band outer = <color>
```

```
(<mask band>!82!<mask>)
```

```
\begin{tikzpicture}
  \pingu[mask, mask band outer=green]
\end{tikzpicture}
```



A.3.53 The blush

```
/pingu/blush = <color>
```

```
(pingu@red)
```

Make it cute:

```
\begin{tikzpicture}
  \pingu[eyes wink, blush=pingu@purple]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/blush` is active.

```
/pingu/blush second = <color>
```

```
(<b blush>)
```

```
\begin{tikzpicture}
  \pingu[blush, blush second=green]
\end{tikzpicture}
```



```
/pingu/blush opacity = <factor>
```

```
(.35)
```

```
\begin{tikzpicture}
  \pingu[blush, blush opacity=.86]
\end{tikzpicture}
```



A.3.54 The banner

```
/pingu/banner = <text>
```

```
(Bannertext)
```

Give the penguin a banner to hold (it adapts to the wing positions):

```
\begin{tikzpicture}
  \pingu[left wing wave, banner>Hello]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/banner` is active.

```
/pingu/banner band = <color>
```

```
(pingu@white!91!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[banner, banner band=green]
\end{tikzpicture}
```



```
/pingu/banner text color = <color>
```

```
(pingu@black)
```

```
\begin{tikzpicture}
  \pingu[wings wave, banner, banner text color=green]
\end{tikzpicture}
```



```
/pingu/banner stick left color = <color>
```

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[banner, banner stick left color=green]
\end{tikzpicture}
```



```
/pingu/banner stick right color = <color>
```

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[banner, banner stick right color=green]
\end{tikzpicture}
```



```
/pingu/banner sticks color = <color>
```

(pingu@bronze)

Calls `/pingu/banner stick left color` and `/pingu/banner stick right color` with the same color:

```
\begin{tikzpicture}
  \pingu[banner, banner sticks color=green]
\end{tikzpicture}
```



```
/pingu/banner stick left length = <length>
```

(20mm)

Changes the banners left stick length:

```
\begin{tikzpicture}
  \pingu[banner, banner stick left length=5mm]
\end{tikzpicture}
```



```
/pingu/banner stick right length = <length>
```

(20mm)

Changes the banners right stick length:

```
\begin{tikzpicture}
  \pingu[banner, banner stick right length=2mm]
\end{tikzpicture}
```



```
/pingu/banner sticks length = <length>
```

(20mm)

Calls `/pingu/banner stick left length` and `/pingu/banner stick right length` with the same length:

```
\begin{tikzpicture}
  \pingu[banner, banner sticks length=9mm]
\end{tikzpicture}
```



```
/pingu/banner raise = <length>
```

(-1mm)

Change the raise of the banner text:

```
\begin{tikzpicture}
  \pingu[banner, banner raise=2mm]
\end{tikzpicture}
```



```
/pingu/banner height = <length>
```

(4mm)

Change the height of the banner (this modifies the half):

```
\begin{tikzpicture}
  \pingu[banner, banner height=6mm]
\end{tikzpicture}
```



```
/pingu/banner font = <font>
```

(\bfseries)

Change the height of the banner:

```
\begin{tikzpicture}
  \pingu[banner, banner font=\itshape]
\end{tikzpicture}
```



```
/pingu/banner bent = <angle>
```

(30)

Change the bending of the banner:

```
\begin{tikzpicture}
  \pingu[banner, banner bent=0]
\end{tikzpicture}
```



A.4 Wing Items

Most wing items created have a two variants: one for the left and one for the right wing. For consistency, both of them are represented in the documentation — many times, they are not just mirrored but two different shapes that appear to be mirrored with special care.

```
/pingu/left wing item angle = <angle>
```

(0)

Relative rotation of the wing items placed in the left wing:

```
\begin{tikzpicture}
  \pingu[cane left, cane right,
    left wing item angle=70]
\end{tikzpicture}
```



also written as `left item angle`

```
/pingu/left wing item flip = <true/false>
```

(false)

Some wing items do have a different style, depending on the wing they are in (e.g. they are mirrored). This option toggles the style for the left wing.

```
\begin{tikzpicture}
  \pingu[flag left, flag right,
    left wing item flip]
\end{tikzpicture}
```



also written as `left item flip`

`/pingu/right wing item angle = <angle>`

(0)

Relative rotation of the `wing items` placed in the right wing:

```
\begin{tikzpicture}
  \pingu[cane left, cane right,
    right wing item angle=70]
\end{tikzpicture}
```



also written as `right item angle`

`/pingu/right wing item flip = <true/false>`

(false)

Some `wing items` do have a different style, depending on the wing they are in (e.g. they are mirrored). This option toggles the style for the right wing.

```
\begin{tikzpicture}
  \pingu[flag left, flag right,
    right wing item flip]
\end{tikzpicture}
```



also written as `right item flip`

A.4.1 The lollipop

`/pingu/lollipop left = <color>`

(pingu@green)

Enable the left lollipop for the penguin:

```
\begin{tikzpicture}
  \pingu[lollipop left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/lollipop left` is active.

`/pingu/lollipop left handle = <color>`

(pingu@bronze)

Change the handle color of the left lollipop:

```
\begin{tikzpicture}
  \pingu[lollipop left, lollipop left handle=green]
\end{tikzpicture}
```



`/pingu/lollipop left second = <color>`

(<lollipop left>!86!white)

Change the second color of the left lollipop, used for the ring:

```
\begin{tikzpicture}
  \pingu[lollipop left, lollipop left second=blue]
\end{tikzpicture}
```



```
/pingu/lollipop right = <color>
```

(pingu@green)

Enable the right lollipop for the penguin:

```
\begin{tikzpicture}
  \pingu[lollipop right=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/lollipop right` is active.

```
/pingu/lollipop right handle = <color>
```

(pingu@bronze)

Change the handle color of the right lollipop:

```
\begin{tikzpicture}
  \pingu[lollipop right, lollipop right handle=green]
\end{tikzpicture}
```



```
/pingu/lollipop right second = <color>
```

(<lollipop right>!86!white)

Change the second color of the right lollipop, used for the ring:

```
\begin{tikzpicture}
  \pingu[lollipop right, lollipop right second=blue]
\end{tikzpicture}
```



A.4.2 The fish

```
/pingu/fish left = <color>
```

(pingu@blue!50!pingu@green!50!black!90!gray)

Enable the left fish for the penguin:

```
\begin{tikzpicture}
  \pingu[fish left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/fish left` is active.

```
/pingu/fish left scales = <color>
```

(white!90!pingu@blue!80!black)

Change the scales color of the left fish:

```
\begin{tikzpicture}
  \pingu[fish left, fish left scales=green]
\end{tikzpicture}
```



```
/pingu/fish left eye = <color>
```

(pingu@white)

Change the eye color of the left fish:

```
\begin{tikzpicture}
  \pingu[fish left, fish left eye=green]
\end{tikzpicture}
```



```
/pingu/fish right = <color>
```

```
(pingu@blue!50!pingu@green!50!black!90!gray)
```

Enable the right fish for the penguin:

```
\begin{tikzpicture}
  \pingu[fish right=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/fish right` is active.

```
/pingu/fish right scales = <color>
```

```
(white!90!pingu@blue!80!black)
```

Change the scales color of the right fish:

```
\begin{tikzpicture}
  \pingu[fish right, fish right scales=green]
\end{tikzpicture}
```



```
/pingu/fish right eye = <color>
```

```
(pingu@white)
```

Change the eye color of the right fish:

```
\begin{tikzpicture}
  \pingu[fish right, fish right eye=green]
\end{tikzpicture}
```



A.4.3 The flower

```
/pingu/flower left = <color>
```

```
(pingu@purple!80!pingu@yellow)
```

```
\begin{tikzpicture}
  \pingu[flower left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/flower left` is active.

```
/pingu/flower left size = <length>
```

```
(.3mm)
```

```
\begin{tikzpicture}
  \pingu[flower left, flower left size=12mm]
\end{tikzpicture}
```



```
/pingu/flower left center = <color>
```

```
(brown!50!pingu@yellow)
```

```
\begin{tikzpicture}
  \pingu[flower left, flower left center=green]
\end{tikzpicture}
```



```
/pingu/flower left center size = <length>
```

```
(.65mm)
```

```
\begin{tikzpicture}
  \pingu[flower left, flower left center size=1cm]
\end{tikzpicture}
```



/pingu/*flower left stem* = <color>

(green!90!pingu@yellow!70!pingu@black)

```
\begin{tikzpicture}
  \pingu[flower left, flower left stem=purple]
\end{tikzpicture}
```

/pingu/*flower left stem length* = <length>

(6mm)

```
\begin{tikzpicture}
  \pingu[flower left, flower left stem length=1cm]
\end{tikzpicture}
```

/pingu/*flower left stem width* = <length>

(.15mm)

```
\begin{tikzpicture}
  \pingu[flower left, flower left stem width=1cm]
\end{tikzpicture}
```

/pingu/*flower left stem tilt* = <length>

(-.5mm)

```
\begin{tikzpicture}
  \pingu[flower left, flower left stem tilt=-1cm]
\end{tikzpicture}
```

/pingu/*flower left petal count* = <count>

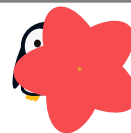
(5)

```
\begin{tikzpicture}
  \pingu[flower left, flower left petal count=8]
\end{tikzpicture}
```

/pingu/*flower left petal width* = <length>

(4.5pt)

```
\begin{tikzpicture}
  \pingu[flower left, flower left petal width=1cm]
\end{tikzpicture}
```

/pingu/*flower left petal start angle* = <length>

(27)

```
\begin{tikzpicture}
  \pingu[flower left, flower left petal start angle=-40]
\end{tikzpicture}
```

/pingu/*flower left petal peak* = <factor>

(2)

```
\begin{tikzpicture}
  \pingu[flower left, flower left petal peak=12]
\end{tikzpicture}
```



/pingu/*flower right* = *<color>*

(pingu@purple!80!pingu@yellow)

```
\begin{tikzpicture}
  \pingu[flower right=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*flower right* is active.

/pingu/*flower right size* = *<length>*

(.3mm)

```
\begin{tikzpicture}
  \pingu[flower right, flower right size=12mm]
\end{tikzpicture}
```



/pingu/*flower right center* = *<color>*

(brown!50!pingu@yellow)

```
\begin{tikzpicture}
  \pingu[flower right, flower right center=green]
\end{tikzpicture}
```



/pingu/*flower right center size* = *<length>*

(.65mm)

```
\begin{tikzpicture}
  \pingu[flower right, flower right center size=1cm]
\end{tikzpicture}
```



/pingu/*flower right stem* = *<color>*

(green!90!pingu@yellow!70!pingu@black)

```
\begin{tikzpicture}
  \pingu[flower right, flower right stem=purple]
\end{tikzpicture}
```



/pingu/*flower right stem length* = *<length>*

(6mm)

```
\begin{tikzpicture}
  \pingu[flower right, flower right stem length=1cm]
\end{tikzpicture}
```



/pingu/*flower right stem width* = *<length>*

(.15mm)

```
\begin{tikzpicture}
  \pingu[flower right, flower right stem width=1cm]
\end{tikzpicture}
```



/pingu/*flower right stem tilt* = *<length>*

(-.5mm)

```
\begin{tikzpicture}
  \pingu[flower right, flower right stem tilt=-1cm]
\end{tikzpicture}
```



/pingu/*flower right petal count* = <count>

(5)

```
\begin{tikzpicture}
  \pingu[flower right, flower right petal count=8]
\end{tikzpicture}
```



/pingu/*flower right petal width* = <length>

(4.5pt)

```
\begin{tikzpicture}
  \pingu[flower right, flower right petal width=1cm]
\end{tikzpicture}
```



/pingu/*flower right petal start angle* = <length>

(27)

```
\begin{tikzpicture}
  \pingu[flower right, flower right petal start angle=-40]
\end{tikzpicture}
```



/pingu/*flower right petal peak* = <factor>

(2)

```
\begin{tikzpicture}
  \pingu[flower right, flower right petal peak=12]
\end{tikzpicture}
```



A.4.4 The cane

/pingu/*cane left* = <color>

(pingu@bronze)

Enable the left cane for the penguin:

```
\begin{tikzpicture}
  \pingu[cane left=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*cane left* is active.

/pingu/*cane left raise* = <length>

(0mm)

Raise the cane of the pingu:

```
\begin{tikzpicture}
  \pingu[cane left, cane left raise=5mm]
\end{tikzpicture}
```



/pingu/*cane right* = <color>

(pingu@bronze)

Enable the right cane for the penguin:

```
\begin{tikzpicture}
  \pingu[cane right=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/cane right` is active.

```
/pingu/cane right raise = <length>
```

(0mm)

Raise the cane of the pingu:

```
\begin{tikzpicture}
  \pingu[cane right,cane right raise=5mm]
\end{tikzpicture}
```



A.4.5 The hand cast

```
/pingu/hand cast left = <text>
```

(X)

Show a symbol above the left wing of the penguin:

```
\begin{tikzpicture}
  \pingu[hand cast left=ABCDEFG]
\end{tikzpicture}
```



also written as `handcast left`

The following keys are only in effect if `/pingu/hand cast left` is active.

```
/pingu/hand cast left color = <color>
```

(pingu@purple)

Change the color of the left hand cast:

```
\begin{tikzpicture}
  \pingu[hand cast left,
    hand cast left color=green]
\end{tikzpicture}
```



also written as `handcast left color`

```
/pingu/hand cast right = <text>
```

(X)

Show a symbol above the right wing of the penguin:

```
\begin{tikzpicture}
  \pingu[hand cast right=ABCDEFG]
\end{tikzpicture}
```



also written as `handcast right`

The following keys are only in effect if `/pingu/hand cast right` is active.

```
/pingu/hand cast right color = <color>
```

(pingu@purple)

Change the color of the right hand cast:

```
\begin{tikzpicture}
  \pingu[hand cast right,
    hand cast right color=green]
\end{tikzpicture}
```



also written as `handcast right color`

A.4.6 The sign post

```
/pingu/sign post left = <text>
```

```
\begin{tikzpicture}
  \pingu[sign post left=ABC]
\end{tikzpicture}
```



also written as `signpost left`

The following keys are only in effect if `/pingu/sign post left` is active.

```
/pingu/sign post left color = <color>
```

```
(brown!70!black)
```

Change the color of the sign post:

```
\begin{tikzpicture}
  \pingu[sign post left, sign post left color=green]
\end{tikzpicture}
```



also written as `signpost left color`

```
/pingu/sign post left font color = <color>
```

```
(white!90!brown)
```

Change the font color of the sign post:

```
\begin{tikzpicture}
  \pingu[sign post left=ABCD,
        sign post left font color=green]
\end{tikzpicture}
```



also written as `signpost left fontcolor`

```
/pingu/sign post right = <text>
```

```
\begin{tikzpicture}
  \pingu[sign post right=ABC]
\end{tikzpicture}
```



also written as `signpost right`

The following keys are only in effect if `/pingu/sign post right` is active.

```
/pingu/sign post right color = <color>
```

```
(brown!70!black)
```

Change the color of the sign post:

```
\begin{tikzpicture}
  \pingu[sign post right, sign post right color=green]
\end{tikzpicture}
```



also written as `signpost right color`

```
/pingu/sign post right font color = <color>
```

```
(white!90!brown)
```

Change the font color of the sign post:

```
\begin{tikzpicture}
  \pingu[sign post right=ABCD,
    sign post right font color=green]
\end{tikzpicture}
```



also written as `signpost right fontcolor`

A.4.7 The lightsaber

```
\begin{tikzpicture}
  \pingu[lightsaber left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/lightsaber left` is active.

```
/pingu/lightsaber left handle = <color>
```

```
(pingu@silver)
```

Change the color of the penguins lightsabers' handle:

```
\begin{tikzpicture}
  \pingu[lightsaber left,
    lightsaber left handle=green]
\end{tikzpicture}
```



```
/pingu/lightsaber left deco = <color>
```

```
(pingu@silver!12!pingu@black)
```

Change the color of the penguins lightsabers' decoration elements:

```
\begin{tikzpicture}
  \pingu[lightsaber left,
    lightsaber left deco=green]
\end{tikzpicture}
```



```
/pingu/lightsaber left ribbs = <color>
```

```
(pingu@silver!50!pingu@black)
```

Change the color of the penguins lightsabers' ribbs:

```
\begin{tikzpicture}
  \pingu[lightsaber left,
    lightsaber left ribbs=green]
\end{tikzpicture}
```



```
/pingu/lightsaber left button = <color>
```

```
(pingu@red!85!pingu@black)
```

Change the color of the penguins lightsabers' first button:

```
\begin{tikzpicture}
  \pingu[lightsaber left,
    lightsaber left button=green]
\end{tikzpicture}
```



```
/pingu/lightsaber left button b = <color>
```

```
(pingu@red!85!pingu@black)
```

Change the color of the penguins lightsabers' second button:

```
\begin{tikzpicture}
  \pingu[lightsaber left,
    lightsaber left button b=green]
\end{tikzpicture}
```



```
/pingu/lightsaber left double = <true/false>
```

```
(false)
```

Toggle the visibility of the second lightsaber:

```
\begin{tikzpicture}
  \pingu[lightsaber left, lightsaber left double,
    left wing item angle=90]
\end{tikzpicture}
```



```
/pingu/lightsaber left color b = <color>
```

```
(<lightsaber left>)
```

Change the color of the penguins second lightsaber, which is only shown if `/pingu/lightsaber left double` is enabled:

```
\begin{tikzpicture}
  \pingu[lightsaber left, lightsaber left double,
    lightsaber left color b=green,
    left wing item angle=90]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[lightsaber left, lightsaber left length=6mm]
\end{tikzpicture}
```



```
/pingu/lightsaber left length b = <length>
```

```
(<lightsaber left length>)
```

Change the length of the penguins second lightsaber (active with `/pingu/lightsaber left double`):

```
\begin{tikzpicture}
  \pingu[lightsaber left, lightsaber left double,
    lightsaber left length b=6mm]
\end{tikzpicture}
```



```
/pingu/lightsaber left yshift = <length>
```

(0pt)

Shift the penguins lightsaber in the y direction:

```
\begin{tikzpicture}
  \pingu[lightsaber left, lightsaber left yshift=12mm,
    lightsaber left length=5mm]
\end{tikzpicture}
```



```
/pingu/lightsaber left glow = <true/false>
```

(\ifpengu@setup@glows@ true\else false\fi)

Toggle the glow of the lightsaber. The glow is enabled by default; use the noglows package option to disable it.

```
\begin{tikzpicture}
  \pingu[lightsaber left, lightsaber left glow=false]
\end{tikzpicture}
```



```
/pingu/lightsaber left solid
```

Disables the /pingu/lightsaber left glow:

```
\begin{tikzpicture}
  \pingu[lightsaber left, lightsaber left solid]
\end{tikzpicture}
```



```
/pingu/lightsaber left glow core = <color>
```

(white)

Change the color of the lightsabers glow core:

```
\begin{tikzpicture}
  \pingu[lightsaber left, lightsaber left glow=true,
    lightsaber left glow core=cyan]
\end{tikzpicture}
```

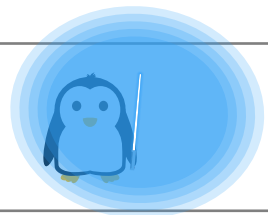


```
/pingu/lightsaber left outer glow factor = <factor>
```

(\pingulightsaberfactor)

Modify the glow factor of the left lightsaber:

```
\begin{tikzpicture}
  \pingu[lightsaber left,
    lightsaber left outer glow factor=.3]
\end{tikzpicture}
```



```
/pingu/lightsaber left disabled
```

Disables the lightsaber so only the handle is visible:

```
\begin{tikzpicture}
  \pingu[lightsaber left, lightsaber left disabled]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[lightsaber right=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/lightsaber right` is active.

```
/pingu/lightsaber right handle = <color>
```

```
(pingu@silver)
```

Change the color of the penguins lightsabers' handle:

```
\begin{tikzpicture}
  \pingu[lightsaber right,
    lightsaber right handle=green]
\end{tikzpicture}
```



```
/pingu/lightsaber right deco = <color>
```

```
(pingu@silver!12!pingu@black)
```

Change the color of the penguins lightsabers' decoration elements:

```
\begin{tikzpicture}
  \pingu[lightsaber right,
    lightsaber right deco=green]
\end{tikzpicture}
```



```
/pingu/lightsaber right ribbs = <color>
```

```
(pingu@silver!50!pingu@black)
```

Change the color of the penguins lightsabers' ribbs:

```
\begin{tikzpicture}
  \pingu[lightsaber right,
    lightsaber right ribbs=green]
\end{tikzpicture}
```



```
/pingu/lightsaber right button = <color>
```

```
(pingu@red!85!pingu@black)
```

Change the color of the penguins lightsabers' first button:

```
\begin{tikzpicture}
  \pingu[lightsaber right,
    lightsaber right button=green]
\end{tikzpicture}
```



```
/pingu/lightsaber right button b = <color>
```

```
(pingu@red!85!pingu@black)
```

Change the color of the penguins lightsabers' second button:

```
\begin{tikzpicture}
  \pingu[lightsaber right,
    lightsaber right button b=green]
\end{tikzpicture}
```



```
/pingu/lightsaber right double = <true/false>
```

(false)

Toggle the visibility of the second lightsaber:

```
\begin{tikzpicture}
  \pingu[lightsaber right, lightsaber right double,
    right wing item angle=90]
\end{tikzpicture}
```



```
/pingu/lightsaber right color b = <color>
```

(<lightsaber right>)

Change the color of the penguins second lightsaber, which is only shown if `/pingu/lightsaber right double` is enabled:

```
\begin{tikzpicture}
  \pingu[lightsaber right, lightsaber right double,
    lightsaber right color b=green,
    right wing item angle=90]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[lightsaber right,
    lightsaber right length=6mm]
\end{tikzpicture}
```



```
/pingu/lightsaber right length b = <length>
```

(<lightsaber right length>)

Change the length of the penguins second lightsaber (active with `/pingu/lightsaber right double`):

```
\begin{tikzpicture}
  \pingu[lightsaber right, lightsaber right double,
    lightsaber right length b=6mm]
\end{tikzpicture}
```



```
/pingu/lightsaber right yshift = <length>
```

(0pt)

Shift the penguins lightsaber in the y direction:

```
\begin{tikzpicture}
  \pingu[lightsaber right,
    lightsaber right yshift=12mm,
    lightsaber right length=5mm]
\end{tikzpicture}
```



```
/pingu/lightsaber right glow = <true/false>
```

(\ifpengu@setup@glows@ true\else false\fi)

Toggle the glow of the lightsaber. The glow is enabled by default; use the `noglows` package option to disable it.

```
\begin{tikzpicture}
  \pingu[lightsaber right,
    lightsaber right glow=false]
\end{tikzpicture}
```



```
/pingu/lightsaber right solid
```

Disables the `/pingu/lightsaber right glow`:

```
\begin{tikzpicture}
  \pingu[lightsaber right, lightsaber right solid]
\end{tikzpicture}
```



```
/pingu/lightsaber right glow core = <color>
```

```
(white)
```

Change the color of the lightsabers glow core:

```
\begin{tikzpicture}
  \pingu[lightsaber right, lightsaber right glow=true,
  lightsaber right glow core=cyan]
\end{tikzpicture}
```

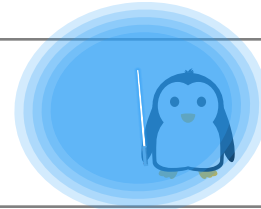


```
/pingu/lightsaber right outer glow factor = <factor>
```

```
(\pingulightsaberfactor)
```

Modify the glow factor of the right lightsaber:

```
\begin{tikzpicture}
  \pingu[lightsaber right,
  lightsaber right outer glow factor=.3]
\end{tikzpicture}
```



```
/pingu/lightsaber right disabled
```

Disables the lightsaber so only the handle is visible:

```
\begin{tikzpicture}
  \pingu[lightsaber right, lightsaber right disabled]
\end{tikzpicture}
```



A.4.8 The lightstaff

```
/pingu/light-staff left = <color>
```

```
(pingu@green)
```

Color similar to `/pingu/light-staff left head`:

```
\begin{tikzpicture}
  \pingu[light-staff left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/light-staff left` is active.

```
\begin{tikzpicture}
  \pingu[light-staff left,
  light-staff left length=18mm]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[light-staff left,
    light-staff left glow length=16mm]
\end{tikzpicture}
```



```
/pingu/light-staff left head = <color>
```

```
(pingu@green)
```

Same as assigning the color to `/pingu/light-staff left`:

```
\begin{tikzpicture}
  \pingu[light-staff left,
    light-staff left head=pingu@purple]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[light-staff left,
    light-staff left staff=green]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[light-staff left,
    light-staff left core=green]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[light-staff left,
    light-staff left core width=2mm]
\end{tikzpicture}
```



```
/pingu/light-staff left outer glow factor = <factor>
```

```
(.022)
```

Similar to `/pingu/lightsaber left outer glow factor`:

```
\begin{tikzpicture}
  \pingu[light-staff left,
    light-staff left outer glow factor=.5]
\end{tikzpicture}
```



```
/pingu/light-staff right = <color>
```

```
(pingu@green)
```

Color similar to `/pingu/light-staff right head`:

```
\begin{tikzpicture}
  \pingu[light-staff right=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/light-staff right` is active.

```
\begin{tikzpicture}
  \pingu[light-staff right,
    light-staff right length=18mm]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[light-staff right,
    light-staff right glow length=16mm]
\end{tikzpicture}
```



```
/pingu/light-staff right head = <color>
```

```
(pingu@green)
```

Same as assigning the color to `/pingu/light-staff right`:

```
\begin{tikzpicture}
  \pingu[light-staff right,
    light-staff right head=pingu@purple]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[light-staff right,
    light-staff right staff=green]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[light-staff right,
    light-staff right core=green]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[light-staff right,
    light-staff right core width=2mm]
\end{tikzpicture}
```



```
/pingu/light-staff right outer glow factor = <factor>
```

```
(.022)
```

Similar to `/pingu/lightsaber right outer glow factor`:

```
\begin{tikzpicture}
  \pingu[light-staff right,
    light-staff right outer glow factor=.5]
\end{tikzpicture}
```



A.4.9 The broom

```
\begin{tikzpicture}
  \pingu[broom left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/broom left` is active.

`/pingu/broom left length = <length>`

(22mm)

```
\begin{tikzpicture}
  \pingu[broom left, broom left length=17mm]
\end{tikzpicture}
```



`/pingu/broom left shift = <length>`

(8mm)

```
\begin{tikzpicture}
  \pingu[broom left, broom left shift=4mm]
\end{tikzpicture}
```



`/pingu/broom left band = <color>`

(pingu@red!80!pingu@black!80!pingu@bronze)

```
\begin{tikzpicture}
  \pingu[broom left, broom left band=pingu@green]
\end{tikzpicture}
```



`/pingu/broom left bristles a = <color> (pingu@bronze!16!pingu@yellow!41!pingu@white!80!brown!95!<broom left>)`

```
\begin{tikzpicture}
  \pingu[broom left, broom left bristles a=pingu@green]
\end{tikzpicture}
```



`/pingu/broom left bristles b = <color>`

(<broom left bristles a>!83!pingu@white)

```
\begin{tikzpicture}
  \pingu[broom left, broom left bristles b=pingu@green]
\end{tikzpicture}
```



`/pingu/broom left bristles c = <color>`

(<broom left bristles b>!83!pingu@white)

```
\begin{tikzpicture}
  \pingu[broom left, broom left bristles c=pingu@green]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[broom right=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/broom right` is active.

/pingu/*broom right length* = *<length>*

(22mm)

```
\begin{tikzpicture}
  \pingu[broom right, broom right length=17mm]
\end{tikzpicture}
```



/pingu/*broom right shift* = *<length>*

(8mm)

```
\begin{tikzpicture}
  \pingu[broom right, broom right shift=4mm]
\end{tikzpicture}
```



/pingu/*broom right band* = *<color>*

(pingu@red!80!pingu@black!80!pingu@bronze)

```
\begin{tikzpicture}
  \pingu[broom right, broom right band=pingu@green]
\end{tikzpicture}
```



/pingu/*broom right bristles a* = *<color>*

(pingu@bronze!16!pingu@yellow!41!pingu@white!80!brown!95!

<broom right>)

```
\begin{tikzpicture}
  \pingu[broom right, broom right bristles a=pingu@green]
\end{tikzpicture}
```



/pingu/*broom right bristles b* = *<color>*

(*<broom right bristles a>!83!pingu@white*)

```
\begin{tikzpicture}
  \pingu[broom right, broom right bristles b=pingu@green]
\end{tikzpicture}
```



/pingu/*broom right bristles c* = *<color>*

(*<broom right bristles b>!83!pingu@white*)

```
\begin{tikzpicture}
  \pingu[broom right, broom right bristles c=pingu@green]
\end{tikzpicture}
```



A.4.10 The guitar

```
\begin{tikzpicture}
  \pingu[guitar left=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*guitar left* is active.

/pingu/*guitar left side color* = *<color>*

(*<guitar left>!80!pingu@white*)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left side color=green]
\end{tikzpicture}
```



/pingu/guitar left tint color = <color> (white!99.5!<guitar left>)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left tint color=green]
\end{tikzpicture}
```



/pingu/guitar left outline color = <color> (black)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left outline color=green]
\end{tikzpicture}
```



/pingu/guitar left fret color = <color> (brown!80!black)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left fret color=green]
\end{tikzpicture}
```



/pingu/guitar left fret side color = <color> (<guitar left fret color>!80!pingu@white)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left fret side color=green]
\end{tikzpicture}
```



/pingu/guitar left head color = <color> (brown!89!white)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left head color=green]
\end{tikzpicture}
```



/pingu/guitar left strings color = <color> (pingu@silver!70!pingu@black!50!white)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left strings color=green]
\end{tikzpicture}
```



/pingu/guitar left peg color = <color> (pingu@black)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left peg color=green]
\end{tikzpicture}
```



/pingu/guitar left strings end color = <color> (pingu@black!80!<guitar left>)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left strings end color=green]
\end{tikzpicture}
```



/pingu/guitar left pickup color = <color>

(white!99!pingu@black)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left pickup color=green]
\end{tikzpicture}
```



/pingu/guitar left fret lines color = <color>

(pingu@silver!70!brown!30!white)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left fret lines color=green]
\end{tikzpicture}
```



/pingu/guitar left strings shift = <length>

(Opt)

```
\begin{tikzpicture}
  \pingu[guitar left, guitar left strings shift=2mm]
\end{tikzpicture}
```



```
\begin{tikzpicture}
  \pingu[guitar right=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/guitar right is active.

/pingu/guitar right side color = <color>

(<guitar right>!80!pingu@white)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right side color=green]
\end{tikzpicture}
```



/pingu/guitar right tint color = <color>

(white!99.5!<guitar right>)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right tint color=green]
\end{tikzpicture}
```



/pingu/guitar right outline color = <color>

(black)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right outline color=green]
\end{tikzpicture}
```



/pingu/guitar right fret color = <color>

(brown!80!black)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right fret color=green]
\end{tikzpicture}
```



/pingu/*guitar right fret side color* = *<color>*

(*<guitar right fret color>*!80!pingu@white)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right fret side color=green]
\end{tikzpicture}
```



/pingu/*guitar right head color* = *<color>*

(brown!89!white)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right head color=green]
\end{tikzpicture}
```



/pingu/*guitar right strings color* = *<color>*

(pingu@silver!70!pingu@black!50!white)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right strings color=green]
\end{tikzpicture}
```



/pingu/*guitar right peg color* = *<color>*

(pingu@black)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right peg color=green]
\end{tikzpicture}
```



/pingu/*guitar right strings end color* = *<color>*

(pingu@black!80!<guitar right>)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right strings end color=green]
\end{tikzpicture}
```



/pingu/*guitar right pickup color* = *<color>*

(white!99!pingu@black)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right pickup color=green]
\end{tikzpicture}
```



/pingu/*guitar right fret lines color* = *<color>*

(pingu@silver!70!brown!30!white)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right fret lines color=green]
\end{tikzpicture}
```



/pingu/*guitar right strings shift* = *<length>*

(Opt)

```
\begin{tikzpicture}
  \pingu[guitar right, guitar right strings shift=2mm]
\end{tikzpicture}
```



A.4.11 The flag

The flag is special in that it is meant to be customized by commands so that the visible insignia is to the users liking.

```
/pingu/flag left = <color>
```

(pingu@purple)

```
\begin{tikzpicture}
  \pingu[flag left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/flag left` is active.

```
/pingu/flag left pole = <color>
```

(pingu@bronze)

Change the color of the flag pole:

```
\begin{tikzpicture}
  \pingu[flag left, flag left pole=green]
\end{tikzpicture}
```



```
/pingu/flag left bobble = <color>
```

(pingu@bronze)

Change the color of the flag poles top bobble:

```
\begin{tikzpicture}
  \pingu[flag left, flag left bobble=green]
\end{tikzpicture}
```



```
/pingu/flag left code = <TeX-code> ({ \noexpand\node[/pingu/@flag@first,/pingu/@flag={<flag left>}
{9.75mm}] (a) at (0,0) {};} )
```

Set the flag code which is effectively the drawing code of the flag. You can use the styles `/pingu/@flag@first` and `/pingu/@flag` to inherit the default flag styles and to stay compliant with the modifications of the other macros:

```
\begin{tikzpicture}
  \pingu[flag left, flag left code={
    \node[/pingu/@flag@first,
      /pingu/@flag={blue}{5mm}]
      (upper) at (0,0) {};
    \node[below,/pingu/@flag={black}{4mm}]
      (lower) at (upper.south) {};
  }]
\end{tikzpicture}
```



Note that `/pingu/@flag` expects two arguments: the color of the flag segment and its thickness.

`/pingu/pride flag left = <color>`

(pingu@bronze)

Uses `/pingu/flag left`, `/pingu/flag left code`, and `/pingu/flag left pole` to set a pride flag. The color argument is passed to `/pingu/flag left pole`.

```
\begin{tikzpicture}
  \pingu[pride flag left=green]
\end{tikzpicture}
```



`/pingu/german flag left = <color>`

(pingu@bronze)

Uses `/pingu/flag left`, `/pingu/flag left code`, and `/pingu/flag left pole` to set a german flag. The color argument is passed to `/pingu/flag left pole`.

```
\begin{tikzpicture}
  \pingu[german flag left=green]
\end{tikzpicture}
```



`/pingu/flag right = <color>`

(pingu@purple)

```
\begin{tikzpicture}
  \pingu[flag right=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/flag right` is active.

`/pingu/flag right pole = <color>`

(pingu@bronze)

Change the color of the flag pole:

```
\begin{tikzpicture}
  \pingu[flag right, flag right pole=green]
\end{tikzpicture}
```



`/pingu/flag right bobble = <color>`

(pingu@bronze)

Change the color of the flag poles top bobble:

```
\begin{tikzpicture}
  \pingu[flag right, flag right bobble=green]
\end{tikzpicture}
```



```
/pingu/flag right code = <TeX-code> ({ \noexpand\node[/pingu/@flag@first,/pingu/@flag={<flag right>}{9.75mm}}] (a) at (0,0) {};} )
```

Set the flag code which is effectively the drawing code of the flag. You can use the styles `/pingu/@flag@first` and `/pingu/@flag` to inherit the default flag styles and to stay compliant with the modifications of the other macros:

```
\begin{tikzpicture}
  \pingu[flag right, flag right code={
    \node[/pingu/@flag@first,
      /pingu/@flag={blue}{5mm}]
      (upper) at (0,0) {};
    \node[below,/pingu/@flag={black}{4mm}]
      (lower) at (upper.south) {};
  }]
\end{tikzpicture}
```



Note that `/pingu/@flag` expects two arguments: the color of the flag segment and its thickness.

```
/pingu/pride flag right = <color>
```

(pingu@bronze)

Uses `/pingu/flag right`, `/pingu/flag right code`, and `/pingu/flag right pole` to set a pride flag. The color argument is passed to `/pingu/flag right pole`.

```
\begin{tikzpicture}
  \pingu[pride flag right=green]
\end{tikzpicture}
```



```
/pingu/german flag right = <color>
```

(pingu@bronze)

Uses `/pingu/flag right`, `/pingu/flag right code`, and `/pingu/flag right pole` to set a german flag. The color argument is passed to `/pingu/flag right pole`.

```
\begin{tikzpicture}
  \pingu[german flag right=green]
\end{tikzpicture}
```



A.4.12 The present

```
/pingu/present left = <color>
```

(pingu@red!50!pingu@black!60!red)

```
\begin{tikzpicture}
  \pingu[present left=blue]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/present left` is active.

```
/pingu/present left width = <length>
```

(7.75mm)

```
\begin{tikzpicture}
  \pingu[present left, present left width=12mm]
\end{tikzpicture}
```



/pingu/*present left height* = *<length>*

(7mm)

```
\begin{tikzpicture}
  \pingu[present left, present left height=12mm]
\end{tikzpicture}
```

/pingu/*present left ribbon* = *<color>*(*<present left band>!97!pingu@black*)

```
\begin{tikzpicture}
  \pingu[present left, present left ribbon=blue]
\end{tikzpicture}
```

/pingu/*present left band* = *<color>*

(pingu@green!70!pingu@black!90!pingu@white)

```
\begin{tikzpicture}
  \pingu[present left, present left band=blue]
\end{tikzpicture}
```

/pingu/*present left band width* = *<length>*

(1.75mm)

```
\begin{tikzpicture}
  \pingu[present left, present left band width=4mm]
\end{tikzpicture}
```

/pingu/*present left xshift* = *<length>*

(-.25*(<present left width>))

```
\begin{tikzpicture}
  \pingu[present left, present left xshift=3mm]
\end{tikzpicture}
```

/pingu/*present left yshift* = *<length>*

(-.25*(<present left height>))

```
\begin{tikzpicture}
  \pingu[present left, present left yshift=3mm]
\end{tikzpicture}
```

/pingu/*present left band second height* = *<length>*

(<present left band width>)

```
\begin{tikzpicture}
  \pingu[present left, present left band second height=3mm]
\end{tikzpicture}
```

/pingu/*present left band second yshift* = *<length>*

(.5*(<present left height>))

```
\begin{tikzpicture}
  \pingu[present left, present left band second yshift=3mm]
\end{tikzpicture}
```



/pingu/*present left lid* = *<color>*(*<present left>!92!pingu@black*)

```
\begin{tikzpicture}
  \pingu[present left, present left lid=green]
\end{tikzpicture}
```

/pingu/*present left lid height* = *<length>*

(1.33mm)

```
\begin{tikzpicture}
  \pingu[present left, present left lid height=3mm]
\end{tikzpicture}
```

/pingu/*present left lid overhang* = *<length>*

(.125mm)

```
\begin{tikzpicture}
  \pingu[present left, present left lid overhang=1mm]
\end{tikzpicture}
```

/pingu/*present left lid yshift* = *<length>*

(0)

```
\begin{tikzpicture}
  \pingu[present left, present left lid yshift=8mm]
\end{tikzpicture}
```

/pingu/*present left lid band* = *<color>*(*<present left band>!98!pingu@black*)

```
\begin{tikzpicture}
  \pingu[present left, present left lid band=blue]
\end{tikzpicture}
```

/pingu/*present right* = *<color>*

(pingu@red!50!pingu@black!60!red)

```
\begin{tikzpicture}
  \pingu[present right=blue]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*present right* is active.

/pingu/*present right width* = *<length>*

(7.75mm)

```
\begin{tikzpicture}
  \pingu[present right, present right width=12mm]
\end{tikzpicture}
```

/pingu/*present right height* = *<length>*

(7mm)

```
\begin{tikzpicture}
  \pingu[present right, present right height=12mm]
\end{tikzpicture}
```



/pingu/*present right ribbon* = *<color>*

(*<present right band>!97!pingu@black*)

```
\begin{tikzpicture}
  \pingu[present right, present right ribbon=blue]
\end{tikzpicture}
```



/pingu/*present right band* = *<color>*

(pingu@green!70!pingu@black!90!pingu@white)

```
\begin{tikzpicture}
  \pingu[present right, present right band=blue]
\end{tikzpicture}
```



/pingu/*present right band width* = *<length>*

(1.75mm)

```
\begin{tikzpicture}
  \pingu[present right, present right band width=4mm]
\end{tikzpicture}
```



/pingu/*present right xshift* = *<length>*

(-.25*(*<present right width>*))

```
\begin{tikzpicture}
  \pingu[present right, present right xshift=3mm]
\end{tikzpicture}
```



/pingu/*present right yshift* = *<length>*

(-.25*(*<present right height>*))

```
\begin{tikzpicture}
  \pingu[present right, present right yshift=3mm]
\end{tikzpicture}
```



/pingu/*present right band second height* = *<length>*

(*<present right band width>*)

```
\begin{tikzpicture}
  \pingu[present right, present right band second height=3mm]
\end{tikzpicture}
```



/pingu/*present right band second yshift* = *<length>*

(.5*(*<present right height>*))

```
\begin{tikzpicture}
  \pingu[present right, present right band second yshift=3mm]
\end{tikzpicture}
```



/pingu/*present right lid* = *<color>*

(*<present right>!92!pingu@black*)

```
\begin{tikzpicture}
  \pingu[present right, present right lid=green]
\end{tikzpicture}
```



/pingu/*present right lid height* = *<length>*

(1.33mm)

```
\begin{tikzpicture}
  \pingu[present right, present right lid height=3mm]
\end{tikzpicture}
```

/pingu/*present right lid overhang* = *<length>*

(.125mm)

```
\begin{tikzpicture}
  \pingu[present right, present right lid overhang=1mm]
\end{tikzpicture}
```

/pingu/*present right lid yshift* = *<length>*

(0)

```
\begin{tikzpicture}
  \pingu[present right, present right lid yshift=8mm]
\end{tikzpicture}
```

/pingu/*present right lid band* = *<color>*

(<present right band>!98!pingu@black)

```
\begin{tikzpicture}
  \pingu[present right, present right lid band=blue]
\end{tikzpicture}
```



A.4.13 The staff

/pingu/*staff left* = *<color>*

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[staff left=green, left item angle=70]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*staff left* is active.

/pingu/*staff left length* = *<length>*

(28mm)

Change the staff length:

```
\begin{tikzpicture}
  \pingu[staff left, staff left length=20mm]
\end{tikzpicture}
```

/pingu/*staff left raise* = *<length>*

(-11mm)

```
\begin{tikzpicture}
  \pingu[staff left, staff left raise=-2mm]
\end{tikzpicture}
```



```
/pingu/staff left code = <tikz-code>
```

```
\begin{tikzpicture}
  \pingu[staff left, staff left code={\fill circle[radius=2mm];}]
\end{tikzpicture}
```



```
/pingu/staff left code app = <tikz-code>
```

Similar to `/pingu/staff left code` but does not override but (locally) append to previous calls.

```
/pingu/staff left code pre = <tikz-code>
```

Similar to `/pingu/staff left code` but does not override but (locally) prepend to previous calls.

```
/pingu/spear left = <color>
```

(gray)

Example of `/pingu/staff left code app` that renders a little spear-tip:

```
\begin{tikzpicture}
  \pingu[spear left=cyan]
\end{tikzpicture}
```



```
/pingu/cross left = <color>
```

(pingu@bronze)

Example of `/pingu/staff left code app` that renders a little cross:

```
\begin{tikzpicture}
  \pingu[cross left]
\end{tikzpicture}
```



```
/pingu/staff right = <color>
```

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[staff right=green, right item angle=70]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/staff right` is active.

```
/pingu/staff right length = <length>
```

(28mm)

Change the staff length:

```
\begin{tikzpicture}
  \pingu[staff right, staff right length=20mm]
\end{tikzpicture}
```



```
/pingu/staff right raise = <length>
```

(-11mm)

```
\begin{tikzpicture}
  \pingu[staff right, staff right raise=-2mm]
\end{tikzpicture}
```



```
/pingu/staff right code = <tikz-code>
```

```
\begin{tikzpicture}
  \pingu[staff right, staff right code={\fill circle[radius=2mm];}]
\end{tikzpicture}
```



```
/pingu/staff right code app = <tikz-code>
```

Similar to `/pingu/staff right code` but does not override but (locally) append to previous calls.

```
/pingu/staff right code pre = <tikz-code>
```

Similar to `/pingu/staff right code` but does not override but (locally) prepend to previous calls.

```
/pingu/spear right = <color>
```

(gray)

Example of `/pingu/staff right code app` that renders a little spear-tip:

```
\begin{tikzpicture}
  \pingu[spear right=cyan]
\end{tikzpicture}
```



```
/pingu/cross right = <color>
```

(pingu@bronze)

Example of `/pingu/staff right code app` that renders a little cross:

```
\begin{tikzpicture}
  \pingu[cross right]
\end{tikzpicture}
```



A.4.14 The hammer

```
/pingu/hammer left = <color>
```

(pingu@silver)

```
\begin{tikzpicture}
  \pingu[hammer left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/hammer left` is active.

```
/pingu/hammer left handle length = <length>
```

(9mm)

```
\begin{tikzpicture}
  \pingu[hammer left, hammer left handle length=20mm]
\end{tikzpicture}
```



```
/pingu/hammer left handle shift = <length>
```

(-4mm)

```
\begin{tikzpicture}
  \pingu[hammer left, hammer left handle shift=2mm]
\end{tikzpicture}
```



/pingu/hammer left width = <length>

(7mm)

```
\begin{tikzpicture}
  \pingu[hammer left, hammer left width=2mm]
\end{tikzpicture}
```



/pingu/hammer left height = <length>

(3.5mm)

```
\begin{tikzpicture}
  \pingu[hammer left, hammer left height=2mm]
\end{tikzpicture}
```



/pingu/hammer left xshift = <length>

(-0.5*<hammer left width>)

```
\begin{tikzpicture}
  \pingu[hammer left, hammer left xshift=2mm]
\end{tikzpicture}
```



/pingu/hammer left yshift = <length>

(-1mm)

```
\begin{tikzpicture}
  \pingu[hammer left, hammer left yshift=2mm]
\end{tikzpicture}
```



/pingu/hammer left handle = <color>

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[hammer left, hammer left handle=green]
\end{tikzpicture}
```



/pingu/hammer left with knob = <color>

(pingu@silver)

```
\begin{tikzpicture}
  \pingu[hammer left=green]
\end{tikzpicture}
```



/pingu/hammer right = <color>

(pingu@silver)

```
\begin{tikzpicture}
  \pingu[hammer right=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/hammer right is active.

/pingu/hammer right handle length = <length>

(9mm)

```
\begin{tikzpicture}
  \pingu[hammer right, hammer right handle length=20mm]
\end{tikzpicture}
```



/pingu/hammer right handle shift = <length>

(-4mm)

```
\begin{tikzpicture}
  \pingu[hammer right, hammer right handle shift=2mm]
\end{tikzpicture}
```



/pingu/hammer right width = <length>

(7mm)

```
\begin{tikzpicture}
  \pingu[hammer right, hammer right width=2mm]
\end{tikzpicture}
```



/pingu/hammer right height = <length>

(3.5mm)

```
\begin{tikzpicture}
  \pingu[hammer right, hammer right height=2mm]
\end{tikzpicture}
```



/pingu/hammer right xshift = <length>

(-0.5*<hammer right width>)

```
\begin{tikzpicture}
  \pingu[hammer right, hammer right xshift=2mm]
\end{tikzpicture}
```



/pingu/hammer right yshift = <length>

(-1mm)

```
\begin{tikzpicture}
  \pingu[hammer right, hammer right yshift=2mm]
\end{tikzpicture}
```



/pingu/hammer right handle = <color>

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[hammer right, hammer right handle=green]
\end{tikzpicture}
```



/pingu/hammer right with knob = <color>

(pingu@silver)

```
\begin{tikzpicture}
  \pingu[hammer right=green]
\end{tikzpicture}
```



A.4.15 The drum stick

/pingu/drum stick left = <color>

(pingu@bronze!90!white)

```
\begin{tikzpicture}
  \pingu[drum stick left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/drum stick left` is active.

<pre>/pingu/drum stick left handle length = <length></pre> <pre>\begin{tikzpicture}</pre> <pre> \pingu[drum stick left, drum stick left handle length=2mm]</pre> <pre>\end{tikzpicture}</pre>	(4mm) 
<pre>/pingu/drum stick left handle shift = <length></pre> <pre>\begin{tikzpicture}</pre> <pre> \pingu[drum stick left, drum stick left handle shift=2mm]</pre> <pre>\end{tikzpicture}</pre>	(-2mm) 
<pre>/pingu/drum stick left xshift = <length></pre> <pre>\begin{tikzpicture}</pre> <pre> \pingu[drum stick left, drum stick left xshift=2mm]</pre> <pre>\end{tikzpicture}</pre>	$(-0.5 * \text{pingu@drumstick@left@width})$ 
<pre>/pingu/drum stick left yshift = <length></pre> <pre>\begin{tikzpicture}</pre> <pre> \pingu[drum stick left, drum stick left yshift=2mm]</pre> <pre>\end{tikzpicture}</pre>	(-1mm) 
<pre>/pingu/drum stick left handle = <color></pre> <pre>\begin{tikzpicture}</pre> <pre> \pingu[drum stick left, drum stick left handle=green]</pre> <pre>\end{tikzpicture}</pre>	(pingu@bronze) 
<pre>/pingu/drum stick right = <color></pre> <pre>\begin{tikzpicture}</pre> <pre> \pingu[drum stick right=green]</pre> <pre>\end{tikzpicture}</pre>	(pingu@bronze!90!white) 

The following keys are only in effect if `/pingu/drum stick right` is active.

<pre>/pingu/drum stick right handle length = <length></pre> <pre>\begin{tikzpicture}</pre> <pre> \pingu[drum stick right, drum stick right handle length=2mm]</pre> <pre>\end{tikzpicture}</pre>	(4mm) 
<pre>/pingu/drum stick right handle shift = <length></pre> <pre>\begin{tikzpicture}</pre> <pre> \pingu[drum stick right, drum stick right handle shift=2mm]</pre> <pre>\end{tikzpicture}</pre>	(-2mm) 

/pingu/drum stick right xshift = <length>

(-0.5*\pingu@drumstick@right@width)

```
\begin{tikzpicture}
  \pingu[drum stick right, drum stick right xshift=2mm]
\end{tikzpicture}
```



/pingu/drum stick right yshift = <length>

(-1mm)

```
\begin{tikzpicture}
  \pingu[drum stick right, drum stick right yshift=2mm]
\end{tikzpicture}
```



/pingu/drum stick right handle = <color>

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[drum stick right, drum stick right handle=green]
\end{tikzpicture}
```



A.4.16 The microphone

/pingu/microphone left = <color>

(pingu@silver!90!white)

```
\begin{tikzpicture}
  \pingu[microphone left=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/microphone left is active.

/pingu/microphone left handle length = <length>

(5.5mm)

```
\begin{tikzpicture}
  \pingu[microphone left, microphone left handle length=2mm]
\end{tikzpicture}
```



/pingu/microphone left handle shift = <length>

(-1.15mm)

```
\begin{tikzpicture}
  \pingu[microphone left, microphone left handle shift=2mm]
\end{tikzpicture}
```



/pingu/microphone left xshift = <length>

(-0.5*\pingu@microphone@left@width)

```
\begin{tikzpicture}
  \pingu[microphone left, microphone left xshift=2mm]
\end{tikzpicture}
```



/pingu/microphone left yshift = <length>

(-1mm)

```
\begin{tikzpicture}
  \pingu[microphone left, microphone left yshift=2mm]
\end{tikzpicture}
```



/pingu/ <i>microphone left handle</i> = <i><color></i>	(pingu@black!80!pingu@silver)
<pre>\begin{tikzpicture} \pingu[microphone left, microphone left handle=green] \end{tikzpicture}</pre>	

/pingu/ <i>microphone right</i> = <i><color></i>	(pingu@silver!90!white)
<pre>\begin{tikzpicture} \pingu[microphone right=green] \end{tikzpicture}</pre>	

The following keys are only in effect if /pingu/*microphone right* is active.

/pingu/ <i>microphone right handle length</i> = <i><length></i>	(5.5mm)
<pre>\begin{tikzpicture} \pingu[microphone right, microphone right handle length=2mm] \end{tikzpicture}</pre>	

/pingu/ <i>microphone right handle shift</i> = <i><length></i>	(-1.15mm)
<pre>\begin{tikzpicture} \pingu[microphone right, microphone right handle shift=2mm] \end{tikzpicture}</pre>	

/pingu/ <i>microphone right xshift</i> = <i><length></i>	(-0.5*\pingu@microphone@right@width)
<pre>\begin{tikzpicture} \pingu[microphone right, microphone right xshift=2mm] \end{tikzpicture}</pre>	

/pingu/ <i>microphone right yshift</i> = <i><length></i>	(-1mm)
<pre>\begin{tikzpicture} \pingu[microphone right, microphone right yshift=2mm] \end{tikzpicture}</pre>	

/pingu/ <i>microphone right handle</i> = <i><color></i>	(pingu@black!80!pingu@silver)
<pre>\begin{tikzpicture} \pingu[microphone right, microphone right handle=green] \end{tikzpicture}</pre>	

A.4.17 The magnifier

/pingu/ <i>magnifier left</i> = <i><color></i>	(pingu@bronze)
<pre>\begin{tikzpicture} \pingu[magnifier left=green] \end{tikzpicture}</pre>	

The following keys are only in effect if `/pingu/magnifier left` is active.

<code>/pingu/magnifier left handle length = <length></code>	(3.5mm)
<pre>\begin{tikzpicture} \pingu[magnifier left, magnifier left handle length=4mm] \end{tikzpicture}</pre>	
<code>/pingu/magnifier left handle shift = <length></code>	(-4mm)
<pre>\begin{tikzpicture} \pingu[magnifier left, magnifier left handle shift=2mm] \end{tikzpicture}</pre>	
<code>/pingu/magnifier left size = <length></code>	(2.5mm)
<pre>\begin{tikzpicture} \pingu[magnifier left, magnifier left size=2mm] \end{tikzpicture}</pre>	
<code>/pingu/magnifier left glass = <color></code>	(pingu@blue!90!pingu@yellow)
<pre>\begin{tikzpicture} \pingu[magnifier left, magnifier left glass=pingu@red] \end{tikzpicture}</pre>	
<code>/pingu/magnifier left opacity = <opacity></code>	(0.24)
<pre>\begin{tikzpicture} \pingu[magnifier left, magnifier left opacity=.15] \end{tikzpicture}</pre>	
<code>/pingu/magnifier left thick = <length></code>	(.5mm)
<pre>\begin{tikzpicture} \pingu[magnifier left, magnifier left thick=1mm] \end{tikzpicture}</pre>	
<code>/pingu/magnifier left xshift = <length></code>	(0mm)
<pre>\begin{tikzpicture} \pingu[magnifier left, magnifier left xshift=1mm] \end{tikzpicture}</pre>	
<code>/pingu/magnifier left yshift = <length></code>	(1mm)
<pre>\begin{tikzpicture} \pingu[magnifier left, magnifier left yshift=1mm] \end{tikzpicture}</pre>	

/pingu/*magnifier right* = *<color>*

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[magnifier right=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*magnifier right* is active.

/pingu/*magnifier right handle length* = *<length>*

(3.5mm)

```
\begin{tikzpicture}
  \pingu[magnifier right, magnifier right handle length=4mm]
\end{tikzpicture}
```



/pingu/*magnifier right handle shift* = *<length>*

(-4mm)

```
\begin{tikzpicture}
  \pingu[magnifier right, magnifier right handle shift=2mm]
\end{tikzpicture}
```



/pingu/*magnifier right size* = *<length>*

(2.5mm)

```
\begin{tikzpicture}
  \pingu[magnifier right, magnifier right size=2mm]
\end{tikzpicture}
```



/pingu/*magnifier right glass* = *<color>*

(pingu@blue!90!pingu@yellow)

```
\begin{tikzpicture}
  \pingu[magnifier right, magnifier right glass=pingu@red]
\end{tikzpicture}
```



/pingu/*magnifier right opacity* = *<opacity>*

(0.24)

```
\begin{tikzpicture}
  \pingu[magnifier right, magnifier right opacity=.15]
\end{tikzpicture}
```



/pingu/*magnifier right thick* = *<length>*

(.5mm)

```
\begin{tikzpicture}
  \pingu[magnifier right, magnifier right thick=1mm]
\end{tikzpicture}
```



/pingu/*magnifier right xshift* = *<length>*

(0mm)

```
\begin{tikzpicture}
  \pingu[magnifier right, magnifier right xshift=1mm]
\end{tikzpicture}
```



/pingu/*magnifier right yshift* = *<length>*

(1mm)

```
\begin{tikzpicture}
  \pingu[magnifier right, magnifier right yshift=1mm]
\end{tikzpicture}
```



A.4.18 The sword

/pingu/*sword left* = *<color>*

(pingu@silver!94!blue)

```
\begin{tikzpicture}
  \pingu[sword left=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*sword left* is active.

/pingu/*sword left handle length* = *<length>*

(3.65mm)

```
\begin{tikzpicture}
  \pingu[sword left, sword left handle length=7mm]
\end{tikzpicture}
```



/pingu/*sword left handle shift* = *<length>*

(-2mm)

```
\begin{tikzpicture}
  \pingu[sword left, sword left handle shift=2mm]
\end{tikzpicture}
```



/pingu/*sword left xshift* = *<length>*

(-0.5*<sword left width>)

```
\begin{tikzpicture}
  \pingu[sword left, sword left xshift=2mm]
\end{tikzpicture}
```



/pingu/*sword left yshift* = *<length>*

(-1mm)

```
\begin{tikzpicture}
  \pingu[sword left, sword left yshift=2mm]
\end{tikzpicture}
```



/pingu/*sword left handle* = *<color>*

(pingu@bronze!80!green!40!pingu@black)

```
\begin{tikzpicture}
  \pingu[sword left, sword left handle=green]
\end{tikzpicture}
```



/pingu/*sword left guard* = *<color>*

(<sword left handle>!80!brown!50!black)

```
\begin{tikzpicture}
  \pingu[sword left, sword left guard=green]
\end{tikzpicture}
```



/pingu/sword left shade = <color>

(yellow!10!white)

```
\begin{tikzpicture}
  \pingu[sword left, sword left shade=green]
\end{tikzpicture}
```



/pingu/sword left shade opacity = <factor>

(.3)

```
\begin{tikzpicture}
  \pingu[sword left, sword left shade opacity=.5]
\end{tikzpicture}
```



/pingu/sword left length = <length>

(14mm)

```
\begin{tikzpicture}
  \pingu[sword left, sword left yshift=4mm]
\end{tikzpicture}
```



/pingu/sword left width = <length>

(.5mm)

```
\begin{tikzpicture}
  \pingu[sword left, sword left width=.25mm]
\end{tikzpicture}
```



/pingu/sword right = <color>

(pingu@silver!94!blue)

```
\begin{tikzpicture}
  \pingu[sword right=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/sword right is active.

/pingu/sword right handle length = <length>

(3.65mm)

```
\begin{tikzpicture}
  \pingu[sword right, sword right handle length=7mm]
\end{tikzpicture}
```



/pingu/sword right handle shift = <length>

(-2mm)

```
\begin{tikzpicture}
  \pingu[sword right, sword right handle shift=2mm]
\end{tikzpicture}
```



/pingu/sword right xshift = <length>

(-0.5*<sword right width>)

```
\begin{tikzpicture}
  \pingu[sword right, sword right xshift=2mm]
\end{tikzpicture}
```



/pingu/sword right yshift = <length>

(-1mm)

```
\begin{tikzpicture}
  \pingu[sword right, sword right yshift=2mm]
\end{tikzpicture}
```



/pingu/sword right handle = <color>

(pingu@bronze!80!green!40!pingu@black)

```
\begin{tikzpicture}
  \pingu[sword right, sword right handle=green]
\end{tikzpicture}
```



/pingu/sword right guard = <color>

(<sword right handle>!80!brown!50!black)

```
\begin{tikzpicture}
  \pingu[sword right, sword right guard=green]
\end{tikzpicture}
```



/pingu/sword right shade = <color>

(yellow!10!white)

```
\begin{tikzpicture}
  \pingu[sword right, sword right shade=green]
\end{tikzpicture}
```



/pingu/sword right shade opacity = <factor>

(.3)

```
\begin{tikzpicture}
  \pingu[sword right, sword right shade opacity=.5]
\end{tikzpicture}
```



/pingu/sword right length = <length>

(14mm)

```
\begin{tikzpicture}
  \pingu[sword right, sword right yshift=4mm]
\end{tikzpicture}
```



/pingu/sword right width = <length>

(.5mm)

```
\begin{tikzpicture}
  \pingu[sword right, sword right width=.25mm]
\end{tikzpicture}
```



A.4.19 The plank

/pingu/plank left = <color>

(pingu@bronze!80!pingu@black)

```
\begin{tikzpicture}
  \pingu[plank left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/plank left` is active.

<code>/pingu/plank left width = <length></code>	(7mm)
<pre>\begin{tikzpicture} \pingu[plank left, plank left width=2mm] \end{tikzpicture}</pre>	
<code>/pingu/plank left height = <length></code>	(16.5mm)
<pre>\begin{tikzpicture} \pingu[plank left, plank left height=2mm] \end{tikzpicture}</pre>	
<code>/pingu/plank left xshift = <length></code>	(0mm)
<pre>\begin{tikzpicture} \pingu[plank left, plank left xshift=2mm] \end{tikzpicture}</pre>	
<code>/pingu/plank left yshift = <length></code>	(-1mm)
<pre>\begin{tikzpicture} \pingu[plank left, plank left yshift=2mm] \end{tikzpicture}</pre>	
<code>/pingu/plank right = <color></code>	(pingu@bronze!80!pingu@black)
<pre>\begin{tikzpicture} \pingu[plank right=green] \end{tikzpicture}</pre>	

The following keys are only in effect if `/pingu/plank right` is active.

<code>/pingu/plank right width = <length></code>	(7mm)
<pre>\begin{tikzpicture} \pingu[plank right, plank right width=2mm] \end{tikzpicture}</pre>	
<code>/pingu/plank right height = <length></code>	(16.5mm)
<pre>\begin{tikzpicture} \pingu[plank right, plank right height=2mm] \end{tikzpicture}</pre>	
<code>/pingu/plank right xshift = <length></code>	(0mm)
<pre>\begin{tikzpicture} \pingu[plank right, plank right xshift=2mm] \end{tikzpicture}</pre>	

```
/pingu/plank right yshift = <length>
```

```
(-1mm)
```

```
\begin{tikzpicture}
  \pingu[plank right, plank right yshift=2mm]
\end{tikzpicture}
```



A.4.20 The laptop

```
/pingu/laptop left = <color>
```

```
(gray!80!pingu@white)
```

```
\begin{tikzpicture}
  \pingu[laptop left]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/laptop left` is active.

```
/pingu/laptop left bracket = <color>
```

```
(pingu@black!80!<laptop left>)
```

```
\begin{tikzpicture}
  \pingu[laptop left, laptop left bracket=green]
\end{tikzpicture}
```



```
/pingu/laptop left lower = <color>
```

```
(<laptop left>!95!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[laptop left, laptop left lower=green]
\end{tikzpicture}
```



```
/pingu/laptop left key = <color>
```

```
(<laptop left>!92!pingu@white)
```

```
\begin{tikzpicture}
  \pingu[laptop left, laptop left key=green]
\end{tikzpicture}
```



```
/pingu/laptop left display = <color>
```

```
(<laptop left>!32!pingu@white)
```

```
\begin{tikzpicture}
  \pingu[laptop left, laptop left display=green]
\end{tikzpicture}
```



```
/pingu/laptop left content = <tikz-code>
```

```
\begin{tikzpicture}
  \pingu[laptop left, laptop left content={\draw circle[radius=2mm]
;}]
\end{tikzpicture}
```



```
/pingu/laptop left mid = <code>
```

```
\begin{tikzpicture}
  \pingu[laptop left, laptop left mid={Hey}]
\end{tikzpicture}
```



/pingu/laptop right = <color>

(gray!80!pingu@white)

```
\begin{tikzpicture}
  \pingu[laptop right]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/laptop right is active.

/pingu/laptop right bracket = <color>

(pingu@black!80!<laptop right>)

```
\begin{tikzpicture}
  \pingu[laptop right, laptop right bracket=green]
\end{tikzpicture}
```



/pingu/laptop right lower = <color>

(<laptop right>!95!pingu@black)

```
\begin{tikzpicture}
  \pingu[laptop right, laptop right lower=green]
\end{tikzpicture}
```



/pingu/laptop right key = <color>

(<laptop right>!92!pingu@white)

```
\begin{tikzpicture}
  \pingu[laptop right, laptop right key=green]
\end{tikzpicture}
```



/pingu/laptop right display = <color>

(<laptop right>!32!pingu@white)

```
\begin{tikzpicture}
  \pingu[laptop right, laptop right display=green]
\end{tikzpicture}
```



/pingu/laptop right content = <tikz-code>

```
\begin{tikzpicture}
  \pingu[laptop right, laptop right content={\draw circle[radius=2
mm];}]
\end{tikzpicture}
```



/pingu/laptop right mid = <code>

```
\begin{tikzpicture}
  \pingu[laptop right, laptop right mid={Hey}]
\end{tikzpicture}
```



A.4.21 The devil fork

/pingu/devil fork left = <color>

(pingu@red)

```
\begin{tikzpicture}
  \pingu[devil fork left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/devil fork left` is active.

`/pingu/devil fork left second = <color>`

(pingu@bronze!10!black)

Staff color of the `/pingu/devil fork left`:

```
\begin{tikzpicture}
  \pingu[devil fork left,
    devil fork left second=green]
\end{tikzpicture}
```



`/pingu/devil fork left length = <color>`

(20mm)

```
\begin{tikzpicture}
  \pingu[devil fork left, devil fork left length=7mm]
\end{tikzpicture}
```



`/pingu/devil fork right = <color>`

(pingu@red)

```
\begin{tikzpicture}
  \pingu[devil fork right=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/devil fork right` is active.

`/pingu/devil fork right second = <color>`

(pingu@bronze!10!black)

Staff color of the `/pingu/devil fork right`:

```
\begin{tikzpicture}
  \pingu[devil fork right,
    devil fork right second=green]
\end{tikzpicture}
```



`/pingu/devil fork right length = <color>`

(20mm)

```
\begin{tikzpicture}
  \pingu[devil fork right, devil fork right length=7mm]
\end{tikzpicture}
```



A.4.22 The horse

`/pingu/horse left = <color>`

(pingu@bronze!80!pingu@black)

Give it a horse:

```
\begin{tikzpicture}
  \pingu[horse left=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/horse left` is active.

```
/pingu/horse left flip = <true/false> (\ifxleft\pingu@str@wing@left true\else false\fi)
```

By default, the left horse will be flipped. The right horse won't.

```
\begin{tikzpicture}
  \pingu[horse left,horse left flip=false]
\end{tikzpicture}
```



```
/pingu/horse left donkey = <true/false> (false)
```

Draw the horse with modifications to look more like a donkey:

```
\begin{tikzpicture}
  \pingu[horse left,horse left donkey=true]
\end{tikzpicture}
```



```
/pingu/horse left has base = <true/false> (false)
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left has base]
\end{tikzpicture}
```



```
/pingu/horse left draw = <color> (<horse left>!80!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left draw=green]
\end{tikzpicture}
```



```
/pingu/horse left mane = <color> (<horse left>!86!pingu@white)
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left mane=green]
\end{tikzpicture}
```



```
/pingu/horse left mane draw = <color> (<horse left mane>!80!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left mane draw=green]
\end{tikzpicture}
```



```
/pingu/horse left thatch = <color> (<horse left mane>)
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left thatch=green]
\end{tikzpicture}
```



```
/pingu/horse left thatch draw = <color> (<horse left thatch>!80!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left thatch draw=green]
\end{tikzpicture}
```



/pingu/horse left tail = <color>

(<horse left mane>)

```
\begin{tikzpicture}
  \pingu[horse left, horse left tail=green]
\end{tikzpicture}
```



/pingu/horse left tail draw = <color>

(<horse left tail>!80!pingu@black)

```
\begin{tikzpicture}
  \pingu[horse left, horse left tail draw=green]
\end{tikzpicture}
```



/pingu/horse left eyes = <color>

(<horse left draw>)

```
\begin{tikzpicture}
  \pingu[horse left, horse left eyes=green]
\end{tikzpicture}
```



also written as `horse left eye`

/pingu/horse left mouth = <color>

(<horse left draw>)

```
\begin{tikzpicture}
  \pingu[horse left, horse left mouth=green]
\end{tikzpicture}
```



/pingu/horse left nose = <color>

(<horse left draw>)

```
\begin{tikzpicture}
  \pingu[horse left, horse left nose=green]
\end{tikzpicture}
```



/pingu/horse left ears = <color>

(<horse left>)

```
\begin{tikzpicture}
  \pingu[horse left, horse left ears=green]
\end{tikzpicture}
```



/pingu/horse left base = <color>

(lightgray!90!black)

```
\begin{tikzpicture}
  \pingu[horse left, horse left has base, horse left base=green]
\end{tikzpicture}
```



/pingu/horse left base draw = <color>

(<horse left base>!91!pingu@black)

```
\begin{tikzpicture}
  \pingu[horse left, horse left has base, horse left base draw=
    green]
\end{tikzpicture}
```



```
/pingu/horse left base shade = <color>
```

```
(lightgray!80!black)
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left has base, horse left base shade=
    green]
\end{tikzpicture}
```



```
/pingu/horse left base shade draw = <color>
```

```
(<horse left base shade>!91!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left has base,
    horse left base shade draw=green]
\end{tikzpicture}
```



```
/pingu/horse left front left hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left front left hoof=green]
\end{tikzpicture}
```



```
/pingu/horse left front right hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left front right hoof=green]
\end{tikzpicture}
```



```
/pingu/horse left back left hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left back left hoof=green]
\end{tikzpicture}
```



```
/pingu/horse left back right hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left back right hoof=green]
\end{tikzpicture}
```



```
/pingu/horse left front hoofs = <color>
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left front hoofs=green]
\end{tikzpicture}
```



```
/pingu/horse left back hoofs = <color>
```

```
\begin{tikzpicture}
  \pingu[horse left, horse left back hoofs=green]
\end{tikzpicture}
```



/pingu/*horse left hoofs* = *<color>*

(\@pingu@none)

```
\begin{tikzpicture}
  \pingu[horse left, horse left hoofs=green]
\end{tikzpicture}
```



/pingu/*horse left xshift* = *<length>*

(Opt)

This key reacts with the /pingu/*horse left flip* option!

```
\begin{tikzpicture}
  \pingu[horse left,horse left xshift=1cm]
\end{tikzpicture}
```



/pingu/*horse left yshift* = *<length>*

(Opt)

```
\begin{tikzpicture}
  \pingu[horse left, horse left yshift=1cm]
\end{tikzpicture}
```



/pingu/*horse left on base*

Uses /pingu/*horse left xshift* and /pingu/*horse left yshift* to align a horse on a base to be set on the penguin-wing:

```
\begin{tikzpicture}
  \pingu[horse left,horse left has base,
        horse left on base]
\end{tikzpicture}
```



/pingu/*horse right* = *<color>*

(pingu@bronze!80!pingu@black)

Give it a horse:

```
\begin{tikzpicture}
  \pingu[horse right=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/*horse right* is active.

/pingu/*horse right flip* = *<true/false>*

(\ifxright\pingu@str@wing@left true\else false\fi)

By default, the right horse will be flipped. The right horse won't.

```
\begin{tikzpicture}
  \pingu[horse right,horse right flip=false]
\end{tikzpicture}
```



```
/pingu/horse right donkey = <true/false>
```

(false)

Draw the horse with modifications to look more like a donkey:

```
\begin{tikzpicture}
  \pingu[horse right, horse right donkey=true]
\end{tikzpicture}
```



```
/pingu/horse right has base = <true/false>
```

(false)

```
\begin{tikzpicture}
  \pingu[horse right, horse right has base]
\end{tikzpicture}
```



```
/pingu/horse right draw = <color>
```

(<horse right>!80!pingu@black)

```
\begin{tikzpicture}
  \pingu[horse right, horse right draw=green]
\end{tikzpicture}
```



```
/pingu/horse right mane = <color>
```

(<horse right>!86!pingu@white)

```
\begin{tikzpicture}
  \pingu[horse right, horse right mane=green]
\end{tikzpicture}
```



```
/pingu/horse right mane draw = <color>
```

(<horse right mane>!80!pingu@black)

```
\begin{tikzpicture}
  \pingu[horse right, horse right mane draw=green]
\end{tikzpicture}
```



```
/pingu/horse right thatch = <color>
```

(<horse right mane>)

```
\begin{tikzpicture}
  \pingu[horse right, horse right thatch=green]
\end{tikzpicture}
```



```
/pingu/horse right thatch draw = <color>
```

(<horse right thatch>!80!pingu@black)

```
\begin{tikzpicture}
  \pingu[horse right, horse right thatch draw=green]
\end{tikzpicture}
```



```
/pingu/horse right tail = <color>
```

(<horse right mane>)

```
\begin{tikzpicture}
  \pingu[horse right, horse right tail=green]
\end{tikzpicture}
```



/pingu/horse right tail draw = <color>

(<horse right tail>!80!pingu@black)

```
\begin{tikzpicture}
  \pingu[horse right, horse right tail draw=green]
\end{tikzpicture}
```



/pingu/horse right eyes = <color>

(<horse right draw>)

```
\begin{tikzpicture}
  \pingu[horse right, horse right eyes=green]
\end{tikzpicture}
```



also written as *horse right eye*

/pingu/horse right mouth = <color>

(<horse right draw>)

```
\begin{tikzpicture}
  \pingu[horse right, horse right mouth=green]
\end{tikzpicture}
```



/pingu/horse right nose = <color>

(<horse right draw>)

```
\begin{tikzpicture}
  \pingu[horse right, horse right nose=green]
\end{tikzpicture}
```



/pingu/horse right ears = <color>

(<horse right>)

```
\begin{tikzpicture}
  \pingu[horse right, horse right ears=green]
\end{tikzpicture}
```



/pingu/horse right base = <color>

(lightgray!90!black)

```
\begin{tikzpicture}
  \pingu[horse right, horse right has base, horse right base=green]
\end{tikzpicture}
```



/pingu/horse right base draw = <color>

(<horse right base>!91!pingu@black)

```
\begin{tikzpicture}
  \pingu[horse right, horse right has base, horse right base draw=
    green]
\end{tikzpicture}
```



/pingu/horse right base shade = <color>

(lightgray!80!black)

```
\begin{tikzpicture}
  \pingu[horse right, horse right has base, horse right base shade=
    green]
\end{tikzpicture}
```



```
/pingu/horse right base shade draw = <color>
```

```
(<horse right base shade>!91!pingu@black)
```

```
\begin{tikzpicture}
  \pingu[horse right, horse right has base,
    horse right base shade draw=green]
\end{tikzpicture}
```



```
/pingu/horse right front left hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[horse right, horse right front left hoof=green]
\end{tikzpicture}
```



```
/pingu/horse right front right hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[horse right, horse right front right hoof=green]
\end{tikzpicture}
```



```
/pingu/horse right back left hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[horse right, horse right back left hoof=green]
\end{tikzpicture}
```



```
/pingu/horse right back right hoof = <color>
```

```
\begin{tikzpicture}
  \pingu[horse right, horse right back right hoof=green]
\end{tikzpicture}
```



```
/pingu/horse right front hoofs = <color>
```

```
\begin{tikzpicture}
  \pingu[horse right, horse right front hoofs=green]
\end{tikzpicture}
```



```
/pingu/horse right back hoofs = <color>
```

```
\begin{tikzpicture}
  \pingu[horse right, horse right back hoofs=green]
\end{tikzpicture}
```



```
/pingu/horse right hoofs = <color>
```

```
(\@pingu@none)
```

```
\begin{tikzpicture}
  \pingu[horse right, horse right hoofs=green]
\end{tikzpicture}
```



```
/pingu/horse right xshift = <color>
```

(Opt)

This key reacts with the `/pingu/horse right flip` option!

```
\begin{tikzpicture}
  \pingu[horse right,horse right xshift=1cm]
\end{tikzpicture}
```



```
/pingu/horse right yshift = <color>
```

(Opt)

```
\begin{tikzpicture}
  \pingu[horse right, horse right yshift=1cm]
\end{tikzpicture}
```



```
/pingu/horse right on base
```

Uses `/pingu/horse right xshift` and `/pingu/horse right yshift` to align a horse on a base to be set on the penguin-wing:

```
\begin{tikzpicture}
  \pingu[horse right,horse right has base,
        horse right on base]
\end{tikzpicture}
```



A.5 Clothes

Clothes are currently completely work in progress as the goal is to create an elegant way to offer clothes that adapt to the wing positions of the penguin. Currently there is only one cloth type that may be heavily edited in the course of development...

A.5.1 The cloak

A cloak consists of the back, which is drawn behind the penguin, of the collar around the head, of the lining at the front and of the sleeves which cover the wings. Except for `/pingu/cloak`, all colors are derived from the main color. The keys of the first version (`cloak cap`, `cloak front color`, `cloak bottom color`, `cloak wings color`, and `cloak padding`) still work, they are aliases.

```
/pingu/cloak = <color>
```

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[cloak=green]
\end{tikzpicture}
```



Cloak-Showcase

```

\begin{tikzpicture}
  \pingu[wings wave, cloak]
  \pingu[wings shock, cloak, xshift=3.5cm]
  \pingu[right wing wave, left wing hug, cloak, xshift=7cm]
  \pingu[right wing grab, left wing raise, cloak, xshift=10.5cm]
\end{tikzpicture}

```



The following keys are only in effect if /pingu/cloak is active.

/pingu/cloak collar = <color>

(<cloak>!98!pingu@black)

```

\begin{tikzpicture}
  \pingu[cloak, cloak collar=green]
\end{tikzpicture}

```



/pingu/cloak lining = <color>

(<cloak>!90!white)

The color at the front, the part of the cloak you see when the wings are not covering it:

```

\begin{tikzpicture}
  \pingu[cloak, cloak lining=green]
\end{tikzpicture}

```



/pingu/cloak sleeves = <color>

(<cloak lining>!94!pingu@black)

```

\begin{tikzpicture}
  \pingu[cloak, cloak sleeves=green]
\end{tikzpicture}

```



/pingu/cloak hem = <color>

(<cloak>!90!black)

The back of the cloak fades from the cloak color to this color at its bottom:

```

\begin{tikzpicture}
  \pingu[cloak, cloak hem=green]
\end{tikzpicture}

```



/pingu/cloak spread = <length>

(1.95mm)

```

\begin{tikzpicture}
  \pingu[cloak, cloak spread=13mm]
\end{tikzpicture}

```



```
/pingu/cloak wings = <true/false>
```

(true)

Without the sleeves, the wings stay uncovered. Giving `/pingu/cloak sleeves` the color `\@pingu@none` has the same effect for the color, but this key also removes the shading of the sleeves:

```
\begin{tikzpicture}
  \pingu[cloak, cloak wings=false]
\end{tikzpicture}
```



```
/pingu/cape = <color>
```

(pingu@bronze)

Uses `/pingu/cloak` but disables all parts that are not part of a cape: the collar, the lining and the sleeves.

```
\begin{tikzpicture}
  \pingu[cape=green]
\end{tikzpicture}
```



A.5.2 The shirt

```
/pingu/shirt = <color>
```

(pingu@bronze)

```
\begin{tikzpicture}
  \pingu[shirt=green]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/shirt` is active.

```
/pingu/shirt raise = <length>
```

(2.25mm)

```
\begin{tikzpicture}
  \pingu[shirt, shirt raise=5mm]
\end{tikzpicture}
```



```
/pingu/shirt padding = <length>
```

(0mm)

```
\begin{tikzpicture}
  \pingu[shirt, shirt padding=4mm]
\end{tikzpicture}
```



```
/pingu/shirt buttons = <color>
```

(pingu@black)

Set the color of all buttons. The three buttons (top, middle and bottom) have a key of their own: `shirt button top`, `shirt button middle` and `shirt button bottom`, and the same is true for their shading with the suffix `shade`.

```
\begin{tikzpicture}
  \pingu[shirt, shirt buttons=green]
  \pingu[shirt, shirt button top=green, shirt button bottom=blue,
    xshift=3cm]
\end{tikzpicture}
```



```
/pingu/shirt buttons shade = <color>
```

Set the shading of all buttons:

```
\begin{tikzpicture}
  \pingu[shirt, shirt buttons shade=green]
\end{tikzpicture}
```



```
/pingu/shirt no buttons
```

Disable all buttons (by setting their colors to `!hide`):

```
\begin{tikzpicture}
  \pingu[shirt, shirt no buttons]
\end{tikzpicture}
```



```
/pingu/shirt above
```

(false)

This is interesting in combination with other extras as it allows the `/pingu/shirt` to be drawn above them.

```
\begin{tikzpicture}
  \pingu[shirt, tie, shirt above]
  \pingu[shirt, tie, xshift=3cm]
\end{tikzpicture}
```



A.5.3 The second shirt

```
/pingu/second shirt = <color>
```

(purple)

Display a shirt below the `/pingu/shirt`:

```
\begin{tikzpicture}
  \pingu[second shirt=green, shirt]
\end{tikzpicture}
```



The following keys are only in effect if `/pingu/second shirt` is active.

```
/pingu/second shirt raise = <length>
```

(3.35mm)

```
\begin{tikzpicture}
  \pingu[second shirt, second shirt raise=5mm]
\end{tikzpicture}
```



```
/pingu/second shirt neck = <color>
```

(<second shirt>!32!pingu@white)

```
\begin{tikzpicture}
  \pingu[second shirt, second shirt neck=green]
\end{tikzpicture}
```



A.5.4 The scarf

/pingu/scarf = <color>

(pingu@red!85!pingu@black)

A scarf around the neck. One end hangs down on the left side of the penguin, as seen by you:

```
\begin{tikzpicture}
  \pingu[scarf=green]
\end{tikzpicture}
```



The following keys are only in effect if /pingu/scarf is active.

/pingu/scarf stripes = <color>

(pingu@white!92!<scarf>)

```
\begin{tikzpicture}
  \pingu[scarf, scarf stripes=green]
\end{tikzpicture}
```



/pingu/scarf no stripes

```
\begin{tikzpicture}
  \pingu[scarf=blue, scarf no stripes]
\end{tikzpicture}
```



/pingu/scarf fringe = <color>

(<scarf>)

```
\begin{tikzpicture}
  \pingu[scarf, scarf fringe=green]
\end{tikzpicture}
```



/pingu/scarf end = <left/right/none>

(left)

Which end of the scarf hangs down, none removes it:

```
\begin{tikzpicture}
  \pingu[scarf, scarf end=right]
  \pingu[scarf, scarf end=none, xshift=3cm]
\end{tikzpicture}
```



/pingu/scarf end length = <length>

(8mm)

```
\begin{tikzpicture}
  \pingu[scarf, scarf end length=12mm]
\end{tikzpicture}
```



/pingu/scarf raise = <length>

(0mm)

```
\begin{tikzpicture}
  \pingu[scarf, scarf raise=-1mm]
\end{tikzpicture}
```



```
/pingu/scarf thickness = <length>
```

(2.6mm)

```
\begin{tikzpicture}
  \pingu[scarf, scarf thickness=4mm]
\end{tikzpicture}
```



B Larger Examples

B.1 A chessboard

Let's assume, that we want to have a board of penguins playing chess. For starters, we need the individual pieces.

The pawns

```
\tikz{\pingu[body=pingu@black]}
\tikz{\pingu[:back,xshift=3cm]}
```



The rooks

```
\tikz{\pingu[body=pingu@black,rook]}
\tikz{\pingu[:back,rook,rook hatch=false]}
```



For the knights we make use of the `/pingu/horse behind` extra, which itself is based on the `/pingu/horse left`/`/pingu/horse right` wing item:

The knights

```
\tikz{\pingu[body=pingu@black,horse behind]}
\tikz{\pingu[:back,horse behind=pingu@black,on horse flip]}
```



The queens

```
\tikz{\pingu[body=pingu@black,princess crown=pingu@bronze]}
\tikz{\pingu[:back,princess crown=pingu@bronze]}
```



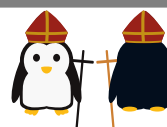
The kings

```
\tikz{\pingu[body=pingu@black,crown]}
\tikz{\pingu[:back,crown]}
```



The bishops

```
\tikz{\pingu[body=pingu@black,cross left=pingu@black,mitra]}
\tikz{\pingu[:back,cross left,mitra]}
```



C The Internals

The goal of this section is to explain the inner workings of the package to aid other people in helping me extending and mainting the penguins. Nothing in this section is copied: the code is read from `tikzpingus.sty` when this document is typeset. Every box names the lines of the file it shows, and the commands and keys inside link to their documentation. The other way round, implementation (lines 649–790) (and similar links) lead from the text to the code.

C.1 The Source at a Glance

The source is organized in *regions*, which an editor can fold (`# region`), and in *chunks*, which are marked with `<name>` and `<@name>` to show them here. The map below is created from these marks, so it is always up to date: regions are

upright, chunks slanted, and a page number means that the code is shown on that page.

whole setup.....	82–646	wings-shock.....	1181–1194
color definitions	84–98 p. 150	wings-hug	1195–1216
lengths and dimens.....	100–125	hairstyle.....	1219–1233
bends.....	127–212	feet.....	1235–1359
pingu@bend	128–146 p. 166	pingu@feet@s	1236–1253 p. 158
pingu@arc	147–211	feet normal	1260–1299 p. 158
selections.....	214–291	feet simple	1316–1330 p. 159
eyes selectors.....	255–264 p. 151	bill.....	1361–1419
bill selector	279–283 p. 153	pingu@bill@s	1363–1372 p. 154
feet selectors	284–287 p. 157	bill normal.....	1374–1382 p. 154
bodytype selector	288–290 p. 155	bill flat	1393–1400 p. 154
anchors	293–402	bill open	1402–1418
pingu@anchor	294–378 p. 163	dots	1422–1884
pingu@relative	380–401 p. 165	extras.....	1465–1676
key setup.....	404–564	extras below.....	1466–1577
pingu@color	405–409 p. 150	cup.....	1493–1517 p. 160
feet conditional.....	418–421 p. 158	eye patch.....	1519–1549
anchor keys	450–457 p. 165	eye patch left.....	1520–1534
feet keys	458–472 p. 157	eye patch right.....	1535–1548
bodytype keys.....	473–479 p. 157	glow	1551–1576
eyes keys	484–503 p. 151	extras above.....	1579–1675
bill keys.....	520–524 p. 153	halo.....	1618–1637 p. 161
subenvs.....	566–569 p. 149	eye frame.....	1639–1652
complete sets and styles.....	574–607 p. 148	sheriff star	1654–1673
mix-all.....	609–635	wingitems	1678–1767
glows.....	637–645	wingitems-setup.....	1679–1716
core.....	648–791	cane.....	1718–1735 p. 162
path commands.....	649–656 p. 146	staff.....	1737–1766
pingu setup.....	658–673 p. 146	clothes	1769–1789
pingu body	674–688 p. 146	bee-lib.....	1792–1883
pingu behind	689–706 p. 146	christmas-lib.....	1886–2035
pingu wings	708–733 p. 147	cloak-lib.....	2078–2160
pingu front.....	735–757 p. 147	devil-lib.....	2194–2283
lower-pingu.....	776–790	emotions-lib.....	2285–2339
main body parts	793–1420	flags-lib.....	2341–2384
body front.....	794–982	formal-lib.....	2386–2500
normal	795–831	fun-lib.....	2502–2897
none	832–892	glasses-lib.....	2899–3002
pingu@x@bodytype@none.....	833–890 p. 156	hats-lib.....	3004–3286
chubby.....	893–929	horse-lib.....	3288–3507
legacy.....	930–981	magic-lib.....	3509–3521
eyes	983–1072	medieval-lib.....	3523–3726
pingu@eyes@s.....	985–1002 p. 152	movement-lib.....	3728–3801
eyes normal.....	1009–1021 p. 152	safe-lib.....	3803–3830
eyes shiny.....	1029–1053 p. 152	science-fiction.....	3832–4020
wings.....	1074–1217	lightsaber base	3859–3910
wings-normal.....	1111–1127	shirts-lib.....	4022–4123
wings-wave.....	1128–1148 p. 160	signs-lib.....	4125–4176
wings-raise.....	1149–1166	sport-lib.....	4178–4284
wings-grab.....	1167–1180	technology-lib.....	4286–4410

C.2 How a Penguin is Drawn

`\pingu` is the only command that draws, and it does so part by part, from the back to the front. Each part is drawn within a scope whose style `/pingu/@<part>` is empty by default, so that the sets of Section C (for example `/pingu/:mix` or `/pingu/:ghost`) can change a single part. The parts of a penguin which are behind or in front of the rest (very-background, background, main, middle, foreground, in this order) are drawn in six layers which are put one above the other when the penguin is finished, so a penguin which is drawn later is above a penguin which is drawn earlier, wings included. The package option `manual-layers` lets all penguins of a picture share the layers instead.

The helpers. Every shape is drawn with one of three helpers: `\pingu@path` (a path), `\pingu@block` (a filled shape) and `\pingu@draw` (a line). They take a color, and draw nothing if it is `\@pingu@none`. This is the mechanism that lets you hide

any part by giving it that color.

tikzpingus.sty, lines 650–655

```

650 \long\def\pingu@path{\@dblarg\pingu@path}
651 \def\pingu@path[#1]#2#3;{\protected@edef\@tmp{#2}\ifx\@tmp\@pingu@none\else\path[#1,#2] #3;\fi}
652 \long\def\pingu@block{\@dblarg\pingu@block}
653 \def\pingu@block@[#1]#2#3;{\protected@edef\@tmp{#2}\ifx\@tmp\@pingu@none\else\path[#1,/pingu/@block=#2] #3;\fi}
654 \long\def\pingu@draw{\@dblarg\pingu@draw}
655 \def\pingu@draw@[#1]#2#3;{\protected@edef\@tmp{#2}\ifx\@tmp\@pingu@none\else\path[#1,/pingu/@draw=#2] #3;\fi}

```

The setup. The group is opened and `/pingu/defaults` followed by the given keys are executed. This is why a key never depends on a former penguin. Then the silhouette (`\pingu@bbox@update`) and the shift of `/pingu/anchor` are calculated, and the outer anchors are placed (`\pingu@place@anchors`).

tikzpingus.sty, lines 659–672

```

659 \newcommand*\pingu[1][]{\%
660 \ifpengu@setup@manual@layers\pgfsetlayers{background,main,middle,foreground}\fi\beginpgroup
661 \pengu@do@back@false
662 \pgfqkeys{/pingu}{defaults,#1}%
663 \pingu@mixall@hook
664 \let\pingu@path\lowerpingu\pingu@default@path\lowerpingu\relax
665 \let\pingu@path\upperpingu\pingu@default@path\upperpingu\relax
666 \ifdim\pingu@side@h@half<10.33mm \PackageWarning{\tikzpingus@filename}{Penguins with a height below 10.33mm may
    have unwanted artifacts.}\fi
667 \pingu@bbox@update
668 \pingu@anchor@shift\pingu@anchor@dx\pingu@anchor@x
669 \pingu@anchor@shift\pingu@anchor@dy\pingu@anchor@y
670 \scope[/pingu/@pingu@place,shift={(\pingu@anchor@dx,\pingu@anchor@dy)}]
671 \pingu@place@anchors
672 \ifpengu@setup@manual@layers\else\pingu@layers@open\fi

```

The body and the face. The body is drawn by the selected body type (`\@pingu@drawer@bodytype@`), and the belly is clipped to the white of the body. The eyes follow.

tikzpingus.sty, lines 675–687

```

675 \scope[/pingu/@pingu,/pingu/@pingu@main]
676 \scope[/pingu/@body]
677 \@pingu@drawer@bodytype@
678 \pingu@body@app
679 \endscope
680 \path(\pingu@name-head-center)++(0,-.05mm-\pingu@outer@off) coordinate (\pingu@name-bill);
681 \scope[shift=(\pingu@name-belly-center),/pingu/@belly]
682 \clip[use path=\pingu@whitepingu]; \pingu@belly@raw
683 \endscope
684 \coordinate(\pingu@name-head) at (\pingu@w@half,.7375cm);
685 \scope[/pingu/@eyes]
686 \@pingu@drawer@lefteye@ \@pingu@drawer@righteye@
687 \endscope

```

Everything behind the wings. The clothes behind the penguin (shirts and the back of the cloak), the extras below the head (implementation (lines 1467–1576)) and the hair come next. The feet are drawn on the background layer unless `/pingu/feet front` is set.

tikzpingus.sty, lines 690–705

```

690 \pingu@layer{background}[]\pingu@draw@clothing@behind\endpingu@layer
691 \pingu@draw@clothing@middle
692 \scope[/pingu/@extras,/pingu/@extras@below]
693 \pingu@draw@extras@below
694 \endscope
695
696 \scope[/pingu/@hair]
697 \@pingu@drawer@hairstyle@
698 \endscope
699
700 \if@pingu@feet@front\else\pingu@layer{background}[]\fi
701 \scope[/pingu/@feet]
702 \@pingu@drawer@leftfoot@ \@pingu@drawer@rightfoot@
703 \endscope
704 \pingu@xif@eb{flipflop}{flipflopstraps}%
705 \if@pingu@feet@front\else\endpingu@layer\fi

```

The wings. The wings are drawn after the body. Afterwards the coordinates for the items of the wings are calculated from the position of the wings, and the front of the clothes follows (implementation (lines 1770–1788)).

tikzpingus.sty, lines 709–732

```

709 \scope[/pingu/@wings]
710 \@pingu@drawer@leftwing@ \@pingu@drawer@rightwing@
711 \endscope
712
713 \path (\pingu@name-wing-left-start) -- (\pingu@name-wing-left) --
714 ([turn]\csname @pingu@wingdown@\@pingu@select@leftwing@ @\endcsname:2.45mm)
715 coordinate (\pingu@name-wing-left-low) -- ([turn]180:4.9mm)
716 coordinate (\pingu@name-wing-left-up);
717 \path (\pingu@name-wing-left-start) -- (\pingu@name-wing-left) --
718 (\pingu@name-wing-left-start) --
719 ([turn]180+\csname @pingu@wingdown@\@pingu@select@leftwing@ @\endcsname:3.45mm)
720 coordinate (\pingu@name-wing-left-start-low) -- ([turn]180:6.9mm)
721 coordinate (\pingu@name-wing-left-start-up);
722 \path (\pingu@name-wing-right-start) -- (\pingu@name-wing-right) --
723 ([turn]-\csname @pingu@wingdown@\@pingu@select@rightwing@ @\endcsname:2.45mm)
724 coordinate (\pingu@name-wing-right-low) -- ([turn]180:4.9mm)
725 coordinate (\pingu@name-wing-right-up);
726 % double path to fix turn
727 \path (\pingu@name-wing-right-start) -- (\pingu@name-wing-right) --
728 (\pingu@name-wing-right-start) --
729 ([turn]180-\csname @pingu@wingdown@\@pingu@select@rightwing@ @\endcsname:3.45mm)
730 coordinate (\pingu@name-wing-right-start-low) -- ([turn]180:6.9mm)
731 coordinate (\pingu@name-wing-right-start-up);
732 \pingu@draw@clothing@main

```

The front. The bill, the extras above the head (implementation (lines 1580–1674)) and the items of the wings are the last parts. At last the meta dots (`/pingu/meta-dots`) and the code of `/pingu@post` are drawn.

tikzpingus.sty, lines 736–756

```

736 \if@pingu@bill@above\pgfonlayer{\pingu@L@foreground}\fi
737 \scope[/pingu/@bill]
738 \@pingu@drawer@bill@
739 \endscope
740 \if@pingu@bill@above\endpgfonlayer\fi
741
742 \scope[/pingu/@extras,/pingu/@extras@above]
743 \pingu@draw@extras@above
744 \endscope
745 \scope[/pingu/@wingitems]
746 \pingu@draw@wing@items
747 \endscope
748 \endscope
749 \if@pingu@draw@metadots \@pingu@draw@dots \fi
750 \pingu@post
751 \ifpingu@setup@manual@layers@\else\pingu@layers@close\fi
752 \endscope
753 \endgroup
754 \ifpingu@setup@manual@layers@\else\pingu@layers@commit\fi
755 \pingu@back@init@call
756 }

```

C.3 Keys, Styles and Defaults

All keys of the package live in `/pingu`. Most of them only store a value, for example a color is stored with `\pingu@color` in a macro and read when it is drawn. The set `/pingu/defaults` puts everything back to the start, and `\pingudefaults` and `\pingudefaultsappend` add your own defaults to it. The composed keys, like `/pingu/:ghost` or `/pingu/:hide`, are styles that change the styles of the parts in `/pingu/@<part>`:

tikzpingus.sty, lines 575–606

```

575 \newif\if@pingu@do@back@
576 \pgfqkeys{/pingu}{
577   :line/.style           = {/pingu/@block/.append style={fill=none,draw=##1!80!black,shade=none,path fading=
    none}},
578   :fill/.style           = {/pingu/body=#1,/pingu/body front=#1,/pingu/feet color=#1, /pingu/bill color=#1,
    /pingu/eyes color=#1, /pingu/eyes second color=#1},
579   :fill/.default         = pingu@main,
580   :pingu/.style          = {/pingu/@body/.style={#1},/pingu/@wings/.style={#1},/pingu/@eyes/.style={#1},/pingu/
    @bill/.style={#1},/pingu/@hair/.style={#1},/pingu/@feet/.style={#1}},
581   :toast/.style          = {/pingu/left wing=none,/pingu/right wing=none,/pingu/feet=none,/pingu/hair=
    \@pingu@none},
582   :mix-draw/.style       = {:\pingu={/pingu/@block/.append style={/pingu/@pingu@all={###1!#1}, fill={###1!#1
    }}}},
583   :mix-draw/.default     = 50!white,
584   :mix-all/.code         = {\ifx\pingu@mixall\@empty\def\pingu@mixall{#1}\else\edef\pingu@mixall{\pingu@mixall
    !#1}\fi},
585   :mix-all/.default     = 50!white,
586   :mix/.style            = {
587     /pingu/@body/.code={\@pingu@color@mixwith\pingu@color@body@main{#1}\@pingu@color@mixwith
    \pingu@color@body@head{#1}},
588     /pingu/@wings/.code={\@pingu@color@mixwith\pingu@color@left@wing{#1}\@pingu@color@mixwith\pingu@color
    @right@wing{#1}},
589     /pingu/@eyes/.code={\@pingu@color@mixwith\pingu@color@eye@left{#1}\@pingu@color@mixwith
    \pingu@color@eye@right{#1}\@pingu@color@mixwith\pingu@color@eye@second@left{#1}\@pingu@color@mixwith

```

```

590 \pingu@color@eye@second@right{#1}},
591 /pingu/@bill/.code={\@pingu@color@mixwith\pingu@color@bill{#1}},
592 \pingu/@hair/.code={\@pingu@color@mixwith\pingu@color@hair@a{#1}\@pingu@color@mixwith\pingu@color@hair@b{#1}
\@pingu@color@mixwith\pingu@color@hair@c{#1}\@pingu@color@mixwith\pingu@color@hair@d{#1}\@pingu@color@
mixwith\pingu@color@hair@e{#1}},
593 /pingu/@feet/.code={\@pingu@color@mixwith\pingu@color@foot@left{#1}\@pingu@color@mixwith
\pingu@color@foot@right{#1}}
594 },
595 :mix/.default = 50!white,
596 :ghost parts/.style = {/pingu/@pingu@all/.append style={opacity={#1}}},
597 :ghost parts/.default = {.5},
598 :ghost/.style = {/pingu/@pingu/.append style={opacity=#1,transparency group}},
599 :ghost/.default = {.5},
600 :ghost glow/.style = {/pingu/@pingu/.append style={opacity=#1,transparency group},/pingu/@block/.append
style={/pingu/@glow}},
601 :ghost glow/.default = {.5},
602 :devil/.style = {/pingu/devil_horns=#1,/pingu/eyes devil=#1},
603 :devil/.default = pingu@purple,
604 :hide/.style = {/pingu/body_type=none,/pingu/hair=\@pingu@none,/pingu/feet=none,/pingu/wings=none,
/pingu/bill=none,/pingu/eyes=none},
605 :back/.code = {\pgfqkeys{/pingu}{eyes=none,bill=none,body front=\pingu@color@body@main,@pingu@main
/.append style={xshift=2\pingu@w@half,xscale=-1},feet back}\gdef\pingu@back@init@call{
\ifpingu@setup@manual@layers@\pgfsetlayers{foreground,middle,main,background}\fi\gdef\pingu@back@init@call
{}}\@pingu@do@back@true}
606 }
\let\pingu@back@init@call\empty

```

A part may need a group of its own, for example to place a part on another layer. This is what `\pingu@layer` is for:

`tikzpingus.sty`, lines 567–568

```

567 \def\pingu@layer#1[#2]{\pgfonlayer{\csname pingu@L@#1\endcsname}\scope[/pingu/@pingu,#2]}
568 \def\endpingu@layer{\endscope\endpgfonlayer}

```

C.4 General Commands

`\pingu@name`

Do **not** redefine this command, it is controlled by `/pingu/name`.

Holds the name of the penguin.

`\pingu@one`

Do not change this dimension as it represents one milli meter as the base unit.

C.4.1 Color Management

There is not a lot of magic in the color handling of this package.

`\pingu@color{name}{color}`

Creates a macro `\pingu@color@<name>` that links to the given color:

`tikzpingus.sty`, lines 406–408

```
406 \def\pingu@color#1#2{%
407     \expandafter\def\csname pingu@color@#1\endcsname{#2}%
408 }
```

I decided against the usage of `\colorlet`, to support `\@pingu@none` and dynamic redefinition of the colors.

`\@pingu@none`

Holds the special color to hide the drawing of specific components as described within [subsection 2.3](#) (holds the value: “!hide”). If you redefine this command, you redefine this hidden color. However, if done outside of the preamble, this may result in weird side effects due to expansions.

As shown in [subsection 2.3](#), this is the definition of the colors, with the helper-macro `\@pingu@color@`, which I created solely for me editor to highlight the colors accordingly:

`tikzpingus.sty`, lines 85–97

```
85 \def\@pingu@color@ #1#2(#3,#4,#5){\definecolor{pingu@#1}{#2}{#3,#4,#5}}
86 \@pingu@color@{main}RGB(3,14,29)
87 \@pingu@color@{black}RGB(23,19,19)
88 \@pingu@color@{silver}RGB(192,192,192)
89 \@pingu@color@{bronze}RGB(205,127,50)
90 \@pingu@color@{white}RGB(248,248,248)
91 \@pingu@color@{yellow}RGB(252,187,21)
92 \@pingu@color@{lightblue}RGB(174,229,243)
93 \@pingu@color@{blue}RGB(48,161,247)
94 \@pingu@color@{green}RGB(129,204,41)
95 \@pingu@color@{red}RGB(217, 40, 28)
96 \@pingu@color@{purple}RGB(244,48,93)
97 \def\@pingu@none{!hide}
```

C.4.2 Selections

A selection is a family of options that exclude each other, like the eyes or the feet: exactly one member is active, and the key that selects it is generated from the name of the member. The code which draws the active member is stored in one macro, and it is called when the part is drawn.

`\pingu@create@selection{name}{none default}`

Creates a new enumeration-like structure that can be used to register a family of options (see [subsection C.5](#) for an example). The *none default* is used for the default member of the selection: *none* which is active, whenever no other key is selected by `\pingu@<selection>@select`. You can add keys by using `\pingu@<selection>@add` and execute the selection by `\@pingu@drawer@<selection>@`.

`\pingu@<selection>@select{select}`

This command is only usable in combination with `\pingu@create@selection`.

`\pingu@<selection>@add{select}`

This command is only usable in combination with `\pingu@create@selection`.

`\@pingu@drawer@<selection>@`

This command is only usable in combination with `\pingu@create@selection`.

C.5 The Eyes

C.5.1 Important Commands

`\pingu@eye@shift`

This dimension holds a hardcoded value to be used as the offset for eyes. Its default value is 2.27621pt. It may be removed in the future (or replaced by an option).

C.5.2 The Code

The eyes of the penguin are controlled by two selections for the left, and the right eye. Because we always want the coordinates for the right and the left eye (as presented in [section 2.1](#)), the *none*-version of `\pingu@create@selection` creates those:

`tikzpingus.sty`, lines 256–263

```

256 \pingu@create@selection{lefteye}{%
257     \path ([yshift=.75mm,xshift=\pingu@eye@shift]\pingu@name-eye-back-left)
258         coordinate (\pingu@name-eye-left);%
259 }
260 \pingu@create@selection{righteye}{%
261     \path ([yshift=.75mm,xshift=-\pingu@eye@shift]\pingu@name-eye-back-right)
262         coordinate (\pingu@name-eye-right);%
263 }
```

With this code, we define the six following commands: `\pingu@lefteye@select`, `\pingu@lefteye@add`, `\@pingu@drawer@lefteye@`, `\pingu@righteye@select`, `\pingu@righteye@add`, and `\@pingu@drawer@righteye@`.

Within the initialization of all penguin options, we use these commands. Furthermore, `/pingu/eyes` acts as a comfort option which operates on both selectors:

`tikzpingus.sty`, lines 485–502

```

485 left eye/.code          = \pingu@lefteye@select{#1},
486 left eye color/.code    = \pingu@color{eye@left}{#1},
487 left eye second color/.code = \pingu@color{eye@second@left}{#1},
488 right eye/.code         = \pingu@righteye@select{#1},
489 right eye color/.code   = \pingu@color{eye@right}{#1},
490 right eye second color/.code = \pingu@color{eye@second@right}{#1},
491 eyes/.style            = {
492     /pingu/left eye = {#1},
493     /pingu/right eye = {#1}
494 },
495 eyes color/.style      = {
496     /pingu/left eye color = {#1},
497     /pingu/right eye color = {#1}
498 },
499 eyes second color/.style = {
500     /pingu/left eye second color = {#1},
501     /pingu/right eye second color = {#1}
502 },
```

Both eyes have two colors to work with: `\pingu@color@eye@left`, `\pingu@color@eye@second@left` for the left as well as `\pingu@color@eye@right` and `\pingu@color@eye@second@right` for the right eye.

C.5.3 Adding new Eyes

Now on how to add new eyes. While it is possible to use `\pingu@lefteye@add` and `\pingu@righteye@add`, setting up the corresponding “eye” and “eyes” comfort-methods is rather tedious. For this, i created the command `\pingu@eyes@s...`

```
\pingu@eyes@s{name}{left-eye}{right-eye} [default color=pingu@black]
```

Allows to easily add new eyes.

Important: Your code for the eyes *must* create the coordinates `\pingu@name-eye-left` and `\pingu@name-eye-right` respectively. This is not checked! It is recommended to use the coordinates `\pingu@name-eye-back-left` and `\pingu@name-eye-back-right` for reference.



The macro `\pingu@eyes@s` is defined with the help of `\pingu@eyes@s@` like this:

```
tikzpingus.sty, lines 986-1001
986 \long\def\pingu@eyes@s#1#2#3{%
987   \ifnextchar[%
988     {\pingu@eyes@s@{#1}{#2}{#3}}%
989     {\pingu@eyes@s@{#1}{#2}{#3}[pingu@black]}%
990 }
991 \long\def\pingu@eyes@s@#1#2#3[#4]{%
992   \pgfqkeys{/pingu}{%
993     left eye #1/.style = {/pingu/left eye={#1}, /pingu/left eye color={##1}},
994     left eye #1/.default = {#4},
995     right eye #1/.style = {/pingu/right eye={#1}, /pingu/right eye color={##1}},
996     right eye #1/.default = {#4},
997     eyes #1/.style = {/pingu/eyes={#1}, /pingu/eyes color={##1}},
998     eyes #1/.default = {#4}
999   }%
1000   \pingu@lefteye@add{#1}{#2}\pingu@righteye@add{#1}{#3}%
1001 }
```

As an example, this is the definition of the “normal eyes” (`/pingu/eyes normal`) in the code:

```
tikzpingus.sty, lines 1010-1020
1010 \pingu@eyes@s{normal}{%
1011   \pingu@block{\pingu@color@eye@left}
1012     ([yshift=.75mm,xshift=\pingu@eye@shift]\pingu@name-eye-back-left)
1013     coordinate (\pingu@name-eye-left)
1014     ellipse [x radius=.1225cm, y radius=.1365cm];
1015 }{%
1016   \pingu@block{\pingu@color@eye@right}
1017     ([yshift=.75mm,xshift=-\pingu@eye@shift]\pingu@name-eye-back-right)
1018     coordinate (\pingu@name-eye-right)
1019     ellipse [x radius=.1225cm, y radius=.1365cm];
1020 }
```

As another example, this is the definition of the “shiny eyes” (`/pingu/eyes shiny`):

tikzpingus.sty, lines 1030–1052

```

1030 \pingu@eyes@s{shiny}{%
1031   \pingu@block{\pingu@color@eye@left}
1032     ([yshift=.75mm,xshift=\pingu@eye@shift]\pingu@name-eye-back-left)
1033     coordinate(\pingu@name-eye-left)
1034     ellipse [x radius=.22cm, y radius=.26cm];
1035   \pingu@block{\pingu@color@eye@second@left}
1036     (\pingu@name-eye-left)++(\eyebaseang:.85mm and \pingu@one)
1037     ellipse [x radius=.8mm, y radius=\pingu@one];
1038   \pingu@block{\pingu@color@eye@second@left}
1039     (\pingu@name-eye-left)++(\eyebaseang+180:.12cm and .14cm)
1040     ellipse [x radius=.25mm, y radius=.35mm];
1041 }{%
1042   \pingu@block{\pingu@color@eye@right}
1043     ([yshift=.75mm,xshift=-\pingu@eye@shift]\pingu@name-eye-back-right)
1044     coordinate(\pingu@name-eye-right)
1045     ellipse [x radius=.22cm, y radius=.26cm];
1046   \pingu@block{\pingu@color@eye@second@right}
1047     (\pingu@name-eye-right)++(\eyebaseang:.85mm and \pingu@one)
1048     ellipse [x radius=.8mm, y radius=\pingu@one];
1049   \pingu@block{\pingu@color@eye@second@right}
1050     (\pingu@name-eye-right)++(\eyebaseang+180:.12cm and .14cm)
1051     ellipse [x radius=.25mm, y radius=.35mm];
1052 }

```

The usage of `\eyebaseang` is outdated and will probably be removed in the future.

C.6 The Bill

C.6.1 The Code

The bill of the penguin is controlled by one selection. As we always want the coordinates for theoretical bill position (as presented in [section 2.1](#)), the *none*-version of `\pingu@create@selection` creates it:

tikzpingus.sty, lines 280–282

```

280 \pingu@create@selection{bill}{%
281   \coordinate (\pingu@name-bill-bottom) at (\pingu@name-bill);%
282 }

```

This code, defines three commands: `\pingu@bill@select`, `\pingu@bill@add`, `\@pingu@drawer@bill@`.

Within the initialization of all penguin options, we use these commands:

tikzpingus.sty, lines 521–523

```

521 bill/.code = \pingu@bill@select{#1},
522 bill above/.is if = @pingu@bill@above,
523 bill color/.code = \pingu@color{bill}{#1},

```

With this, the bill has one color to work with: `\pingu@color@bill`.

C.6.2 Adding new Bills

Now on how to add new bills. While it is possible to use `\pingu@bill@add`, setting up the “bill” comfort-methods is rather tedious. For this, i created the command `\pingu@bill@s...`

`\pingu@bill@s{name}-{bill}`

Allows to easily add new bills.

Important: Your code for the bill *must* create the coordinate `\pingu@name-bill-bottom`. This is not checked! It is recommended to use the coordinate `\pingu@name-bill` for reference.

The macro `\pingu@bill@s` is defined like this:

```
tikzpingus.sty, lines 1364-1371
1364 \long\def\pingu@bill@s#1#2{%
1365     \pgfqkeys{/pingu}{%
1366         bill #1/.style = {/pingu/bill={#1}, /pingu/bill color={#1}},%
1367         bill #1/.default = pingu@yellow
1368     }%
1369     \pingu@bill@add{#1}{#2}%
1370 }
1371 \pgfqkeys{/pingu}{bill none/.style={/pingu/bill=none}}
```

As an example, this is the definition of the “normal bill” (`/pingu/bill normal`) in the code:

```
tikzpingus.sty, lines 1375-1381
1375 \pingu@bill@s{normal}{%
1376     \pingu@block[rounded corners=.1mm]{\pingu@color@bill}
1377     (\pingu@name-bill)++(-.19cm,0) arc (180:360:.19cm and .225cm)
1378     coordinate[pos=.5] (\pingu@name-bill-bottom) -- ++(-.2mm,.05mm)
1379     to[bend right=10] ([xshift=-.17cm,yshift=.05mm]\pingu@name-bill)
1380     -- cycle;
1381 }
```

As another example, this is the definition of the “flat bill” (`/pingu/bill flat`):

```
tikzpingus.sty, lines 1394-1399
1394 \pingu@bill@s{flat}{%
1395     \pingu@block{\pingu@color@bill} (\pingu@name-bill)++(-.19cm,0)
1396     arc (180:360:.19cm and .225cm)
1397     coordinate[pos=.5] (\pingu@name-bill-bottom)
1398     -- cycle;
1399 }
```

C.7 The Body Type

C.7.1 Important Commands

`\pingu@w@half`

This dimension holds a hardcoded value defining the half width of the penguin. Its default value is 26.67445pt.

`\pingu@side@h@half`

This dimension holds a hardcoded value defining the half height of the lower part of the penguin. Its default value is 36.27708pt. See `\pingu@head@h@half` for the additional head-height.

`\pingu@head@h@half`

This dimension holds a hardcoded value defining the half height of the upper (head) part of the penguin. Its default value is 24.00684pt.



`\pingu@bend`

This dimension holds a hardcoded value used as a basis for bend lines. Its default value is 3.55658pt. It may be removed in the future (or replaced by an option).

`\pingu@lower@off`

This dimension holds a hardcoded value used to offset the white inner parts of the penguin (for the bottom part). Its default value is 7.11317pt. It may be removed in the future (or replaced by an option).

`\pingu@outer@off`

This dimension holds a hardcoded value used to offset the lower part of the basic penguin. Its default value is 7.11317pt. It may be removed in the future (or replaced by an option).

`\pingu@foot@outer@w`

This dimension holds a hardcoded value defining the width of the outer feet. Its default value is 12.80373pt. It may be removed in the future (or replaced by an option).

`\pingu@foot@outer@h`

This dimension holds a hardcoded value defining the height of the outer feet. Its default value is 7.11317pt. It may be removed in the future (or replaced by an option).

`\pingu@foot@inner@w`

This dimension holds a hardcoded value defining the width of the inner feet. Its default value is 15.6491pt. It may be removed in the future (or replaced by an option).

`\pingu@foot@inner@h`

This dimension holds a hardcoded value defining the height of the inner feet. Its default value is 3.27222pt. It may be removed in the future (or replaced by an option).

`\pingu@headcon@x`

This dimension holds a hardcoded value defining the x-position offsetting the connection point of the penguin head. Its default value is 11.20335pt. It may be removed in the future (or replaced by an option).

`\pingu@headcon@y`

This dimension holds a hardcoded value defining the y-position offsetting the connection point of the penguin head. Its default value is 13.29729pt. It may be removed in the future (or replaced by an option).

C.7.2 The Code

The body of the penguin is controlled by one selection. There are a lot of coordinates to be created by this command (as presented in [section 2.1](#)). Furthermore, it must save some paths (`\pingu@lowerpingu`, `\pingu@upperpingu` and `\pingu@whitepingu`). Therefore, it is recommended, that you create a new one by first copying an existing definition. Due to its length, the *none*-version of `\pingu@create@selection` uses `\pingu@x@bodytype@none` as a helper:

`tikzpingus.sty`, lines 289–289

```
289 \pingu@create@selection{bodytype}{\pingu@x@bodytype@none}
```

This code, defines three commands: `\pingu@bodytype@select`, `\pingu@bodytype@add`, `\@pingu@drawer@bodytype@`.

The macro `\pingu@x@bodytype@none` is defined as follows:

tikzpingus.sty, lines 834-889

```

834 \def\pingu@x@bodytype@none{%
835     % set the lower half
836     \path[save path=\pingu@lowerpingu] (0,\pingu@outer@off)
837         % left side
838         \pingu@bend@body{-\pingu@side@h@half-\pingu@outer@off}
839         coordinate[pos=.775] (\pingu@name-waist-right)
840         % feety stuff
841         \pingu@arc@aa
842         coordinate (\pingu@name-foot-right)
843         \pingu@arc@ab
844         coordinate (\pingu@name-bottom-center)
845         \pingu@arc@ac
846         coordinate (\pingu@name-foot-left)
847         \pingu@arc@ad
848         % right side
849         \pingu@bend@body{\pingu@side@h@half+\pingu@outer@off}
850         coordinate[pos=.225] (\pingu@name-waist-left)
851         % switch to left
852         coordinate (\pingu@name-wings-side-left)
853         -- (0,\pingu@outer@off)
854         coordinate[pos=.5] (\pingu@name-head-center)
855         coordinate (\pingu@name-wings-side-right)
856         -- cycle;
857     % set the upper half
858     \path[save path=\pingu@upperpingu] (\pingu@name-head-center)
859         ++(0,\pingu@head@h@half)
860         coordinate (\pingu@name-head-top)
861         arc(90:450:\pingu@w@half) and \pingu@head@h@half
862         coordinate[pos=\pingu@anglehr] (\pingu@name-head-right)
863         coordinate[pos=\pingu@anglehl] (\pingu@name-head-left);
864
865     \coordinate (\pingu@name-belly-back) at (\pingu@w@half,-1cm);
866     \coordinate (\pingu@name-belly-center) at (\pingu@w@half,-.68\pingu@side@h@half);
867
868     % set the white above
869     \path[save path=\pingu@whitepingu] (\pingu@name-belly-back)
870         ++(-\pingu@w@half+\pingu@one,\pingu@lower@off)
871         \pingu@bend@belly{1.05cm-\pingu@side@h@half-\pingu@lower@off}
872         \pingu@arc@af
873         \pingu@arc@ag
874         \pingu@arc@ah
875         \pingu@arc@ai
876         \pingu@bend@belly{\pingu@side@h@half-1.05cm+\pingu@lower@off}
877         \pingu@arc@aj
878         -- cycle;
879
880     \coordinate (\pingu@name-eye-back-right)
881         at ([xshift=-.275cm,yshift=.25cm-\pingu@outer@off]\pingu@name-head-center);
882     \coordinate (\pingu@name-eye-back-left)
883         at ([xshift=.275cm,yshift=.25cm-\pingu@outer@off]\pingu@name-head-center);
884

```

```

885 \coordinate (\pingu@name-head-back-con-right)
886   at ([xshift=-\pingu@headcon@x,yshift=-\pingu@headcon@y]\pingu@name-eye-back-right);
887 \coordinate (\pingu@name-head-back-con-left)
888   at ([xshift=\pingu@headcon@x,yshift=-\pingu@headcon@y]\pingu@name-eye-back-left);
889 }

```

Important: Please note, that when writing the initial penguin, I did not intend to allow additional body shapes. Therefore, most of the defaults and “globals” must be defined here (alongside a lot of hardcoding).

Within the initialization of all penguin options, we define the main colors:

tikzpingus.sty, lines 474-478

```

474 body type/.code          = \pingu@bodytype@select{#1},
475 body main/.code          = \pingu@color{body@main}{#1},
476 body head/.code          = \pingu@color{body@head}{#1},
477 body/.style              = {/pingu/body main={#1}, /pingu/body head={#1}},
478 body front/.code         = \pingu@color{body@front}{#1},

```

With this, the body has three colors to work with: `\pingu@color@body@main`, `\pingu@color@body@head`, and `\pingu@color@body@front`.

C.8 The Feet

C.8.1 Important Commands

`\basicfeetbend`

This macro holds a hardcoded value to be used as the angle of rotation of the feet. Its default value is 45. The left foot uses the negative value. It may be removed in the future (or replaced by an option).

The only reason for the command being user accessible is, that it was the primary way to modify the penguin in a previous version of the package.

`\pingu@foot@single@w`

This dimension holds a hardcoded value to be used as the base foot width. Its default value is 2.98741pt. It may be removed in the future (or replaced by an option).

`\pingu@foot@single@h`

This dimension holds a hardcoded value to be used as the base foot height. Its default value is 5.69054pt. It may be removed in the future (or replaced by an option).

C.8.2 The Code

Similarly to the eyes, the feet of the penguin are controlled by two selections for the left, and the right foot. There are no extra coordinates necessary, therefore `\pingu@create@selection` receives no argument for the *none*-version:

tikzpingus.sty, lines 285-286

```

285 \pingu@create@selection{leftfoot}{}
286 \pingu@create@selection{rightfoot}{}

```

This defines the six commands: `\pingu@leftfoot@select`, `\pingu@leftfoot@add`, `\@pingu@drawer@leftfoot@`, `\pingu@rightfoot@select`, `\pingu@rightfoot@add`, and `\@pingu@drawer@rightfoot@`.

Within the initialization of all penguin options, we use these commands. Furthermore, `/pingu/feet` acts as a comfort option which operates on both selectors:



tikzpingus.sty, lines 459–471

```

459 left foot color/.code = \pingu@color{foot@left}{#1},
460 left foot/.code = \pingu@leftfoot@select{#1},
461 right foot color/.code = \pingu@color{foot@right}{#1},
462 right foot/.code = \pingu@rightfoot@select{#1},
463 feet/.style = {
464     /pingu/left foot={#1},
465     /pingu/right foot={#1}
466 },
467 feet color/.style = {
468     /pingu/left foot color={#1},
469     /pingu/right foot color={#1}
470 },
471 feet front/.is if = @pingu@feet@front,

```

The conditional `\if@pingu@feet@front` is defined above and defaults to `false`:

tikzpingus.sty, lines 419–420

```

419 \newif\if@pingu@feet@front
420 \newif\if@pingu@bill@above

```

Each foot has one color to work with: `\pingu@color@foot@left` for the left and `\pingu@color@foot@right` for the right foot.

C.8.3 Adding new Feet

Now on how to add new feet. While it is possible to use `\pingu@leftfoot@add` and `\pingu@rightfoot@add`, setting up the corresponding “foot” and “feet” comfort-methods is rather tedious. For this, i created the command `\pingu@feet@s...`

```
\pingu@feet@s{name}{left-foot}{right-foot} [default color=pingu@yellow]
```

Allows easily adding new feet.

The macro `\pingu@feet@s` is defined with the help of `\pingu@feet@s@` like this:

tikzpingus.sty, lines 1237–1252

```

1237 \long\def\pingu@feet@s#1#2#3{%
1238     \ifnextchar[%
1239         {\pingu@feet@s@{#1}{#2}{#3}}%
1240         {\pingu@feet@s@{#1}{#2}{#3}[pingu@yellow]}%
1241     }
1242 \long\def\pingu@feet@s@#1#2#3[#4]{%
1243     \pgfqkeys{/pingu}{%
1244         left foot #1/.style = {/pingu/left foot={#1}, /pingu/left foot color={##1}},%
1245         left foot #1/.default = {#4},%
1246         right foot #1/.style = {/pingu/right foot={#1}, /pingu/right foot color={##1}},%
1247         right foot #1/.default = {#4},%
1248         feet #1/.style = {/pingu/feet={#1}, /pingu/feet color={##1}},%
1249         feet #1/.default = {#4}%
1250     }%
1251     \pingu@leftfoot@add{#1}{#2}\pingu@rightfoot@add{#1}{#3}%
1252 }

```

As an example, this is the definition of the “normal feet” (`/pingu/feet normal`) in the code:

tikzpingus.sty, lines 1261–1298

```

1261 \pingu@feet@s{normal}{%
1262     \coordinate (@tmp) at ([xshift=-.5\pingu@foot@single@h]\pingu@name-foot-left);

```

```

1263 \pingu@block[rotate around={-25+\basicfeetbend:(@tmp)}]{\pingu@color@foot@left}
1264   (@tmp) ellipse [
1265     x radius=\pingu@foot@single@w,
1266     y radius=\pingu@foot@single@h
1267   ];
1268 \pingu@block[rotate around={\basicfeetbend:(\pingu@name-foot-left)}]
1269   {\pingu@color@foot@left} (\pingu@name-foot-left) ellipse [
1270     x radius=\pingu@foot@single@w,
1271     y radius=\pingu@foot@single@h
1272   ];
1273
1274 \coordinate (@tmp) at ([xshift=.5\pingu@foot@single@h]\pingu@name-foot-left);
1275 \pingu@block[rotate around={25+\basicfeetbend:(@tmp)}]{\pingu@color@foot@left}
1276   (@tmp) ellipse [
1277     x radius=\pingu@foot@single@w,
1278     y radius=\pingu@foot@single@h
1279   ];
1280 }{
1281 \coordinate (@tmp) at ([xshift=-.5\pingu@foot@single@h]\pingu@name-foot-right);
1282 \pingu@block[rotate around={-25-\basicfeetbend:(@tmp)}]{\pingu@color@foot@right}
1283   (@tmp) ellipse [
1284     x radius=\pingu@foot@single@w,
1285     y radius=\pingu@foot@single@h
1286   ];
1287 \pingu@block[rotate around={-\basicfeetbend:(\pingu@name-foot-right)}]
1288   {\pingu@color@foot@right} (\pingu@name-foot-right) ellipse [
1289     x radius=\pingu@foot@single@w,
1290     y radius=\pingu@foot@single@h
1291   ];
1292 \coordinate (@tmp) at ([xshift=.5\pingu@foot@single@h]\pingu@name-foot-right);
1293 \pingu@block[rotate around={25-\basicfeetbend:(@tmp)}]{\pingu@color@foot@right}
1294   (@tmp) ellipse [
1295     x radius=\pingu@foot@single@w,
1296     y radius=\pingu@foot@single@h
1297   ];
1298 }

```

As another example, this is the definition of the “simple feet” (`/pingu/feet simple`):

`tikzpingus.sty`, lines 1317–1329

```

1317 \pingu@feet@simple{%
1318   \pingu@block[rotate around={-\basicfeetbend:(\pingu@name-foot-left)}]
1319     {\pingu@color@foot@left} (\pingu@name-foot-left) ellipse [
1320       x radius=2.66\pingu@foot@single@w,
1321       y radius=\pingu@foot@single@h
1322     ];
1323 }{
1324   \pingu@block[rotate around={\basicfeetbend:(\pingu@name-foot-right)}]
1325     {\pingu@color@foot@right} (\pingu@name-foot-right) ellipse [
1326       x radius=2.66\pingu@foot@single@w,
1327       y radius=\pingu@foot@single@h
1328     ];
1329 }

```

C.9 The Wings

In contrast to the other selections, a wing consists of more than the drawing: the key for the wing, its color, the position of its tip and the angle of the items it holds. The helper `\@pingu@wings@s` registers all of it. Its six arguments are the name, the code for the left wing, the code for the right wing, `\pingu@wi@above` or `\pingu@wi@below` (whether the wing items are drawn above the wing), the angle of the wing items, and the angle with which the wing goes down. It creates the keys `left wing <name>`, `right wing <name>` and `wings <name>`. The position of the tips must be known to calculate the anchors, so they are registered separately with `\pingu@wingend@add`. This is the wave:

tikzpingus.sty, lines 1129–1147

```

1129 \@pingu@wings@s{wave}{%
1130   \coordinate (\pingu@name-wing-left-start) at ([yshift=-5.85mm]\pingu@name-wings-side-left);
1131   \scope[xshift=-\p@] % shift to avoid rounding errors in pdf viewers
1132   \pgfinterruptboundingbox
1133   \path[/pingu/@invclip] \pingu@path@lowerpingu;
1134   \endpgfinterruptboundingbox
1135   \pingu@block[rotate around={-39:(\pingu@name-wing-left-start)}]{\pingu@color@left@wing} (\pingu@name-wing-
left-start) ellipse [x radius=2.2mm, y radius=.935mm] +(90:.22cm and .935cm) coordinate (\pingu@name-wing-
left-tip) +(150:.22cm and .935cm) coordinate (\pingu@name-wing-left-cap-up) +(-3:.22cm and .935cm)
coordinate (\pingu@name-wing-left-cap-down) +(90:.175cm and .75cm) coordinate (\pingu@name-wing-left);
1136   \pingu@wing@compensator{\pingu@block{\pingu@color@left@wing} (\pingu@name-wing-left-start)+(.03mm,4.6mm)
to[out=-80,in=220] (\pingu@name-wing-left-cap-up) -- (\pingu@name-wing-left-start) -- cycle;
1137   \pingu@block{\pingu@color@left@wing} (\pingu@name-wing-left-start)+(.085mm,-4.4mm) to[out=80,in=-131] (
\pingu@name-wing-left-cap-down) -- (\pingu@name-wing-left-start) -- cycle;}
1138   \endscope}{%
1139   \coordinate (\pingu@name-wing-right-start) at ([yshift=-5.85mm]\pingu@name-wings-side-right);
1140   \scope[xshift=\p@] % shift to avoid rounding errors in pdf viewers
1141   \pgfinterruptboundingbox
1142   \path[/pingu/@invclip] \pingu@path@lowerpingu;
1143   \endpgfinterruptboundingbox
1144   \pingu@block[rotate around={39:(\pingu@name-wing-right-start)}]{\pingu@color@right@wing} (\pingu@name-wing-
right-start) ellipse [x radius=.22cm, y radius=.935cm] +(90:.22cm and .935cm) coordinate (\pingu@name-wing-
right-tip) +(30:.22cm and .935cm) coordinate (\pingu@name-wing-right-cap-up) +(177:.22cm and .935cm)
coordinate (\pingu@name-wing-right-cap-down) +(90:.175cm and .775cm) coordinate (\pingu@name-wing-right);
1145   \pingu@wing@compensator{\pingu@block{\pingu@color@right@wing} (\pingu@name-wing-right-start)+(-.03mm,4.6mm)
to[out=-100,in=-45] (\pingu@name-wing-right-cap-up) -- (\pingu@name-wing-right-start) -- cycle;
1146   \pingu@block{\pingu@color@right@wing} (\pingu@name-wing-right-start)+(-.1mm,-4.4mm) to[out=100,in=-50] (
\pingu@name-wing-right-cap-down) -- (\pingu@name-wing-right-start) -- cycle;}
1147   \endscope}{\pingu@wi@above}{-10}{-110}

```

The wings have to start at the coordinate of `\pingu@name-wing-left-start`. The two paths with `\pingu@path@lowerpingu` and the shifts of one point exist to hide the seam between the wing and the body, as explained for the clips in the body.

C.10 Extras

Everything else you can put on a penguin is an *extra*. An extra consists of some keys, a switch `\ifpingu@x@<name>@` and the code to draw. `\pingu@create@extra` creates all of it: it takes the name, the keys (which call `\pingu@set@extra` to switch the extra on) and the defaults, which are appended to `/pingu/extra defaults` so that every penguin starts with a clean set. Extras behind the body are drawn by `\pingu@draw@eb@<name>` and extras in front of it by `\pingu@draw@ea@<name>`. The lists with the order of the extras (implementation (lines 1467–1576) and implementation (lines 1580–1674)) decide what covers what, a later extra is drawn above an earlier one. They are the only place to change the order of two extras, and a new extra has to be added to one of them. The face (glasses, mask, ...) is in front of the headphones, which are behind the hats, and the pants are behind the belt. This is the cup, an extra below the head:

tikzpingus.sty, lines 1494–1516

```

1494 \pingu@create@extra{cup}{%
1495   cup/.code      = {\pingu@set@extra{cup}{true}\pingu@color{cup}{#1}},
1496   cup/.default   = pingu@green,
1497   cup steam/.code = \pingu@color{cup@steam}{#1},%
1498   cup steam/.default = lightgray!90!orange,%
1499   cup steam shift/.code = \def\pingu@x@cup@steam@shift{#1},%
1500   cup steam opacity/.code = \def\pingu@x@cup@steam@opacity{#1},%
1501   cup straw/.code = \pingu@color{cup@straw}{#1}%
1502 }{cup straw=\pingu@color@cup,cup steam=\@pingu@none,cup steam shift=0mm,cup steam opacity=.55}
1503
1504 \def\pingu@draw@eb@cup{%
1505   \pingu@block[rounded corners=.25mm]{\pingu@color@cup} (\pingu@name-belly-center)+(-.325cm,.275cm) to[bend
right=2] ++(.5mm,-.675cm) to[bend right=8] ++(.55cm,\z@) to[bend right=2] ++(.5mm,.675cm);
1506   \path (\pingu@name-belly-center)+(.0,.275cm) -- (\pingu@name-bill) coordinate[pos=.5,xshift=-2mm] (
\pingu@name-cup-straw-outer-angle);
1507   \path (\pingu@name-belly-center)+(-.115cm,.275cm) coordinate (\pingu@name-straw-outer-start);
1508   \pingu@block[rounded corners=.15mm]{\pingu@color@cup@straw} (\pingu@name-straw-outer-start) -- (\pingu@name-
cup-straw-outer-angle) -- (\pingu@name-bill) -- ++(.5mm,0) -- ([xshift=.5mm]\pingu@name-cup-straw-outer-
angle) -- ([xshift=.5mm]\pingu@name-straw-outer-start) -- cycle;
1509   \pingu@block{\pingu@color@cup} (\pingu@name-belly-center)+(.0,.275cm) ellipse [x radius=.325cm,y radius=.5
mm];
1510
1511   \pgfonlayer{\pingu@L@foreground}
1512   \pingu@block[rounded corners=.05mm,opacity=\pingu@x@cup@steam@opacity]{\pingu@color@cup@steam} (\pingu@name-
belly-center)+(.15mm+\pingu@x@cup@steam@shift,4mm) coordinate(@) to[out=100,in=260] ++(.3mm,3.75mm) -- ++(
.1mm,0) to[out=280,in=80] ([xshift=.1mm]@) -- cycle;
1513   \pingu@block[rounded corners=.05mm,opacity=\pingu@x@cup@steam@opacity]{\pingu@color@cup@steam} (\pingu@name-
belly-center)+(0mm+\pingu@x@cup@steam@shift,3.33mm) coordinate(@) to[out=100,in=260] ++(.2mm,2.75mm) --
++(.1mm,0) to[out=280,in=80] ([xshift=.1mm]@) -- cycle;
1514   \pingu@block[rounded corners=.05mm,opacity=\pingu@x@cup@steam@opacity]{\pingu@color@cup@steam} (\pingu@name-
belly-center)+(-1.25mm+\pingu@x@cup@steam@shift,3.65mm) coordinate(@) to[out=100,in=260] ++(.3mm,3.75mm)
-- ++(.1mm,0) to[out=280,in=80] ([xshift=.1mm]@) -- cycle;
1515   \endpgfonlayer
1516 }

```

And the halo above the head, which uses a layer for the glow:

tikzpingus.sty, lines 1619–1636

```

1619 \newif\if@pingu@x@halo@glow
1620 \newif\if@pingu@x@halo@above
1621 \pingu@create@extra{halo}{%
1622     halo/.code      = \pingu@set@extra{halo}{true}\pingu@color{halo}{#1},
1623     halo/.default   = pingu@lightblue,
1624     halo raise/.code = \def\pingu@x@halo@raise{#1},
1625     halo above/.is if = @pingu@x@halo@above,
1626     halo glow/.is if = @pingu@x@halo@glow% only active if glows loaded
1627 }{halo glow=\ifpingu@setup@glows@ true\else false\fi,halo raise=0mm,halo above=false}
1628
1629 \def\pingu@draw@ea@halo{%
1630 \if@pingu@x@halo@above\pingu@layer{foreground}[]\fi
1631 \pingu@block[even odd rule]{\pingu@color@halo} ([yshift=\pingu@x@halo@raise]\pingu@name-head-top) ellipse [x
    radius=\pingu@w@half-\pingu@one,y radius=2.5mm] ellipse [x radius=\pingu@w@half-1.95mm,y radius=1.55mm];
1632 \if@pingu@x@halo@glow
1633 \pingu@block[even odd rule,/pingu/@glow]{\pingu@color@halo} ([yshift=\pingu@x@halo@raise]\pingu@name-head-top)
    ellipse [x radius=(\pingu@w@half-1.95mm)*1.7,y radius=1.55mm*2.35] ellipse [x radius=\pingu@w@half-1.95mm,y
    radius=1.55mm];
1634 \fi
1635 \if@pingu@x@halo@above\endpingu@layer{foreground}\fi
1636 }

```

Extras for the wings are created with `\pingu@create@wing@extra`, which creates the keys for both wings:

tikzpingus.sty, lines 1719–1734

```

1719 \pingu@create@wing@extra{cane}{%
1720     cane \@lr/.code      = {\pingu@set@extra{cane\@lr}{true}\pingu@color{cane\@lr}{##1}},
1721     cane \@lr/.default   = pingu@bronze,
1722     cane \@lr\space raise/.code = {\@namedef{pinug@x@cane@raise@\@lr}{##1}},
1723     cane \@lr\space length/.code = {\@namedef{pinug@x@cane@length@\@lr}{##1}},
1724 }{cane \@lr\space raise=0mm,cane \@lr\space length=15mm}
1725
1726 % #1 wing
1727 \def\pingu@draw@wi@cane#1{%
1728     \pingu@setup@wing{cane}{#1}%
1729     \if@pingu@wing@item@side@left@
1730         \pingu@block[thick,rotate around={\@angle+5:(\pingu@name-wing-#1)}]{\@col}{(\pingu@name-wing-#1)+(0,-7
            mm+\pinug@x@cane@raise@left) -- ++(0,\csname pinug@x@cane@length@#1\endcsname) arc(180:0:2.25mm and 2.2mm)
            arc(360:180:.3mm and .275mm) arc(0:180:1.65mm and 1.65mm) -- ++(0,-\csname pinug@x@cane@length@#1\endcsname
            ) arc(360:180:.3mm and .275mm) -- cycle;
1731     \else
1732         \pingu@block[thick,rotate around={\@angle-5:(\pingu@name-wing-#1)}]{\@col}{(\pingu@name-wing-#1)+(0,-7
            mm+\pinug@x@cane@raise@right) -- ++(0,\csname pinug@x@cane@length@#1\endcsname) arc(0:180:2.25mm and 2.2mm)
            arc(180:360:.3mm and .275mm) arc(180:0:1.65mm and 1.65mm) -- ++(0,-\csname pinug@x@cane@length@#1
            \endcsname) arc(180:360:.3mm and .275mm) -- cycle;
1733     \fi
1734 }

```

All extras are grouped in libraries (christmas-lib, hats-lib, ...) which are regions of the source, as the map in Section C.2 shows. A library is only a convenient place to look: nothing is loaded on demand.

C.11 Anchors and Placement

C.11.1 Important Commands

`\pingu@bbox@left`

Together with `\pingu@bbox@right`, `\pingu@bbox@bottom`, `\pingu@bbox@top`, `\pingu@bbox@midx`, and `\pingu@bbox@midy` this dimension spans the silhouette of the body, the box every anchor refers to.

`\pingu@bbox@right`

See `\pingu@bbox@left`.

`\pingu@bbox@bottom`

See `\pingu@bbox@left`.

`\pingu@bbox@top`

See `\pingu@bbox@left`.

`\pingu@bbox@midx`

See `\pingu@bbox@left`.

`\pingu@bbox@midy`

See `\pingu@bbox@left`.

`\pingu@bbox@update`

Recalculates the six dimensions of the silhouette. Only `/pingu/height` affects them.

`\pingu@anchor@add{name}{x}{y}`

Registers a new named anchor for `/pingu/anchor`.

`\pingu@place@anchors`

Creates the outer coordinates (center, north, ..., and `anchor`) of the current penguin.

C.11.2 The Code

The silhouette and all named anchors:

`tikzpingus.sty`, lines 295–377

```

295 \newdimen\pingu@shoulder@x
296 \newdimen\pingu@bbox@left \newdimen\pingu@bbox@right
297 \newdimen\pingu@bbox@bottom \newdimen\pingu@bbox@top
298 \newdimen\pingu@bbox@midx \newdimen\pingu@bbox@midy
299 % only the height is configurable, so re-assigning is cheap
300 \def\pingu@bbox@update{%
301   \pingu@bbox@left\z@
302   \pingu@bbox@right2\pingu@w@half
303   \pingu@bbox@bottom-\dimexpr\pingu@side@h@half+\pingu@foot@outer@h\relax
304   \pingu@bbox@top\dimexpr\pingu@outer@off+\pingu@head@h@half\relax
305   \pingu@bbox@midx\pingu@w@half
306   \pingu@bbox@midy\dimexpr(\pingu@bbox@top+\pingu@bbox@bottom)/2\relax

```

```

307 \pingu@shoulder@x\dimexpr2\pingu@w@half-.10674pt\relax
308 }
309
310 % An anchor is a point of the penguin that is moved onto the origin.
311 \let\@pingu@anchors@\empty
312 % #1 name | #2 x | #3 y
313 \def\pingu@anchor@add#1#2#3{%
314 \pingu@csappto\@pingu@anchors@{"#1"}%
315 \expandafter\def\csname @pingu@anchor@#1@\endcsname{{#2}{#3}}%
316 }
317 \pingu@anchor@add{origin} \pingu@bbox@left {0pt}
318 \pingu@anchor@add{center} \pingu@bbox@midx \pingu@bbox@midy
319 \pingu@anchor@add{north} \pingu@bbox@midx \pingu@bbox@top
320 \pingu@anchor@add{south} \pingu@bbox@midx \pingu@bbox@bottom
321 \pingu@anchor@add{east} \pingu@bbox@right\pingu@bbox@midy
322 \pingu@anchor@add{west} \pingu@bbox@left \pingu@bbox@midy
323 \pingu@anchor@add{north east} \pingu@bbox@right\pingu@bbox@top
324 \pingu@anchor@add{north west} \pingu@bbox@left \pingu@bbox@top
325 \pingu@anchor@add{south east} \pingu@bbox@right\pingu@bbox@bottom
326 \pingu@anchor@add{south west} \pingu@bbox@left \pingu@bbox@bottom
327 \pingu@anchor@add{head} \pingu@bbox@midx \pingu@outer@off
328 \pingu@anchor@add{head top} \pingu@bbox@midx \pingu@bbox@top
329 \pingu@anchor@add{bill} \pingu@bbox@midx {-0.05mm}
330 \pingu@anchor@add{belly} \pingu@bbox@midx {-0.68*\pingu@side@h@half}
331 \pingu@anchor@add{feet} \pingu@bbox@midx \pingu@bbox@bottom
332 \pingu@anchor@add{left shoulder} \pingu@shoulder@x \pingu@outer@off
333 \pingu@anchor@add{right shoulder}\pingu@bbox@left \pingu@outer@off
334
335 % The wings are the one part whose reach depends on the selected style, so every
336 % style registers where its ends sit relative to the shoulder it grows from.
337 % #1 style | #2/#3 left end | #4/#5 left tip | #6/#7 right end | #8/#9 right tip
338 \def\pingu@wingend@add#1#2#3#4#5#6#7#8#9{%
339 \namedef{\pingu@wingend@#1@leftx}{#2}\namedef{\pingu@wingend@#1@lefty}{#3}%
340 \namedef{\pingu@wingend@#1@lefttipx}{#4}\namedef{\pingu@wingend@#1@lefttipy}{#5}%
341 \namedef{\pingu@wingend@#1@rightx}{#6}\namedef{\pingu@wingend@#1@righty}{#7}%
342 \namedef{\pingu@wingend@#1@righttipx}{#8}\namedef{\pingu@wingend@#1@righttipy}{#9}%
343 }
344 % #1 left/right | #2 x, y, tipx, or tipy
345 \def\pingu@wingend#1#2{\csname @pingu@wingend@\csname @pingu@select@#1wing@\endcsname @#1#2\endcsname}
346 \pingu@anchor@add{left wing} \pingu@shoulder@x+\pingu@wingend{left}{x} \pingu@outer@off+
347 \pingu@wingend{left}{y}
348 \pingu@anchor@add{right wing} \pingu@bbox@left+\pingu@wingend{right}{x} \pingu@outer@off+
349 \pingu@wingend{right}{y}
350
351 \pingu@anchor@add{left wing tip} \pingu@shoulder@x+\pingu@wingend{left}{tipx} \pingu@outer@off+
352 \pingu@wingend{left}{tipy}
353 \pingu@anchor@add{right wing tip} \pingu@bbox@left+\pingu@wingend{right}{tipx} \pingu@outer@off+
354 \pingu@wingend{right}{tipy}
355
356 \newdimen\pingu@anchor@dx \newdimen\pingu@anchor@dy
357 % shift by the negated anchor; a value without a unit is given in |cm|
358 \def\pingu@anchor@shift#1#2{%
359 \pgfmathparse{-1*(#2)}%
360 \ifpgfmathunitsdeclared#1\pgfmathresult pt\relax\else#1\pgfmathresult cm\relax\fi

```

```

356 }
357 \def\pingu@anchor@set#1#2{\def\pingu@anchor@x{#1}\def\pingu@anchor@y{#2}}
358 \def\pingu@anchor@select#1{%
359   \ifcsname @pingu@anchor@#1\endcsname
360     \expandafter\expandafter\expandafter\pingu@anchor@set\csname @pingu@anchor@#1\endcsname
361   \else\pingu@anchor@free{#1}\fi
362 }
363 % any unknown name has to be a point |(<x>,<y>)|, parentheses optional
364 \def\pingu@anchor@free#1{%
365   \pgfutil@in@{,}{#1}%
366   \ifpgfutil@in@
367     \pgfutil@in@{,}{#1}%
368     \ifpgfutil@in@{\let\pingu@anchor@next\pingu@anchor@free@p
369       \else\let\pingu@anchor@next\pingu@anchor@free@c\fi
370     \else\let\pingu@anchor@next\pingu@anchor@err\fi
371     \pingu@anchor@next#1\@nil
372 }
373 \def\pingu@anchor@free@p(#1,#2)#3\@nil{\pingu@anchor@set{#1}{#2}}
374 \def\pingu@anchor@free@c#1,#2\@nil{\pingu@anchor@set{#1}{#2}}
375 \def\pingu@anchor@err#1\@nil{%
376   \PackageError{\tikzpingus@filename}{There is no '#1' anchor. Choose one of '\@pingu@anchors@' or pass a
     point like '(1cm,2mm)'}{Or ask the package maintainer :D}%
377 }

```

Relative placement joins the two anchors facing each other:

[tikzpingus.sty](#), lines 381-400

```

381 % Relative placement joins the two anchors facing each other, e.g. \ |right of|
382 % our |south west| with their |south east|, keeping both on the same ground.
383 \def\pingu@relative#1#2{\def\pingu@rel@dir{#1}\pingu@relative@#2::\@nil}
384 \def\pingu@relative@#1:#2:#3\@nil{%
385   \def\pingu@rel@name{#1}\def\pingu@rel@dist{#2}%
386   \ifx\pingu@rel@dist\empty\def\pingu@rel@dist{0pt}\fi
387   \csname pingu@relative@\pingu@rel@dir\endcsname
388 }
389 % #1 coordinate our own anchor is moved to
390 \def\pingu@relative@to#1{%
391   \edef\pingu@rel@tmp{\noexpand\pgfqkeys{/pingu}{@place={shift={(#1)}}}\pingu@rel@tmp
392 }
393 \def\pingu@relative@right{\pingu@anchor@select{south west}%
394   \pingu@relative@to{[xshift=\pingu@rel@dist]\pingu@rel@name-south east}}
395 \def\pingu@relative@left{\pingu@anchor@select{south east}%
396   \pingu@relative@to{[xshift=-\pingu@rel@dist]\pingu@rel@name-south west}}
397 \def\pingu@relative@above{\pingu@anchor@select{south}%
398   \pingu@relative@to{[yshift=\pingu@rel@dist]\pingu@rel@name-north}}
399 \def\pingu@relative@below{\pingu@anchor@select{north}%
400   \pingu@relative@to{[yshift=-\pingu@rel@dist]\pingu@rel@name-south}}

```

Both are hooked into these keys:

tikzpingus.sty, lines 451–456

```

451 anchor/.code = \pingu@anchor@select{#1},
452 @place/.code = {\pgfqkeys{/pingu}{@pingu@place/.style={#1}}},
453 right of/.code = \pingu@relative{right}{#1},
454 left of/.code = \pingu@relative{left}{#1},
455 above of/.code = \pingu@relative{above}{#1},
456 below of/.code = \pingu@relative{below}{#1},

```

C.12 The Bends

`to[bend right=<angle>]` is expensive, so all body bends use precomputed control points:

tikzpingus.sty, lines 129–145

```

129 % |to[bend right=<angle>]| re-derives its control points on every use. As all
130 % body bends are vertical and their angle is known at load time, the two
131 % factors yielding the same control points are computed once instead.
132 \def\pingu@bend@factor{.3899534}
133 % #1 macro to define | #2 angle, as given to |bend right|
134 \def\pingu@newbend#1#2{%
135   \pgfmathsetmacro\pingu@bend@x{\pingu@bend@factor*sin(#2)}%
136   \pgfmathsetmacro\pingu@bend@y{\pingu@bend@factor*cos(#2)}%
137   \edef#1##1{\noexpand\pingu@bend@v{\pingu@bend@x}{\pingu@bend@y}{##1}}%
138 }
139 % #1 x factor | #2 y factor | #3 signed height of the segment
140 \def\pingu@bend@v#1#2#3{.. controls +({#1*(#3)},{#2*(#3)}) and +({#1*(#3)},{-#2*(#3)}) .. ++(0,#3)}
141
142 \pingu@newbend\pingu@bend@body{\pingu@bend}
143 \pingu@newbend\pingu@bend@belly{.8*\pingu@bend}
144 \pingu@newbend\pingu@bend@chubby{1.65*\pingu@bend}
145 \pingu@newbend\pingu@bend@legacy{1.25*\pingu@bend}

```

Key Overview

This index shows all of the options and commands that a normal user of the package may use! See the [Internals Overview](#) for an overview of keys and commands used behind the scenes. Page numbers on the lowest level are hidden, but they can still be used as clickable hyperlinks.

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